



Ch 2: Knowledge Representation and Reasoning

Artificial Intelligence

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Overview

Ch 2: Knowledge Representation and Reasoning (7 hrs)

1. First-Order Logic (FOL) and reasoning
2. Probabilistic reasoning
 - 2.1.1. Hidden Markov Models
 - 2.1.2. Dynamic Bayesian Networks
3. Ontological Engineering
4. Semantic Networks
5. Description Logic
6. Fuzzy Logic-
 - 6.1.1. Fuzzy Inference Systems
 - 6.1.1.1. Mamdani
 - 6.1.1.2. Sugeno
 - 6.1.1.3. Tsukamoto
 - 6.1.2. Defuzzification techniques

1. Fuzzy Inference Systems

1.1. Mamdani

1.2. Sugeno

1.3. Tsukamoto

2. Defuzzification techniques

FUZZY LOGIC

Fuzzy Logic

F^{uzzy} → un-clear or ambiguous

C^{risp} → **antonym of fuzzy** → clear, clean, and sharp.

- Fuzzy logic was proposed by **Zadeh** in 1965
- It was applied in **steam engine** control by **Mamdani** in 1974.
- Fuzzy logic was made practically useful in Japan in the **1990s**.



[Click Here to open in youtube](#)

2009 Benjamin Franklin Medal
Winner: Lotfi A. Zadeh



2009
Benjamin Franklin Medal
in Electrical Engineering

Lotfi A. Zadeh

The Franklin Institute



Fuzzy Logic

- Fuzzy logic is not fuzzy
- Fuzzy logic is a *precise logic of imprecision*
- Fuzzy logic has the capability to reason and make rational decisions in an environment of
 - imprecision
 - uncertainty
 - partiality of truth.
- Fuzzy logic is designed to deal with *imperfect information*



Fuzzy Logic

From Prof. Lotfi Zadeh Slides

- In bivalent logic, *BL*, (*Classical or Crisp Logic*)
 - *truth is bivalent,*
 - *every proposition, p , is either true or false*
 - *no degrees of truth is allowed*
- In multivalent logic, *ML*, (*Fuzzy Logic*)
 - *truth is a matter of degree*
 - *It is called *fuzzy logic*, *FL*:*

Fuzzy logic is much more general than traditional logical systems.



Fuzzy Logic

What is Fuzzy logic?

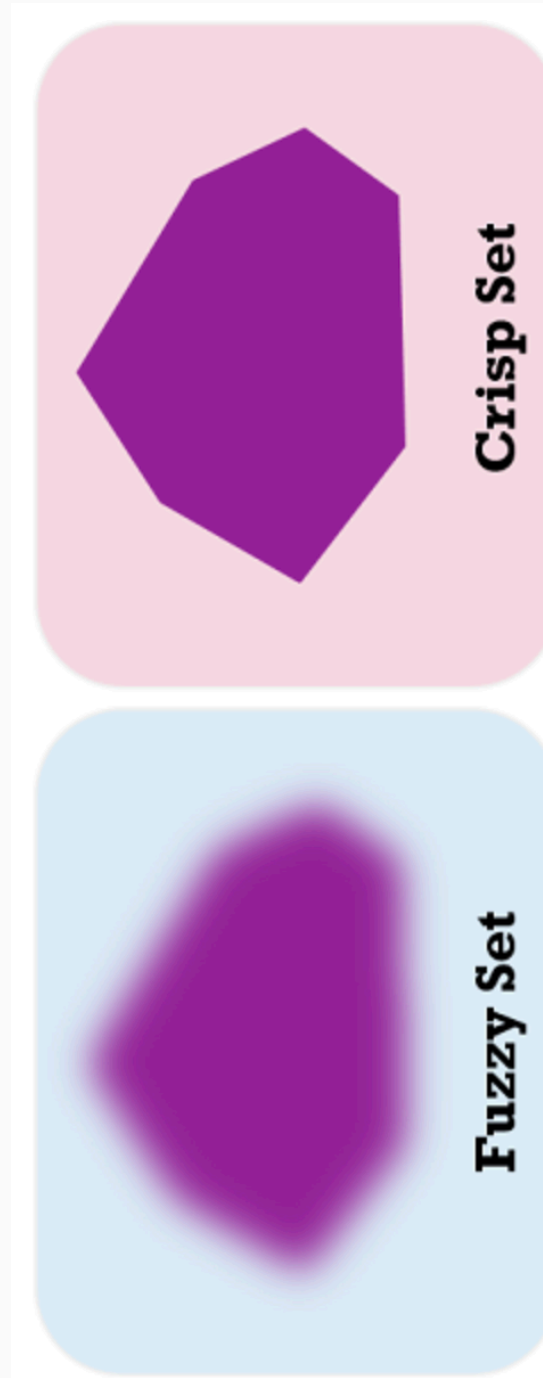
Fuzzy logic (FL) is aimed at a formalization of modes of reasoning which are approximate rather than exact

- Exact (Crisp):**
- All men are mortal
 - Socrates is a man
 - Socrates is mortal**
- Approximate:**
- Most Swedes are tall
 - Magnus is a Swedes
 - It is likely that Magnus is tall**

Fuzzy Logic vs Crisp Logic

Crisp logic is the same as boolean logic (either 0 or 1). Either a statement is true (1) or it is not (0), meanwhile fuzzy logic captures the degree to which something is true.

Difference between crisp logic and fuzzy logic:



Crisp (*Classical*) Logic:

- Binary logic (1 represent absolute truth value and 0 represent absolute false value.)
- shows the state either occur or non occur (membership)

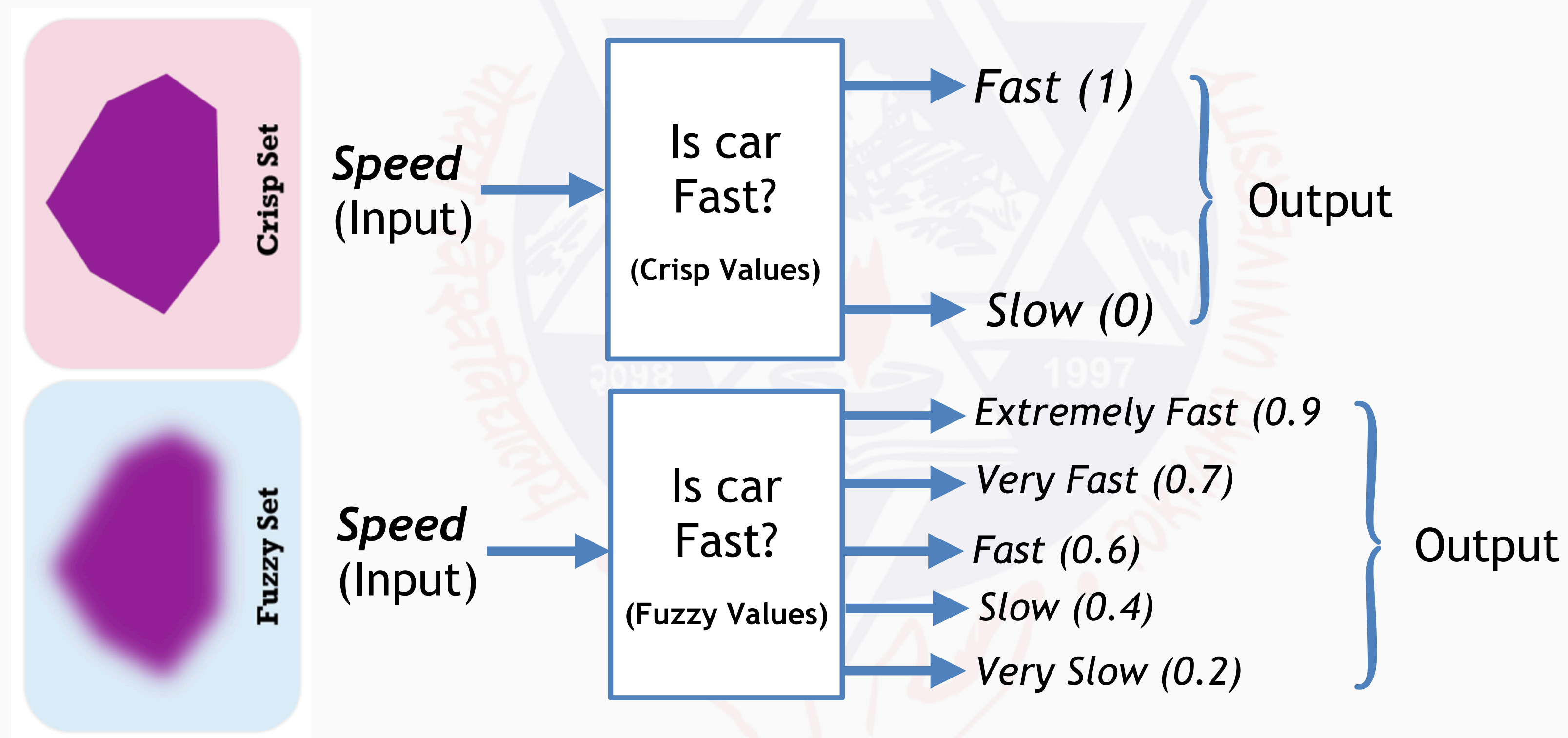
Fuzzy logic :

- Continuous valued logic (there are intermediate values present which are partially true or partially false)
- Consider about degree of membership (Membership function)



Fuzzy Logic vs Crisp Logic

Consider the statement: “*The car is moving fast.*”



Linguistic Variables

- ***Linguistic variables*** describe the system inputs and outputs in natural language terms such as "Cold," "Warm," "Hot" etc. to represent temperature.
- These variables are used in fuzzy sets to represent degree of membership.
- Examples of linguistic variables (for Car speed, temperature and fan speed):
 - **Car Speed:** *Extremely fast, very fast, fast, slow, very slow*
 - **Temperature:** *Cold, Warm, Hot*
 - **Fan Speed:** *Slow, Medium, Fast*



Classical Sets

- A *classical set* is a collection of objects in a given range with a *sharp boundary*.
- An object can either *belong to* the set or *not belong to* the set.
- *Example*: A set of vowel alphabet in English, v .

$$v = \{a, e, i, o, u\}$$

- Each *object* in a set is called a *member* or *element* of the set. For example, object i is the *member* of set v and the object m is not.

$$i \in v$$

$$m \notin v$$

- A classical *set* has two groups: *members* or *non-members*. Hence, *no partial membership* exists in classical *sets*.



Classical Sets

- Suppose we have three sets:

$$X = \{x : x \text{ is a alphabet in English language}\}$$

$$Y = \{1,0\} \text{ and}$$

$$v = \{a, e, i, o, u\}$$

- If an object x belongs to set v , then the **membership function** $\mu_v(x)$ for the set v can be expressed as

$$\mu_v(x) = \begin{cases} 1 & \text{if } x \in v \\ 0 & \text{if } x \notin v \end{cases}$$

- This **membership function** $\mu_v(x)$ for set v , is exactly a mapping from the element x in the universe X to one of two elements in universe $Y = \{1,0\}$



Classical Sets

- Using the *membership function* $\mu_v(x)$, the following objects have the following membership values for set v :

$\mu_v(a) = 1$	$\mu_v(f) = 0$	$\mu_v(k) = 0$	$\mu_v(p) = 0$	$\mu_v(u) = 1$
$\mu_v(b) = 0$	$\mu_v(g) = 0$	$\mu_v(l) = 0$	$\mu_v(q) = 0$	$\mu_v(v) = 0$
$\mu_v(c) = 0$	$\mu_v(h) = 0$	$\mu_v(m) = 0$	$\mu_v(r) = 0$	$\mu_v(w) = 0$
$\mu_v(d) = 0$	$\mu_v(i) = 1$	$\mu_v(n) = 0$	$\mu_v(s) = 0$	$\mu_v(x) = 0$
$\mu_v(e) = 1$	$\mu_v(j) = 0$	$\mu_v(o) = 1$	$\mu_v(t) = 0$	$\mu_v(y) = 0$
				$\mu_v(z) = 0$

- In a crisp set, an object is either a member or not a member (each member in crisp set is mapped either to 1 or 0). There is a *sharp boundary* in crisp set.



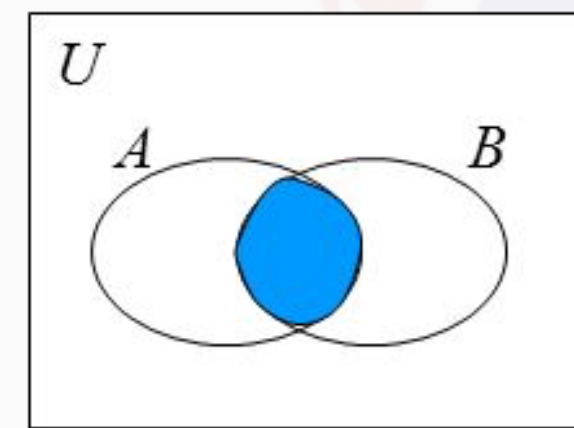
Classical Set Operations

Intersection of A and B $A \cap B = A(x) \cap B(x)$

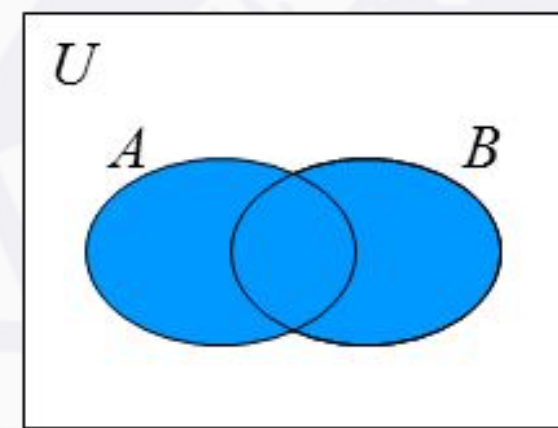
Union of A and B $A \cup B = A(x) \cup B(x)$

Complement of A $A'(x) = 1 - A$

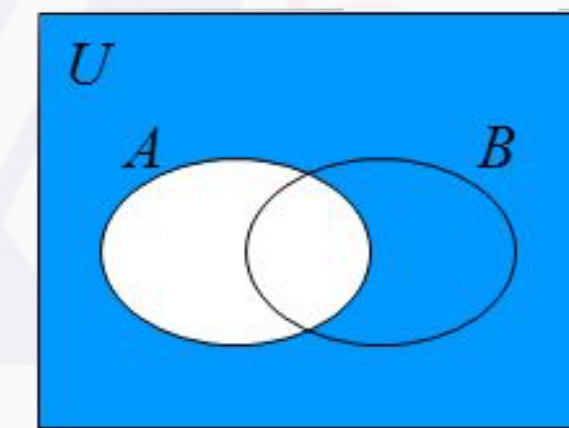
Difference of A and B $A - B = A(x) \cap B'(x)$



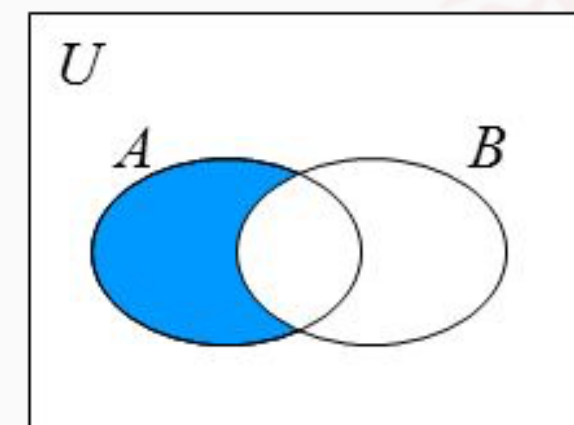
$A \cap B$



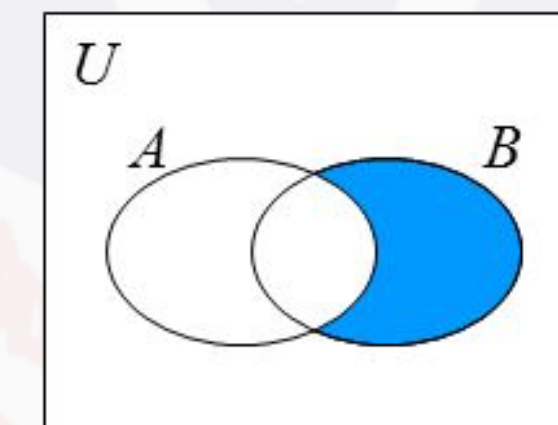
$A \cup B$



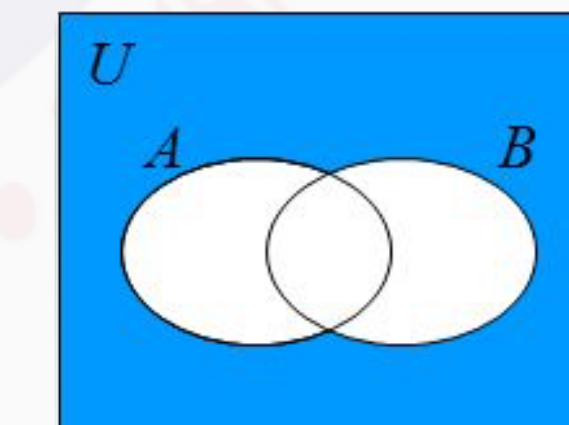
A'



$A \cap (B')$



$(A') \cap B$

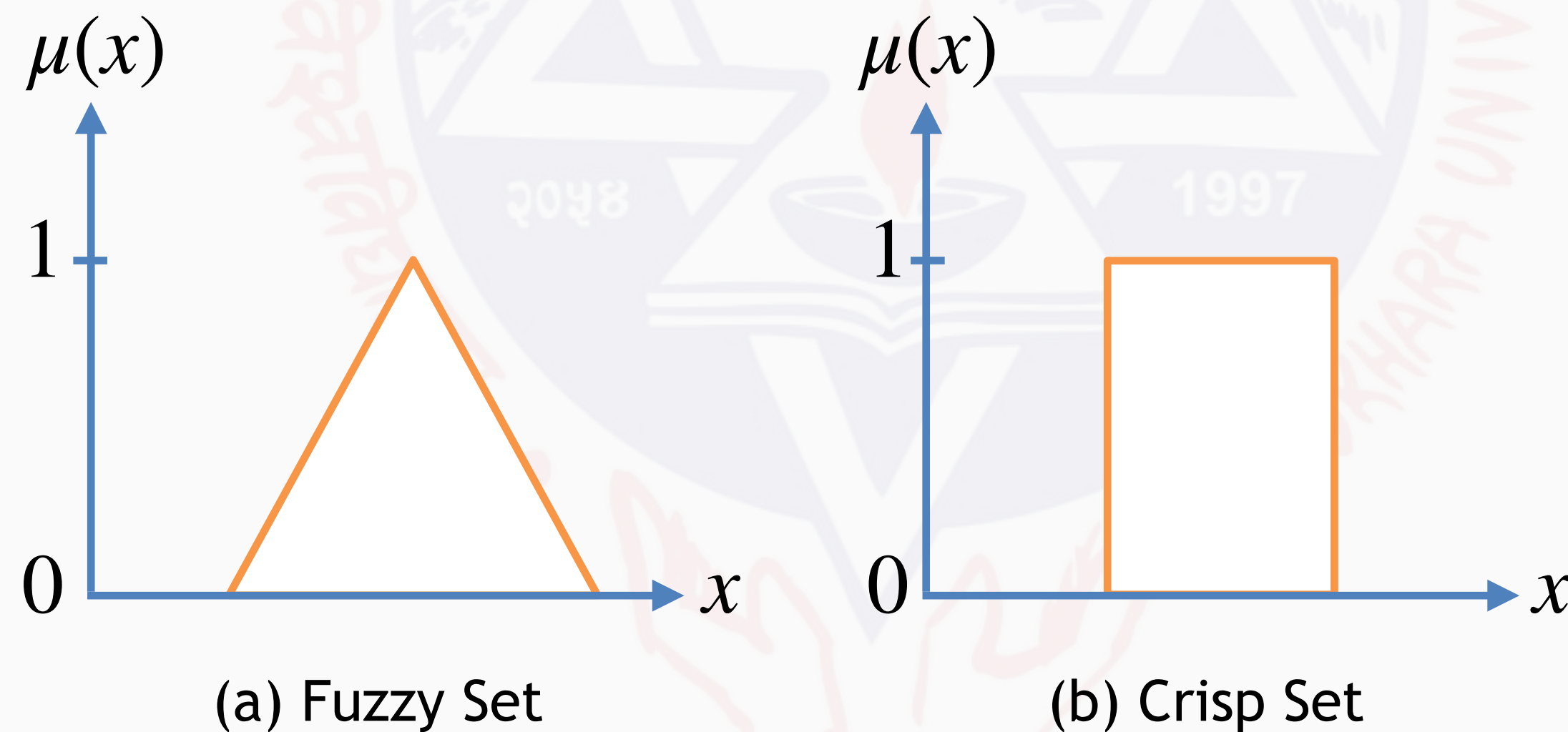


$(A \cup B)'$



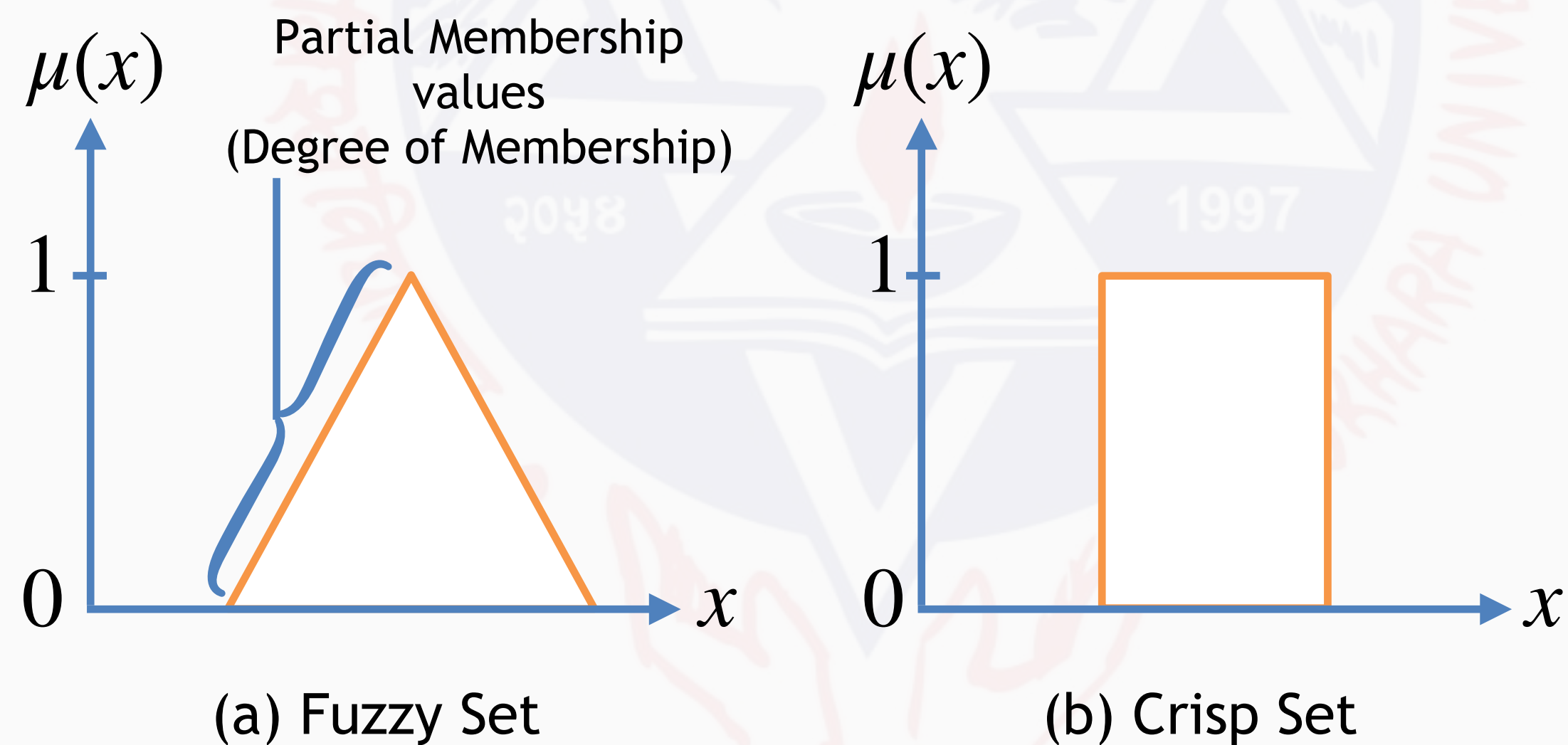
Fuzzy Sets

- A **fuzzy set** is a collection of elements where each element has a degree of membership, ranging from 0 (not a member) to 1 (full member).
- A fuzzy set does not have a clear boundary. It allows partial membership.
- Fuzzy sets are the extension of the classical sets



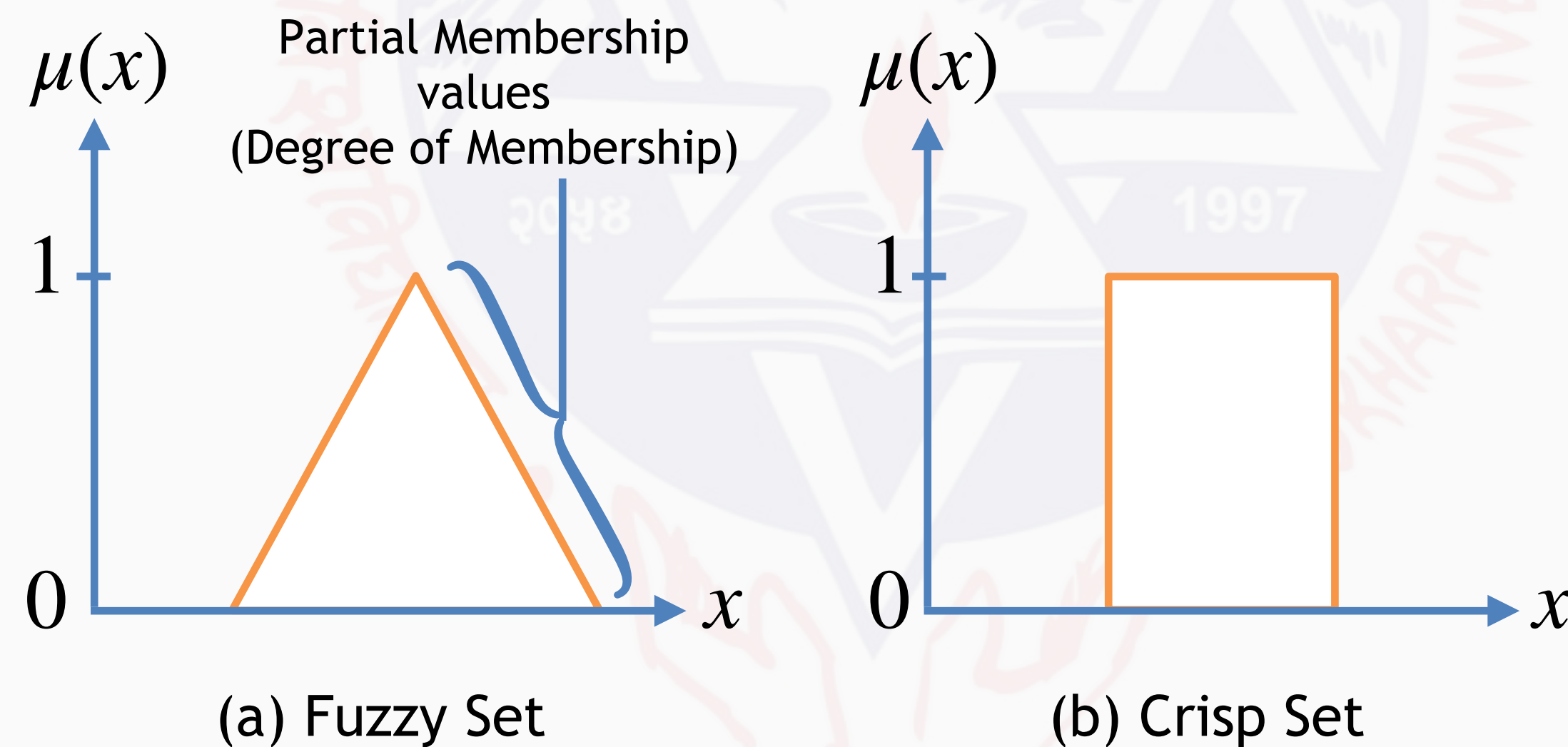
Fuzzy Sets

- **Membership function** describes the membership of an element in a fuzzy set.
- **Membership function** assigns a value between 0 and 1
- The values between 0 and 1 indicates the **degree of membership**.



Fuzzy Sets

- The membership function (μ) defines the degree of membership for each input.
- A member of a fuzzy set can fall under multiple subsets of the fuzzy set with various partial membership values.

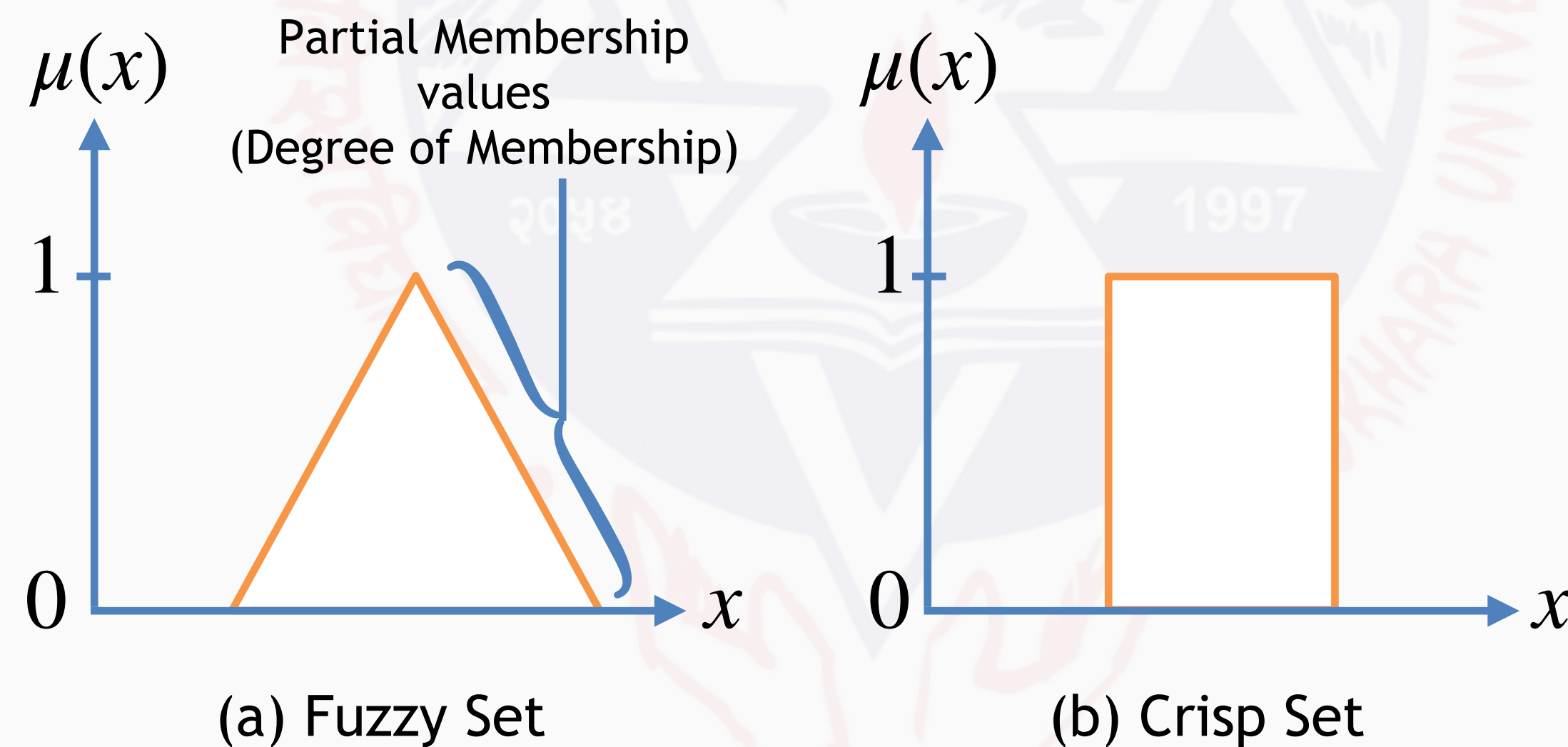


Fuzzy Sets

- Mathematically, a fuzzy set $\bar{A}$ in the universe U can be defined as a set of ordered pair as

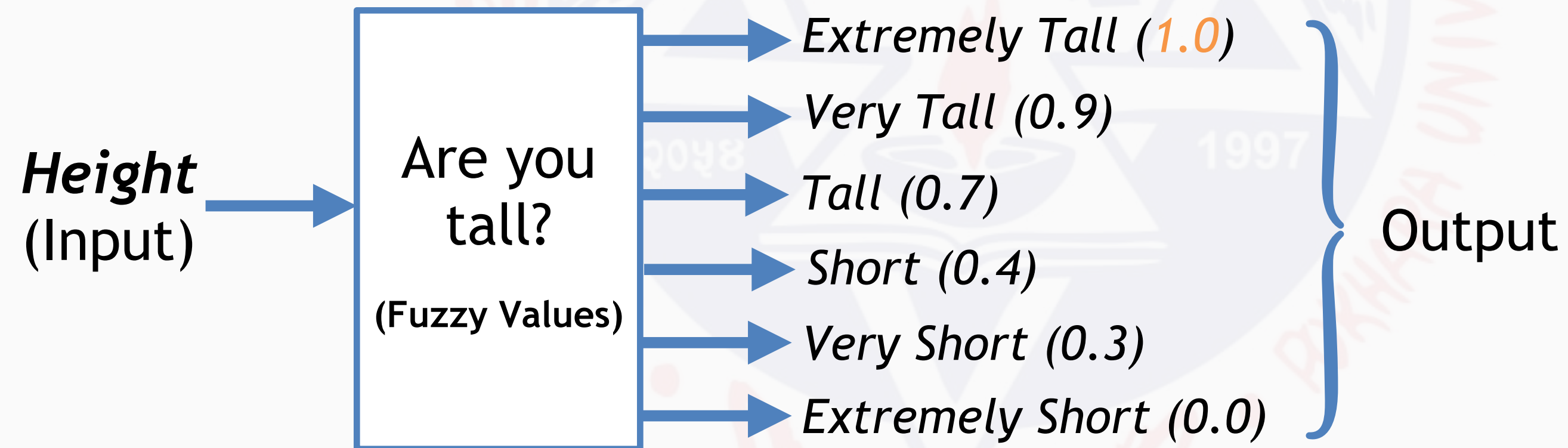
$$\bar{A} = \{(x, \mu_{\bar{A}}(x)) \mid x \in U\}, \text{ where } \mu_{\bar{A}}(x) \in [0,1]$$

The $(x, \mu_{\bar{A}}(x))$ is an order pair of member x and its membership value (the member function $\mu_{\bar{A}}(x)$ assigns a membership value that lies from 0 to 1 for x).



Fuzzy Sets

- A fuzzy set of "Tall people" might include:
 - A person with height 5'5" tall with a membership of 0.7.
 - A person with height 6'0" tall with a membership of 0.9.
 - A person with height 6'5" tall with a membership of 1.0.
 - A person with height 4'0" tall with a membership of 0.3.



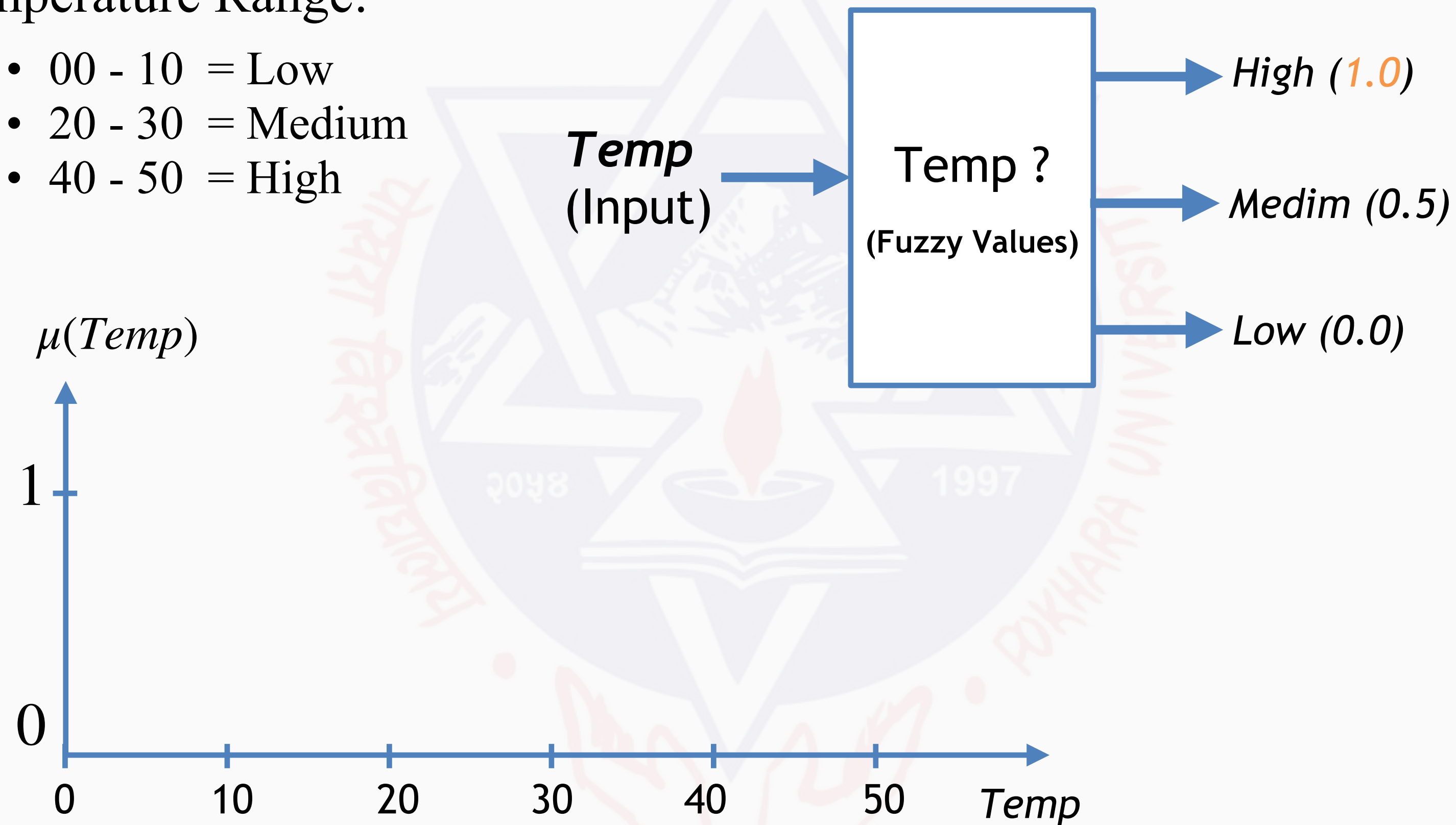
- The membership function defines the degree of membership for each input.



Fuzzy Sets

- Temperature Range:

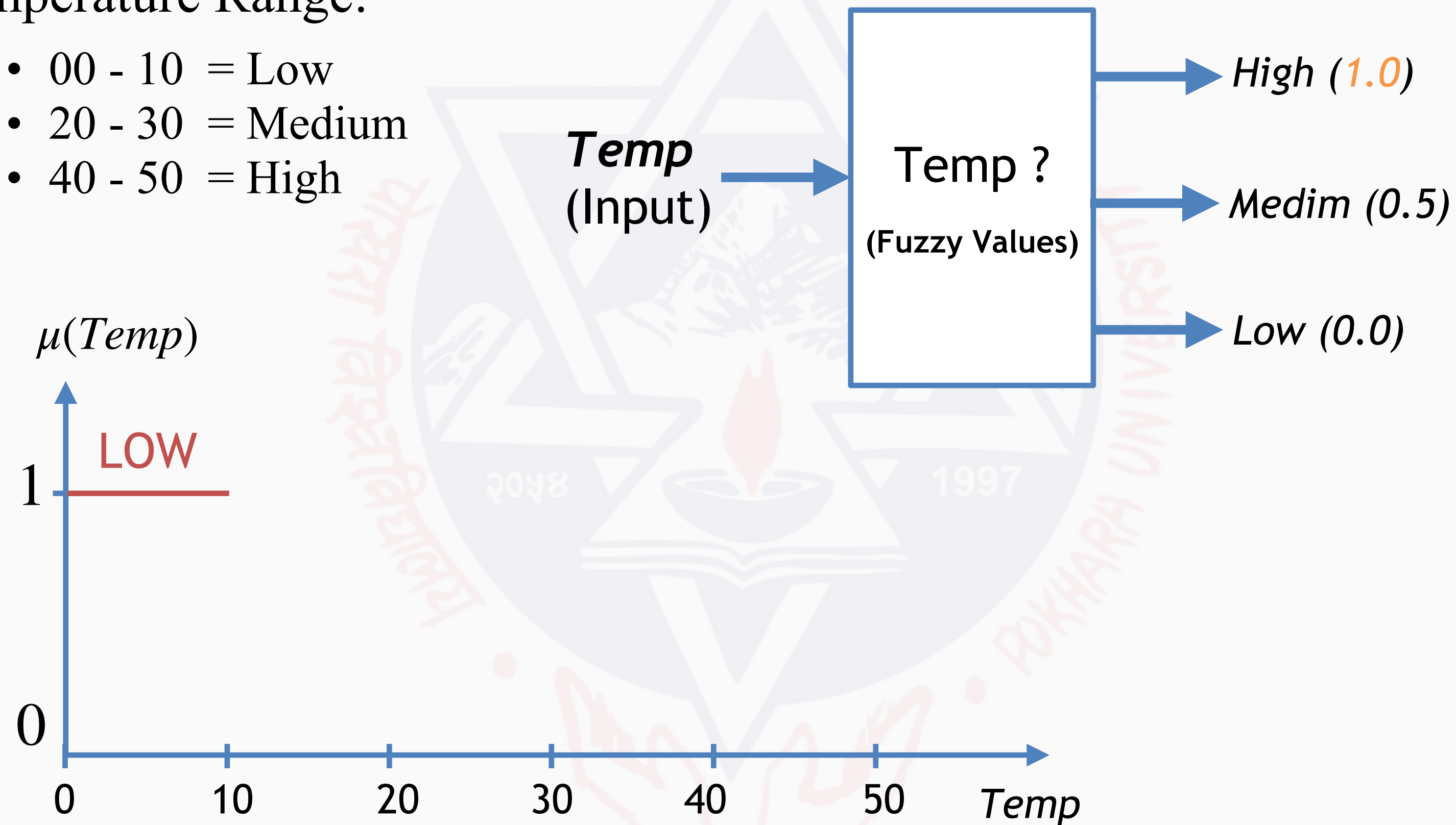
- 00 - 10 = Low
- 20 - 30 = Medium
- 40 - 50 = High



Fuzzy Sets

- Temperature Range:

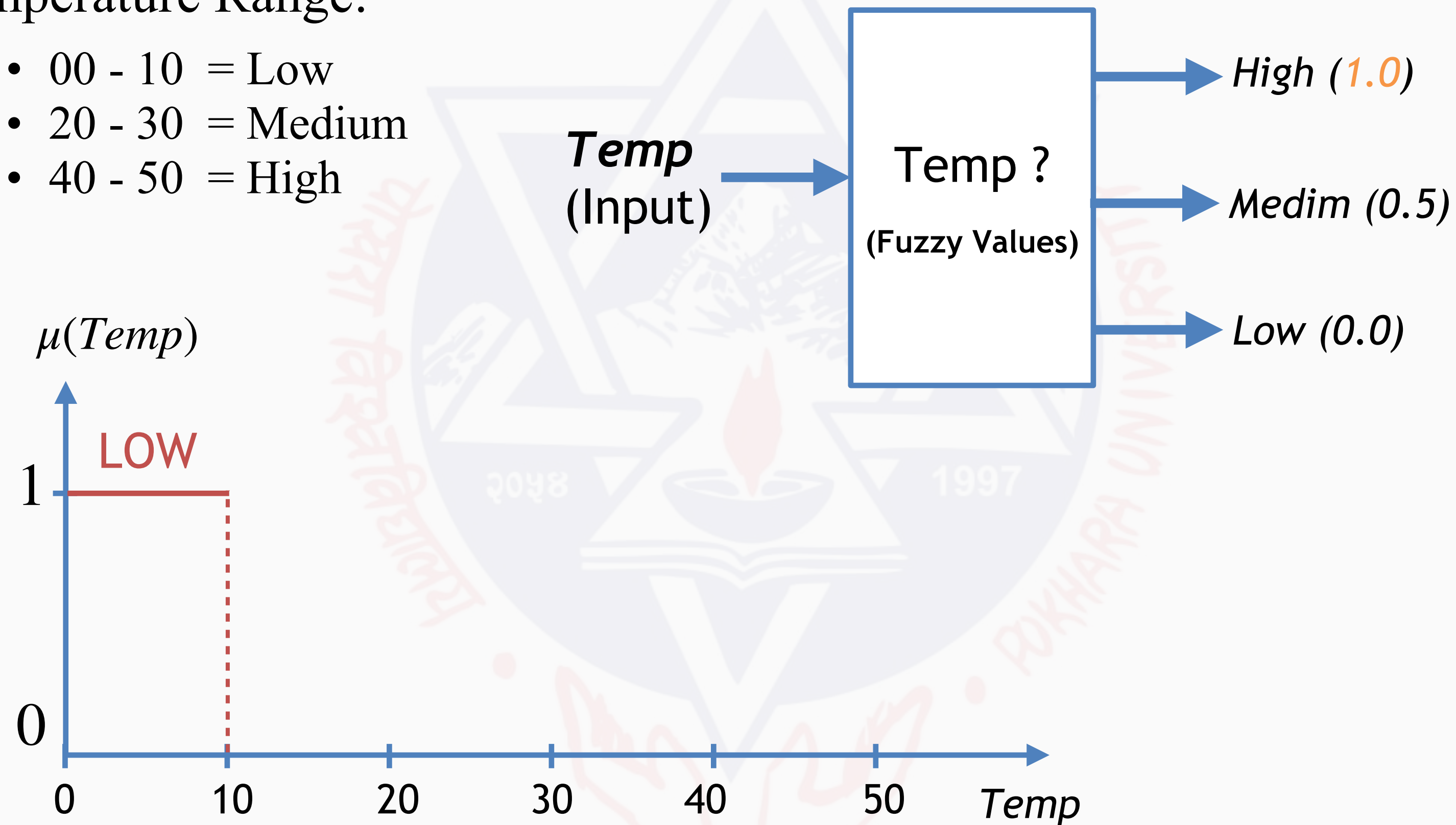
- 00 - 10 = Low
- 20 - 30 = Medium
- 40 - 50 = High



Fuzzy Sets

- Temperature Range:

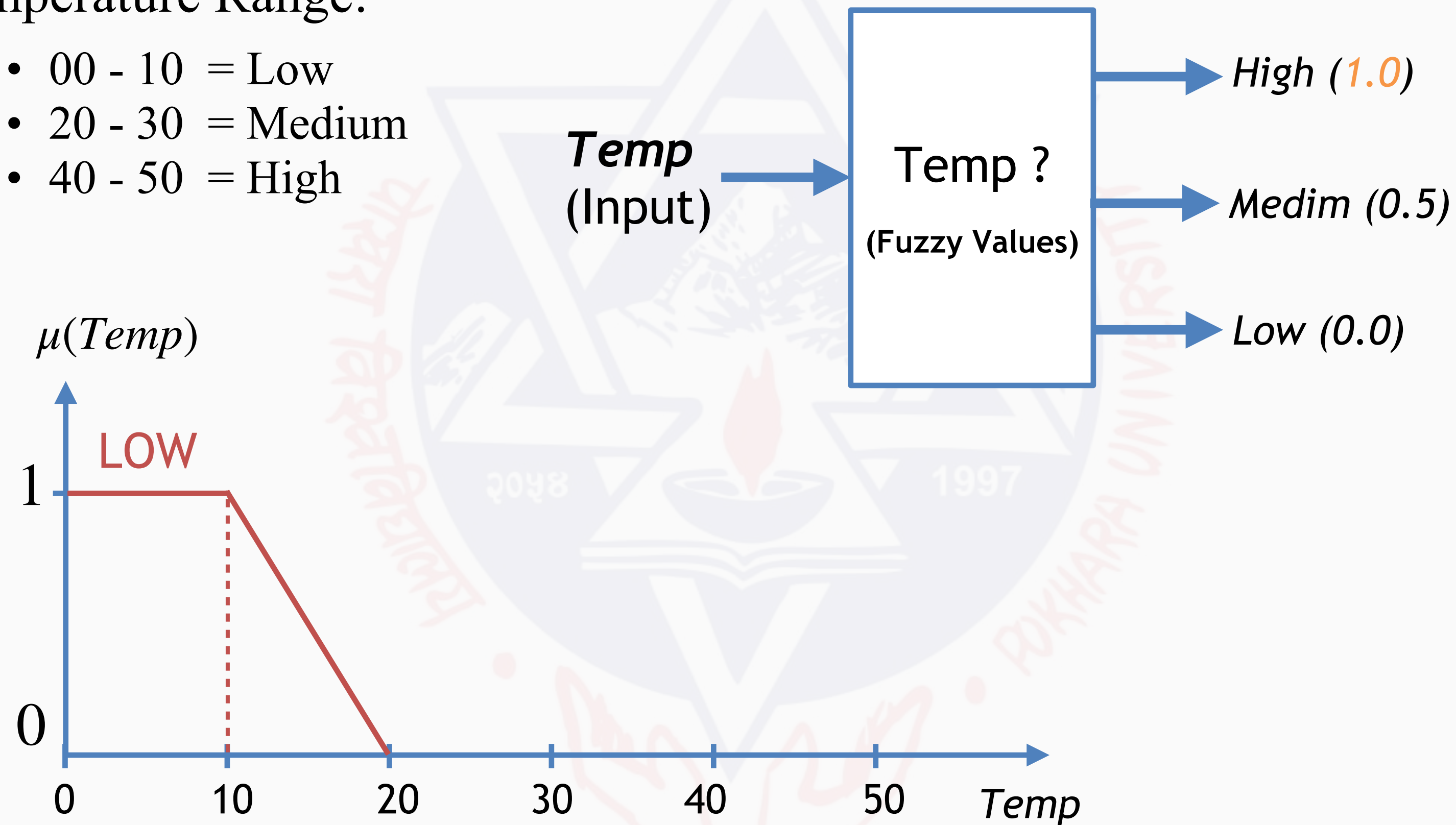
- 00 - 10 = Low
- 20 - 30 = Medium
- 40 - 50 = High



Fuzzy Sets

- Temperature Range:

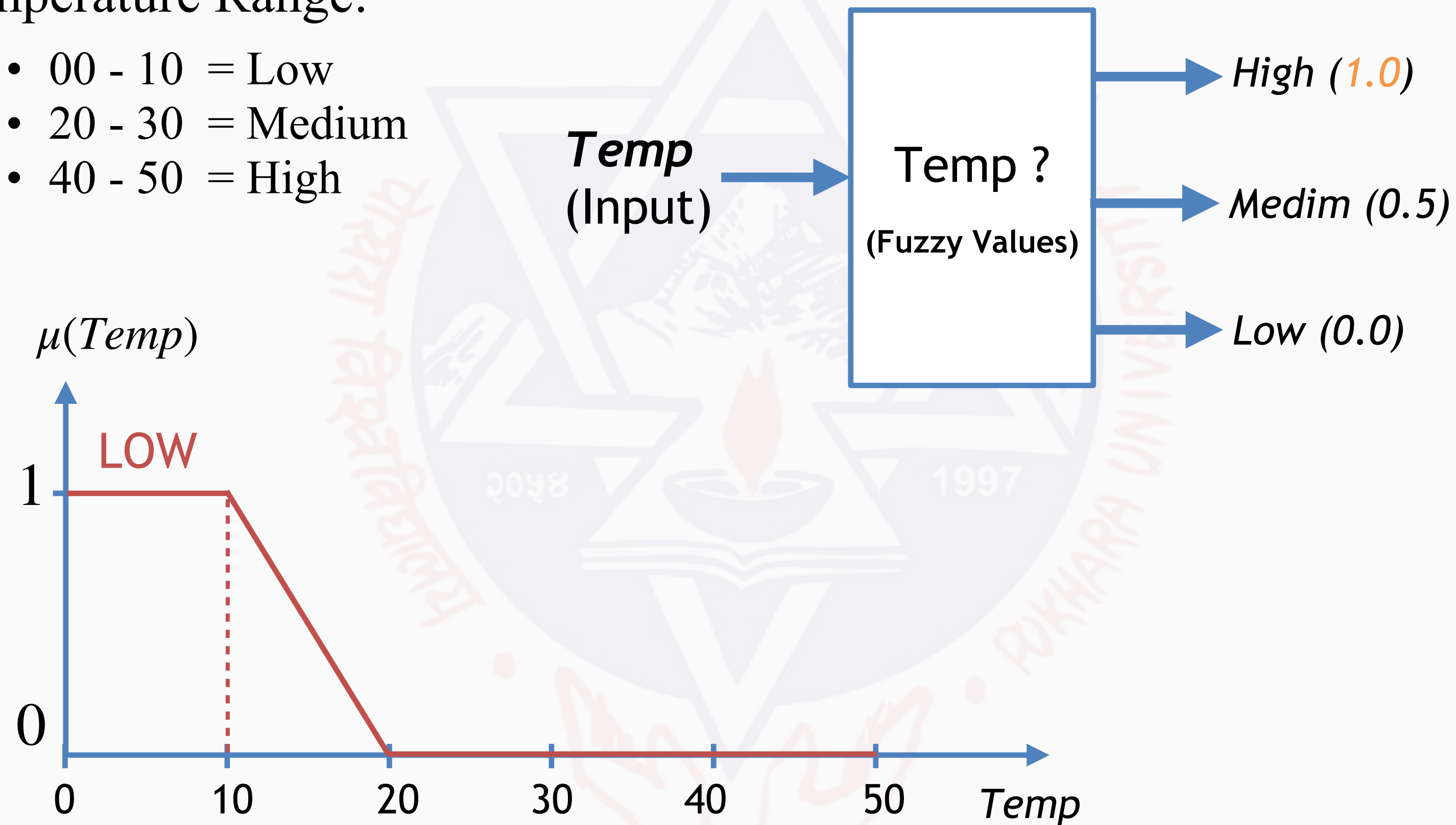
- 00 - 10 = Low
- 20 - 30 = Medium
- 40 - 50 = High



Fuzzy Sets

- Temperature Range:

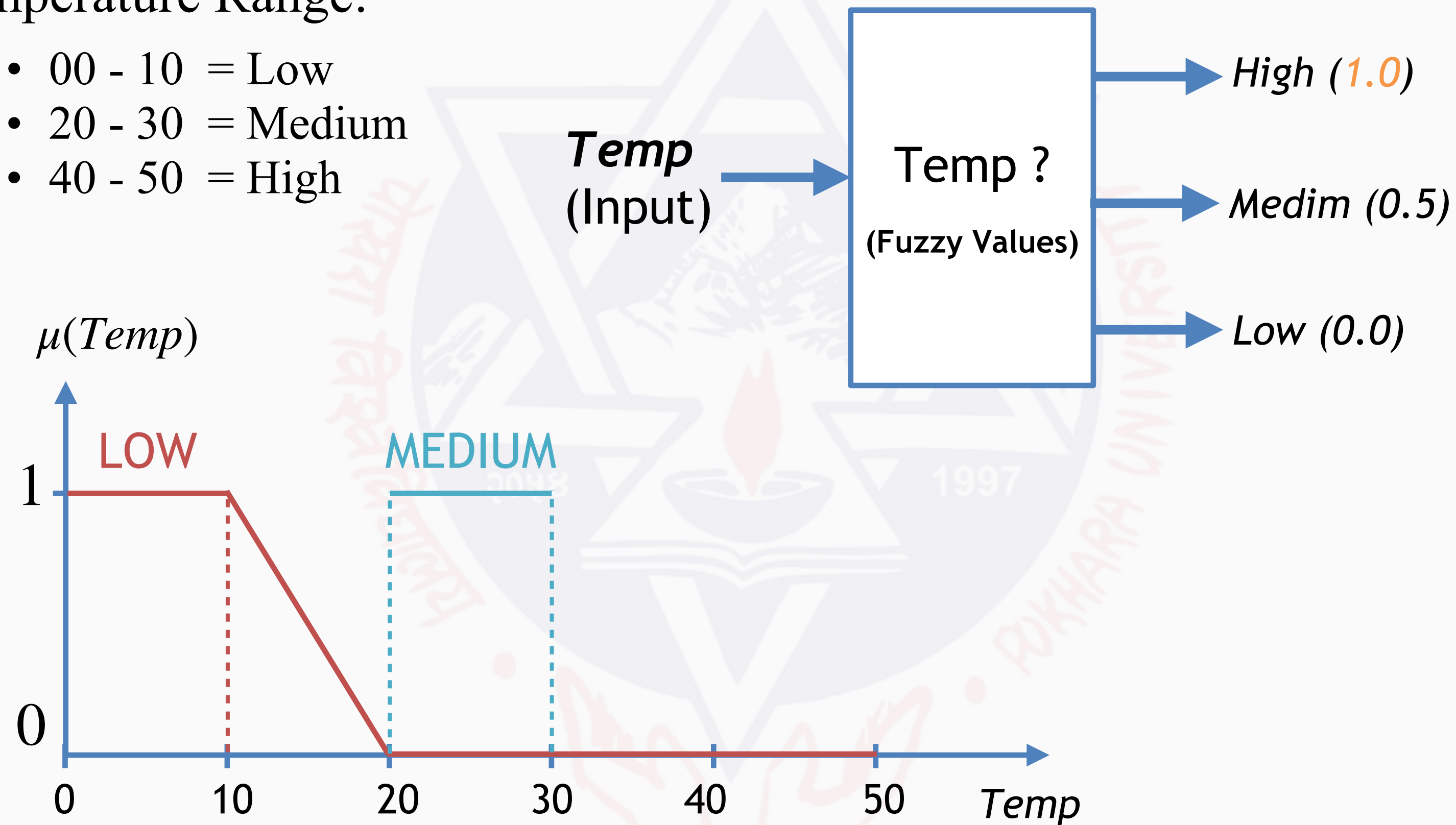
- 00 - 10 = Low
- 20 - 30 = Medium
- 40 - 50 = High



Fuzzy Sets

- Temperature Range:

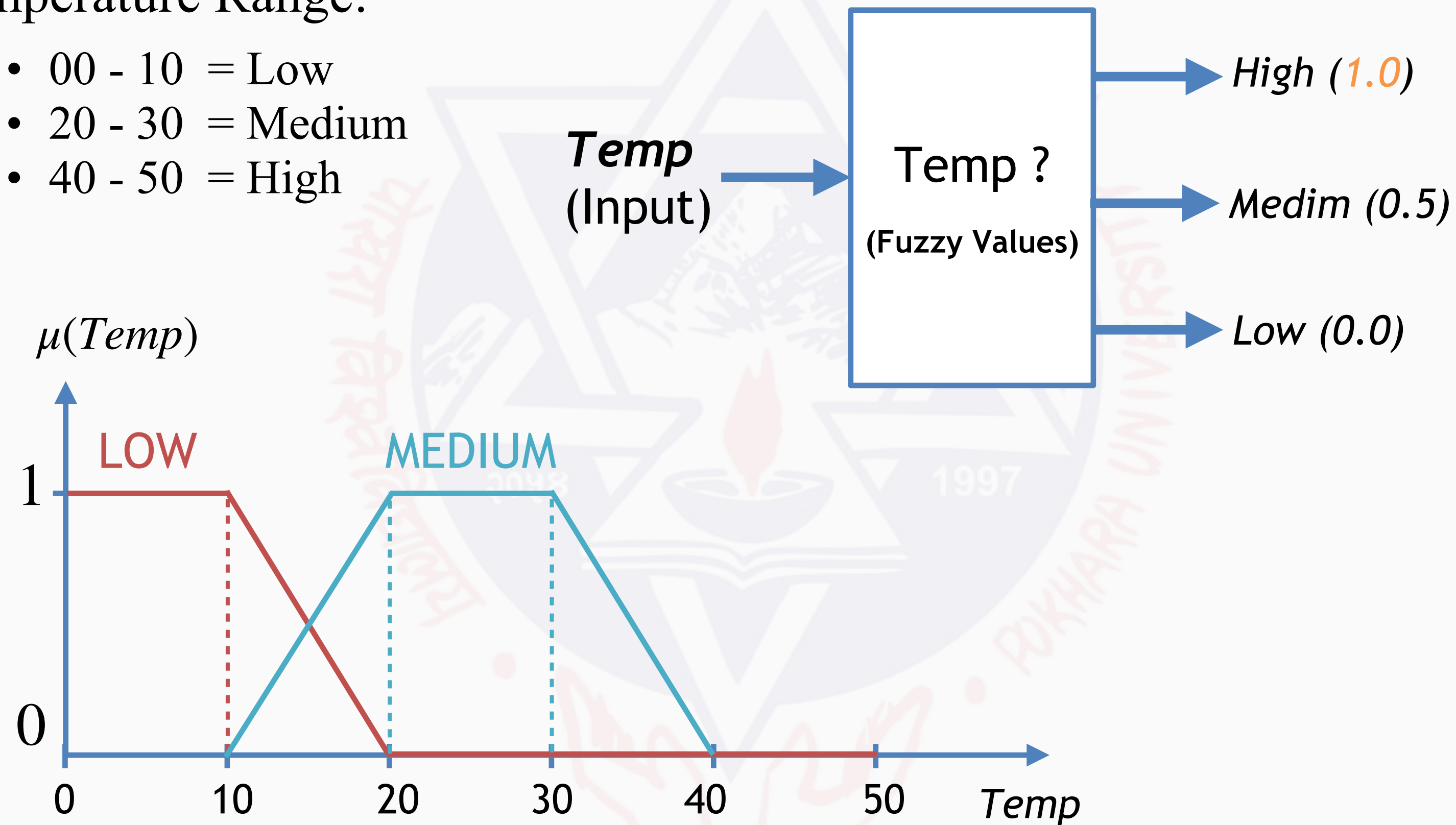
- 00 - 10 = Low
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- 40 - 50 = High



Fuzzy Sets

- Temperature Range:

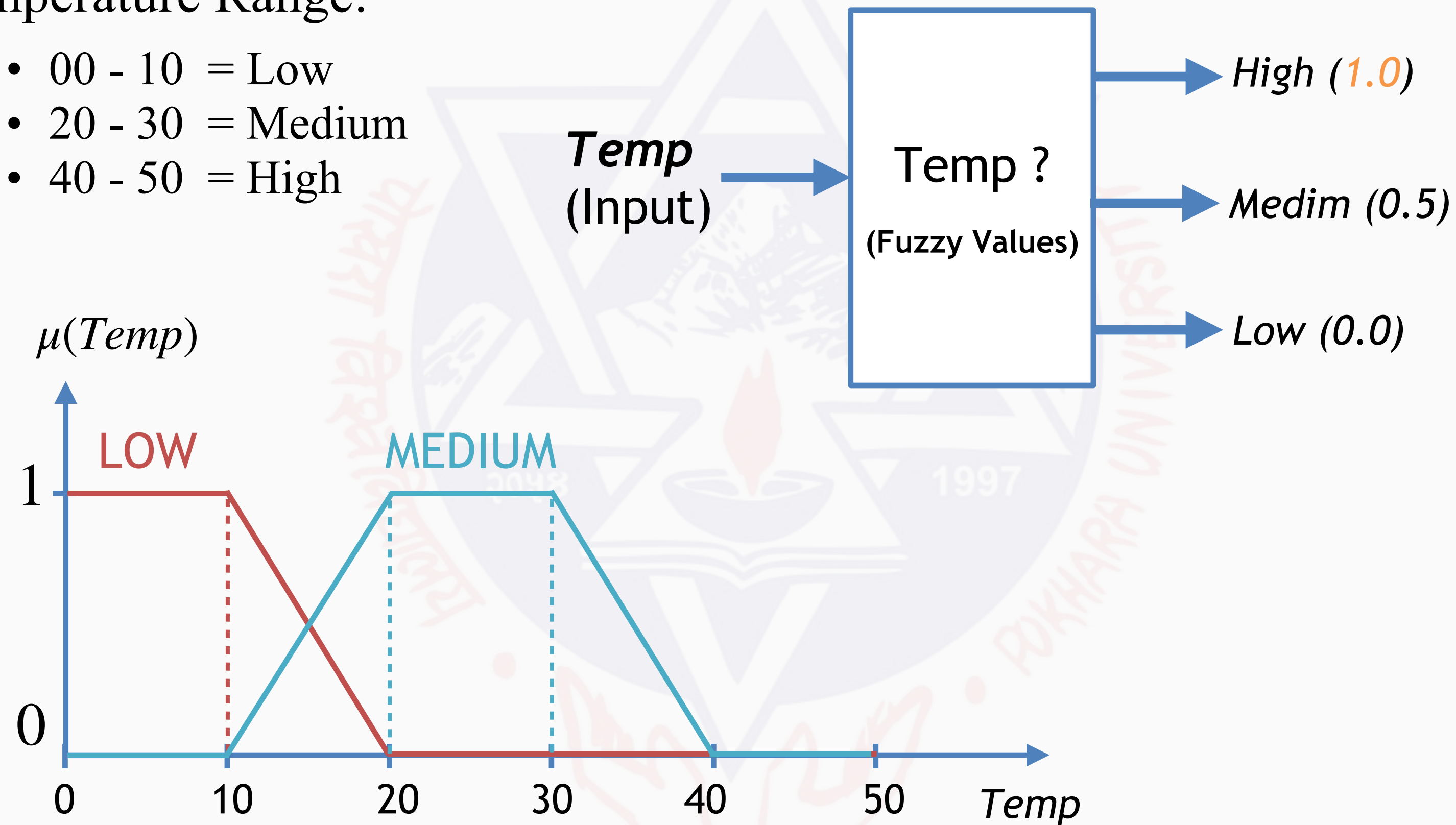
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Fuzzy Sets

- Temperature Range:

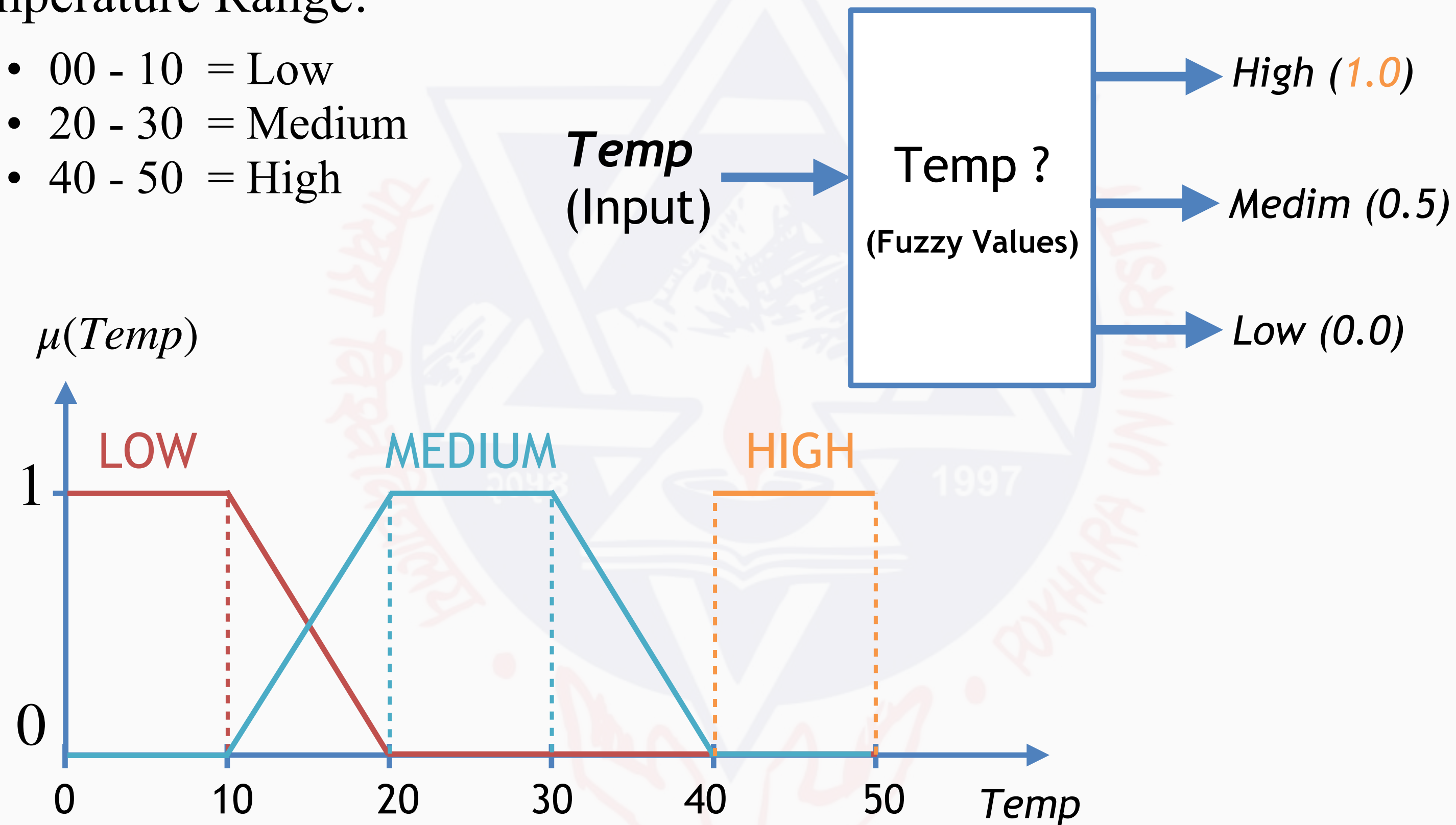
- 00 - 10 = Low
- 20 - 30 = Medium
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Fuzzy Sets

- Temperature Range:

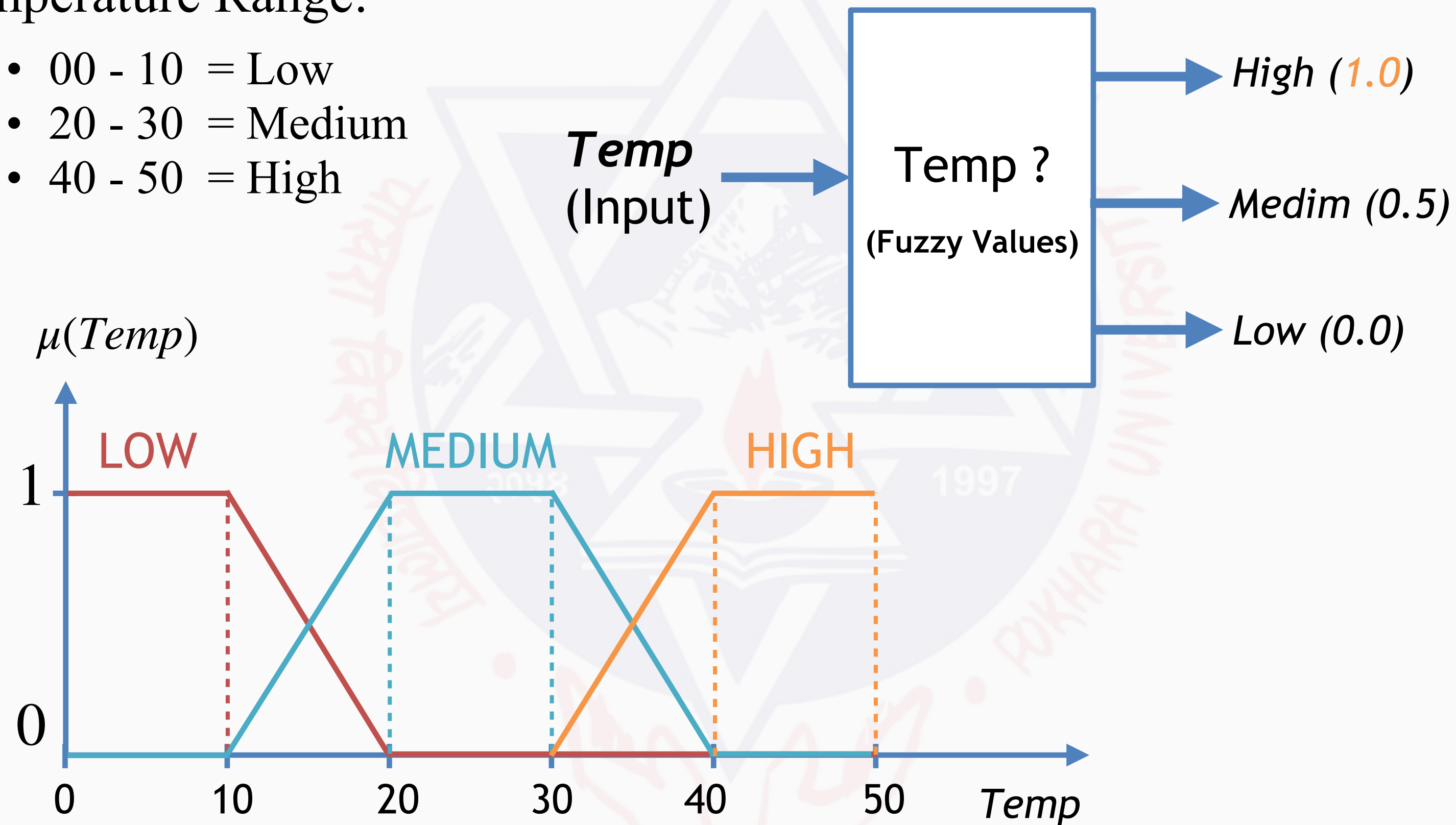
- 00 - 10 = Low
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Fuzzy Sets

- Temperature Range:

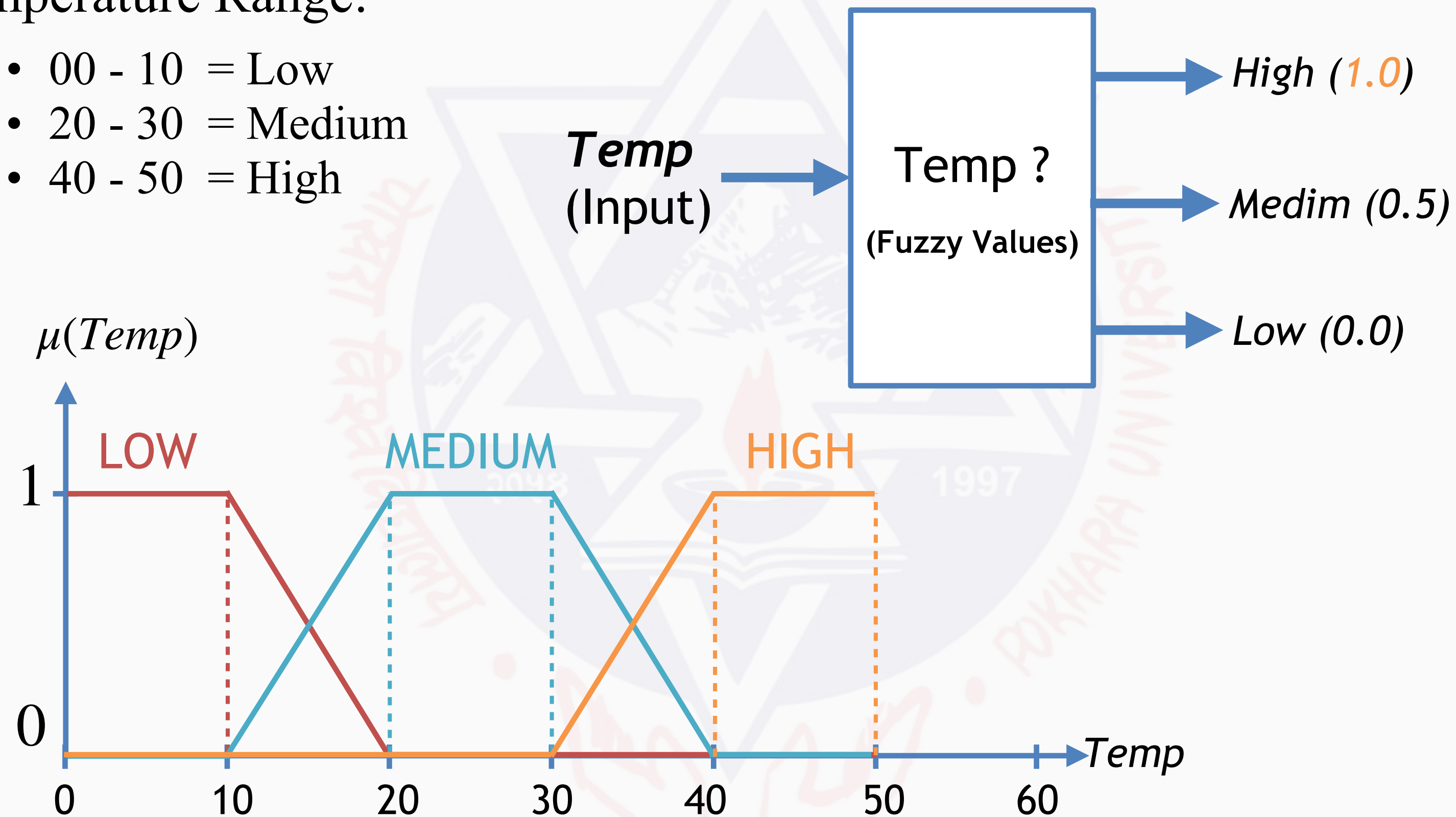
- 00 - 10 = Low
- 20 - 30 = Medium
- 40 - 50 = High



Fuzzy Sets

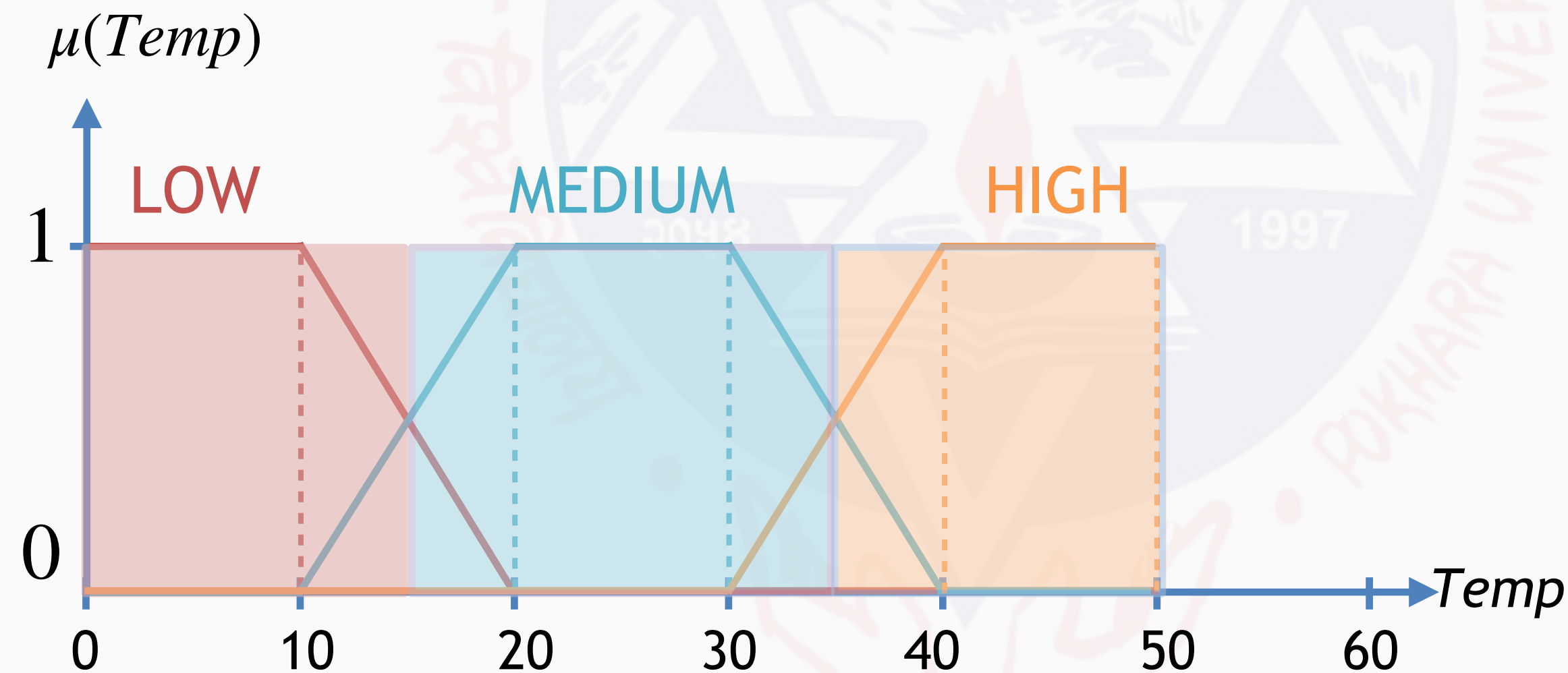
- Temperature Range:

- 00 - 10 = Low
- 20 - 30 = Medium
- 40 - 50 = High



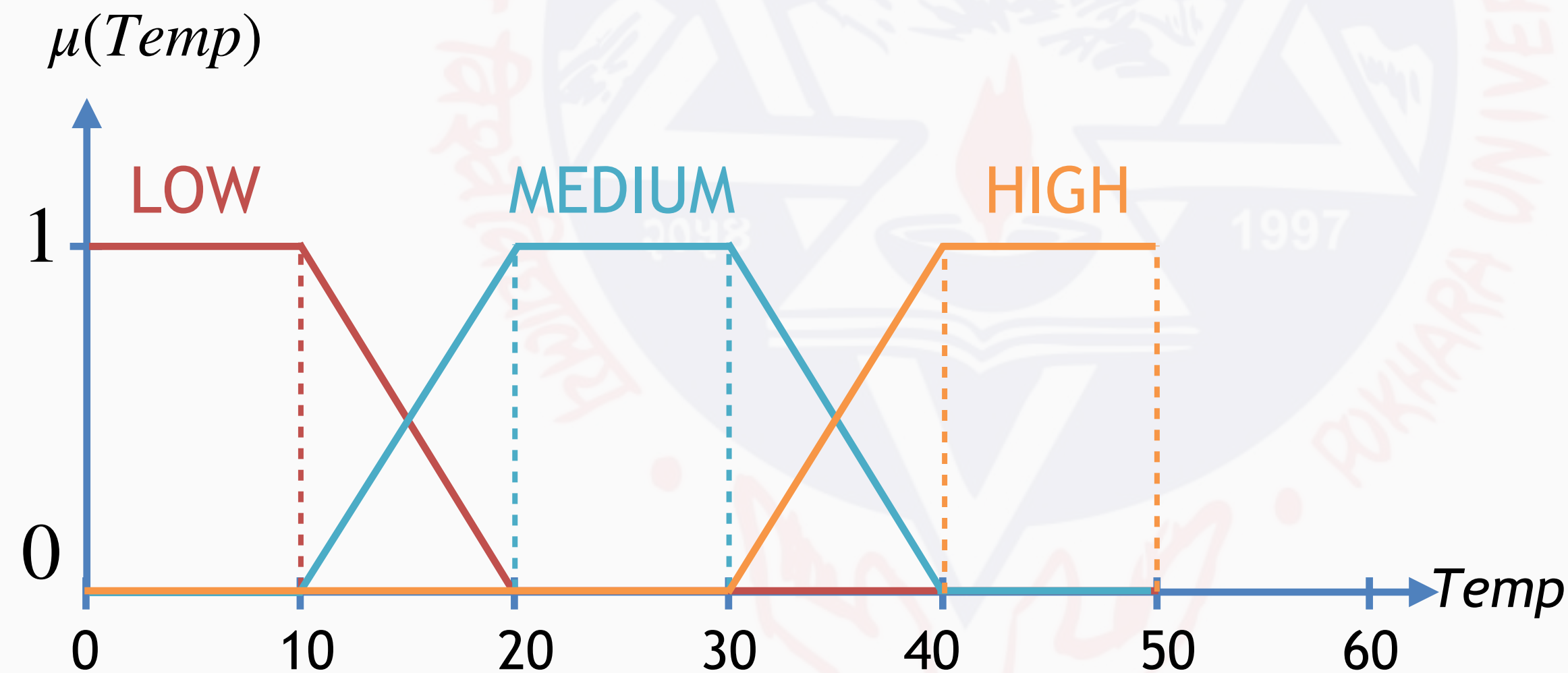
Fuzzy Sets

- In the classical set, any temperature can only be categorized into one subset, either LOW, MEDIUM or HIGH, and the boundary is crystal clear.



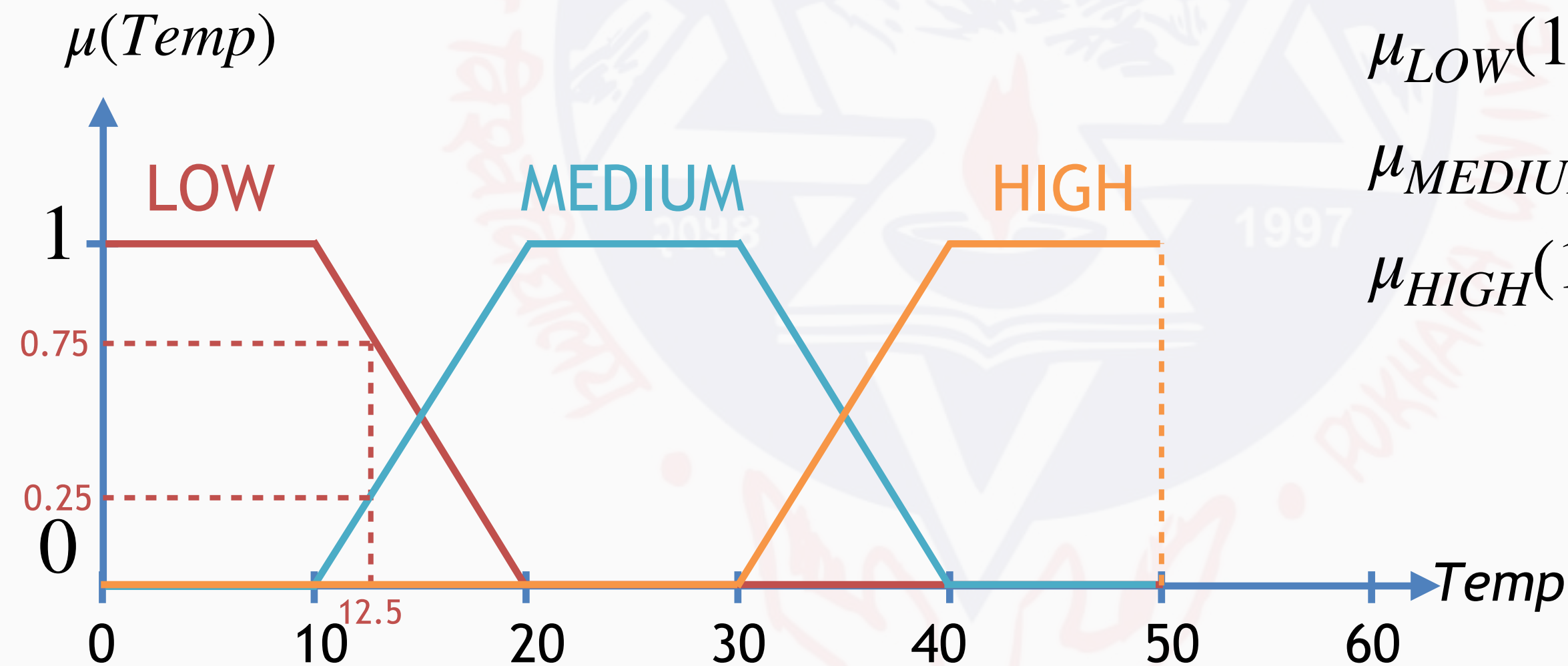
Fuzzy Sets

- But in the fuzzy set, boundaries becomes vague or smooth or ambiguous.
- One temperature can be categorized into two or may be even three subsets simultaneously.



Fuzzy Sets

- The temperature 12.5°C belongs to LOW subset by 0.75 degree ($\mu_{\text{LOW}}(12.5) = 0.75$).
- At the same time it belongs to MEDIUM by 0.25 degree ($\mu_{\text{MEDIUM}}(12.5) = 0.25$).
- It belongs to HIGH by 0 degree ($\mu_{\text{HIGH}}(12.5) = 0$).



$$\mu_{\text{LOW}}(12.5) = 0.75$$

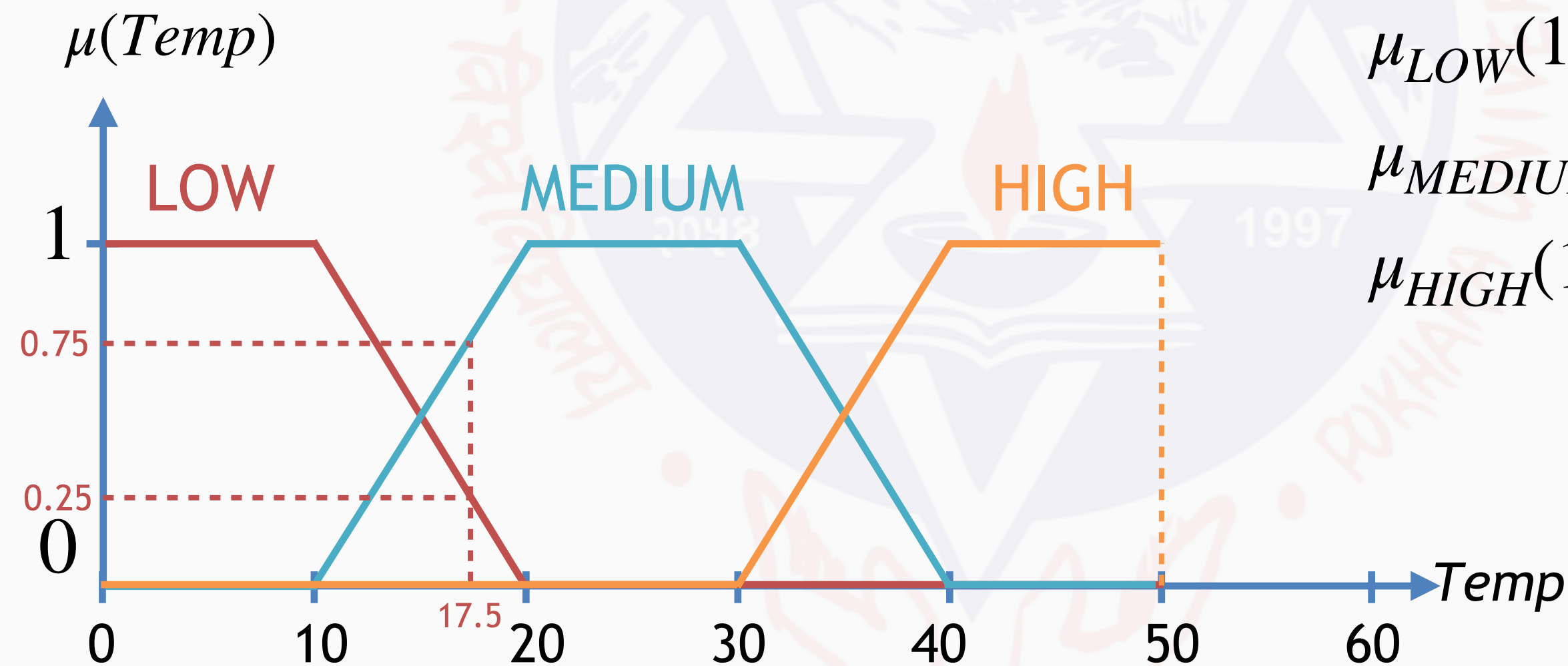
$$\mu_{\text{MEDIUM}}(12.5) = 0.25$$

$$\mu_{\text{HIGH}}(12.5) = 0$$



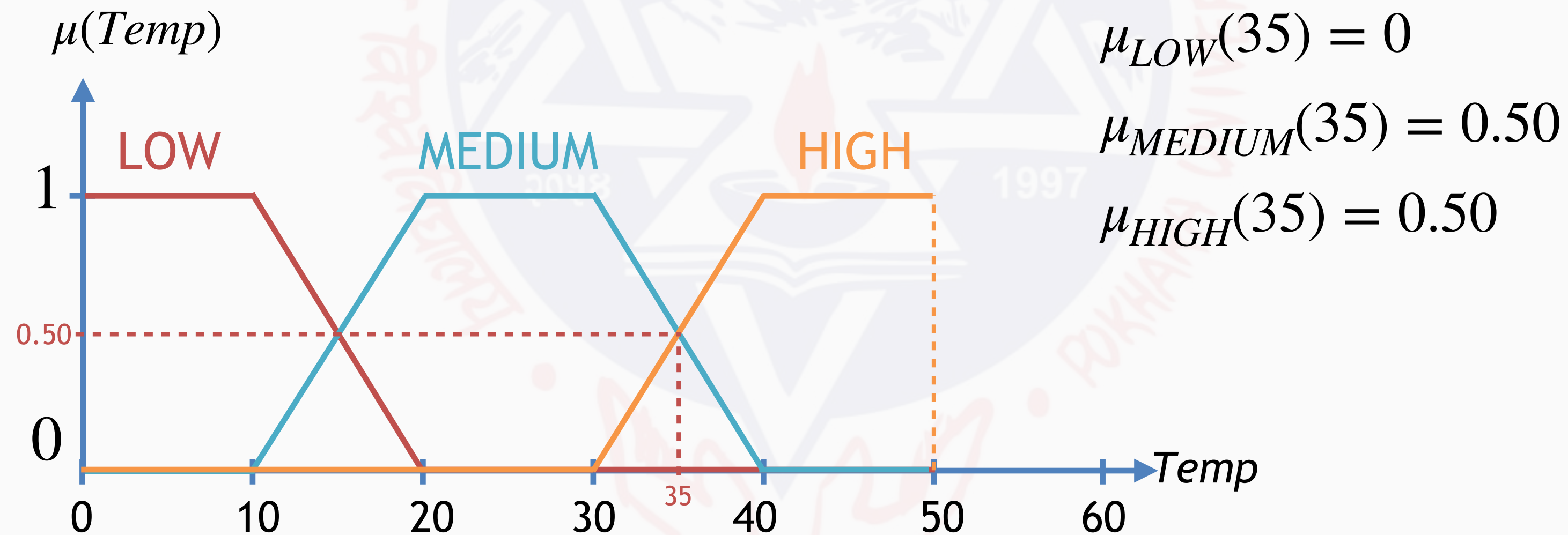
Fuzzy Sets

- The temperature 17.5°C belongs to LOW subset by 0.25 degree ($\mu_{\text{LOW}}(17.5) = 0.25$).
- At the same time it belongs to MEDIUM by 0.75 degree ($\mu_{\text{MEDIUM}}(17.5) = 0.75$).
- It belongs to HIGH by 0 degree ($\mu_{\text{HIGH}}(17.5) = 0$).



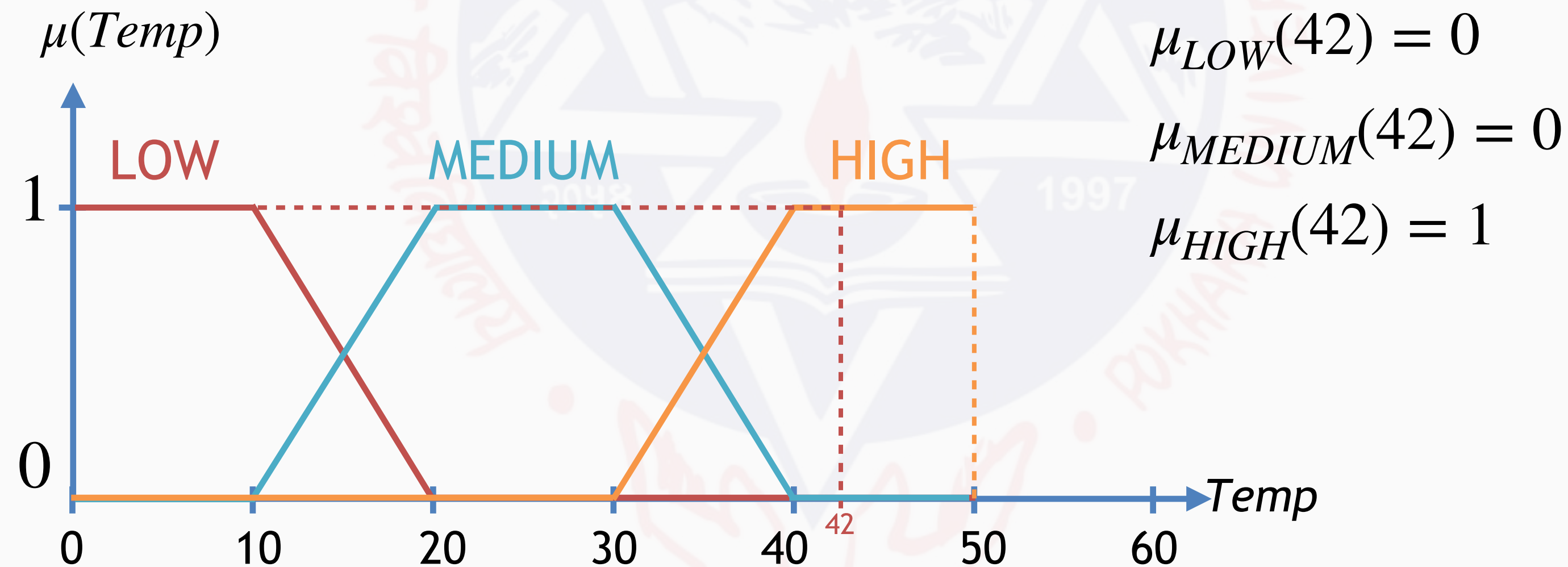
Fuzzy Sets

- The temperature 35°C belongs to LOW subset by 0 degree ($\mu_{\text{LOW}}(35) = 0$)
- At the same time it belongs to MEDIUM by 0.50 degree ($\mu_{\text{MEDIUM}}(35) = 0.50$).
- It belongs to HIGH by 0.50 degree ($\mu_{\text{HIGH}}(35) = 0.50$).

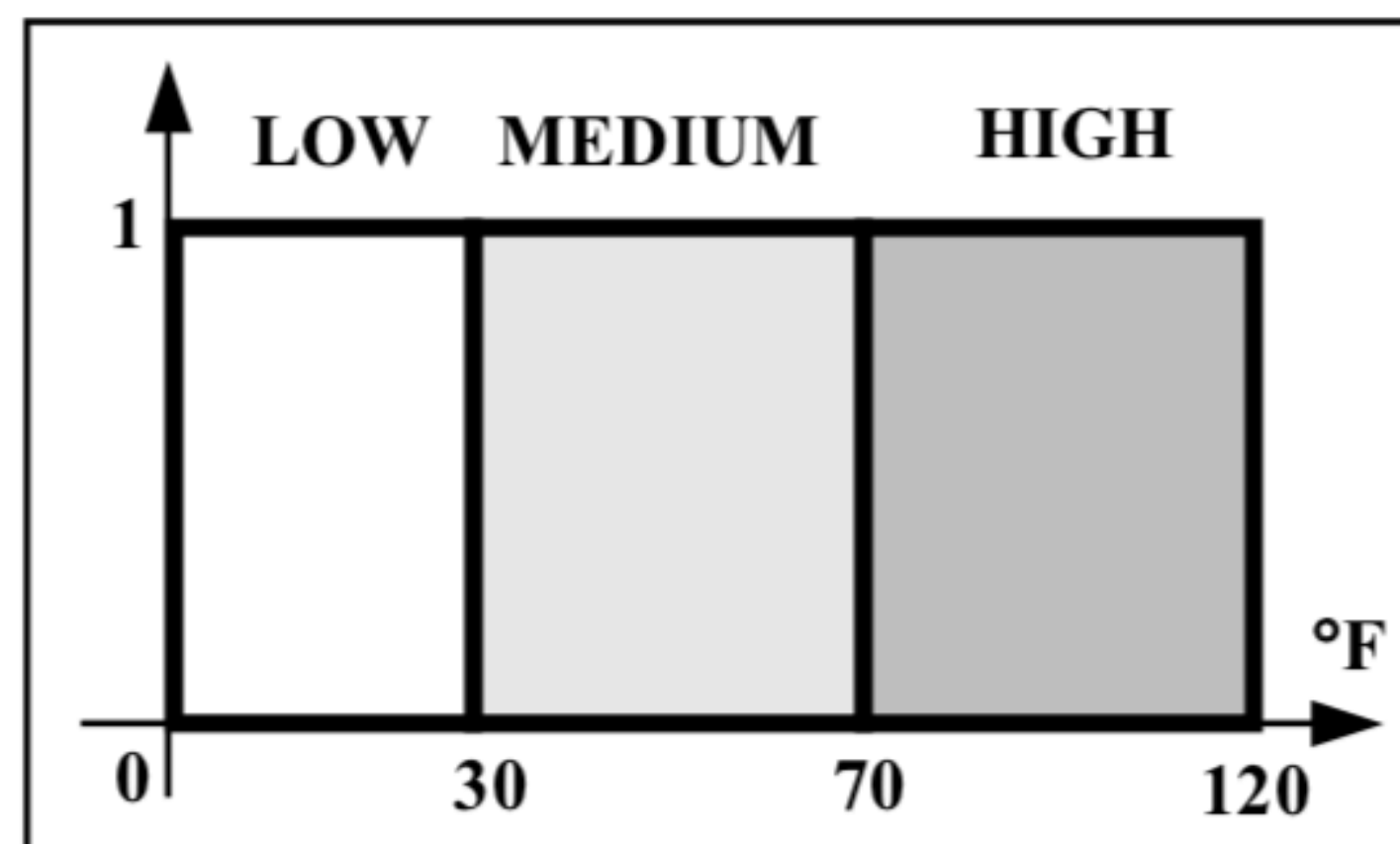


Fuzzy Sets

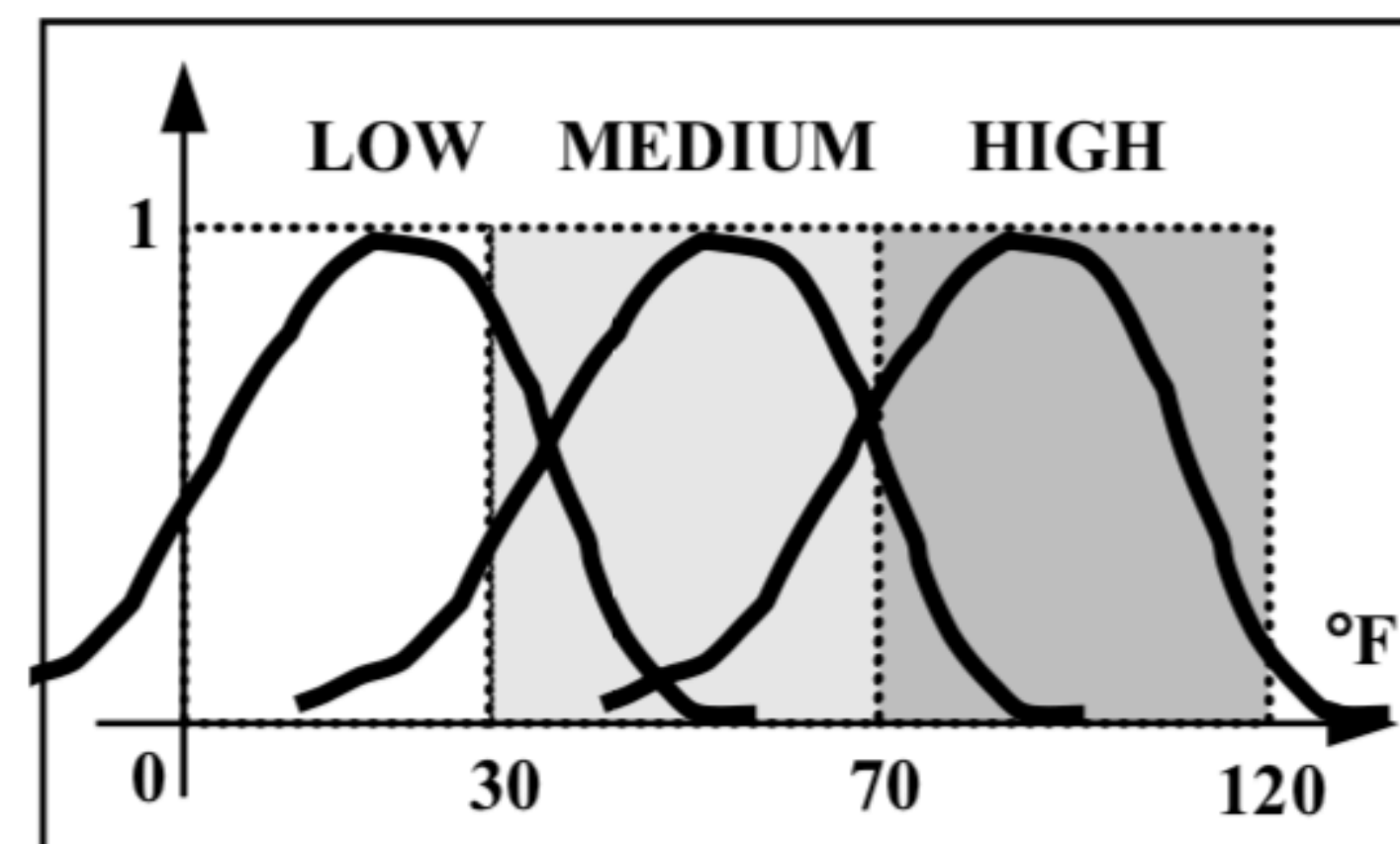
- The temperature 42°C belongs to LOW subset by 0 degree ($\mu_{\text{LOW}}(42) = 0$)
- At the same time it belongs to MEDIUM by 0 degree ($\mu_{\text{MEDIUM}}(42) = 0$).
- It belongs to HIGH by 1.0 degree ($\mu_{\text{HIGH}}(42) = 1$).



Fuzzy Sets



(a) Classical set



(b) Fuzzy set

Figure 2.3. Representations of classical and fuzzy sets



Fuzzy Sets- Membership Functions

- A fuzzy set $\bar{A}$ in the universe U is defined as a set of ordered pair as $\bar{A} = \{(x, \mu_{\bar{A}}(x)) \mid x \in U\}$, where $\mu_{\bar{A}}(x) \in [0,1]$ is called the **membership function** for the fuzzy set $\bar{A}$.
- The $(x, \mu_{\bar{A}}(x))$ is an order pair of member x and its membership value.
- The membership function $\mu_{\bar{A}}(x)$ maps each element $x \in U$ with a value in the interval $[0,1]$. This membership values is between 0 and 1 including 0 and 1.

$$\mu_{\bar{A}}(x) = \begin{cases} 1 & \text{if } x \text{ is totally in } \bar{A} \\ 0 & \text{if } x \text{ is not totally in } \bar{A} \\ 0 < \mu_{\bar{A}}(x) < 1 & \text{Otherwise} \end{cases}$$

- This membership value of an element x represents the *degree of membership* of element in $\bar{A}$.



Fuzzy Sets- Membership Functions

- A fuzzy set $\bar{A}$ in the universe U is defined as a set of ordered pair as $\bar{A} = \{(x, \mu_{\bar{A}}(x)) \mid x \in U\}$, where $\mu_{\bar{A}}(x) \in [0,1]$
- **When U is discrete and finite, $\bar{A}$ can be expressed as**

$$\bar{A} = \frac{\mu_{\bar{A}}(x_1)}{x_1} + \frac{\mu_{\bar{A}}(x_2)}{x_2} + \dots + \frac{\mu_{\bar{A}}(x_n)}{x_n} = \sum_{i=1}^n \frac{\mu_{\bar{A}}(x_i)}{x_i}$$

- **When U is continuous and infinite, $\bar{A}$ can be expressed as**

$$\bar{A} = \int_{i=1}^n \frac{\mu_{\bar{A}}(x_i)}{x_i}$$



Fuzzy Sets- Membership Functions

- We just saw, a discrete fuzzy set $\bar{A}$ can be expressed as

$$\bar{A} = \frac{\mu_{\bar{A}}(x_1)}{x_1} + \frac{\mu_{\bar{A}}(x_2)}{x_2} + \dots + \frac{\mu_{\bar{A}}(x_n)}{x_n} = \sum_{i=1}^n \frac{\mu_{\bar{A}}(x_i)}{x_i}$$

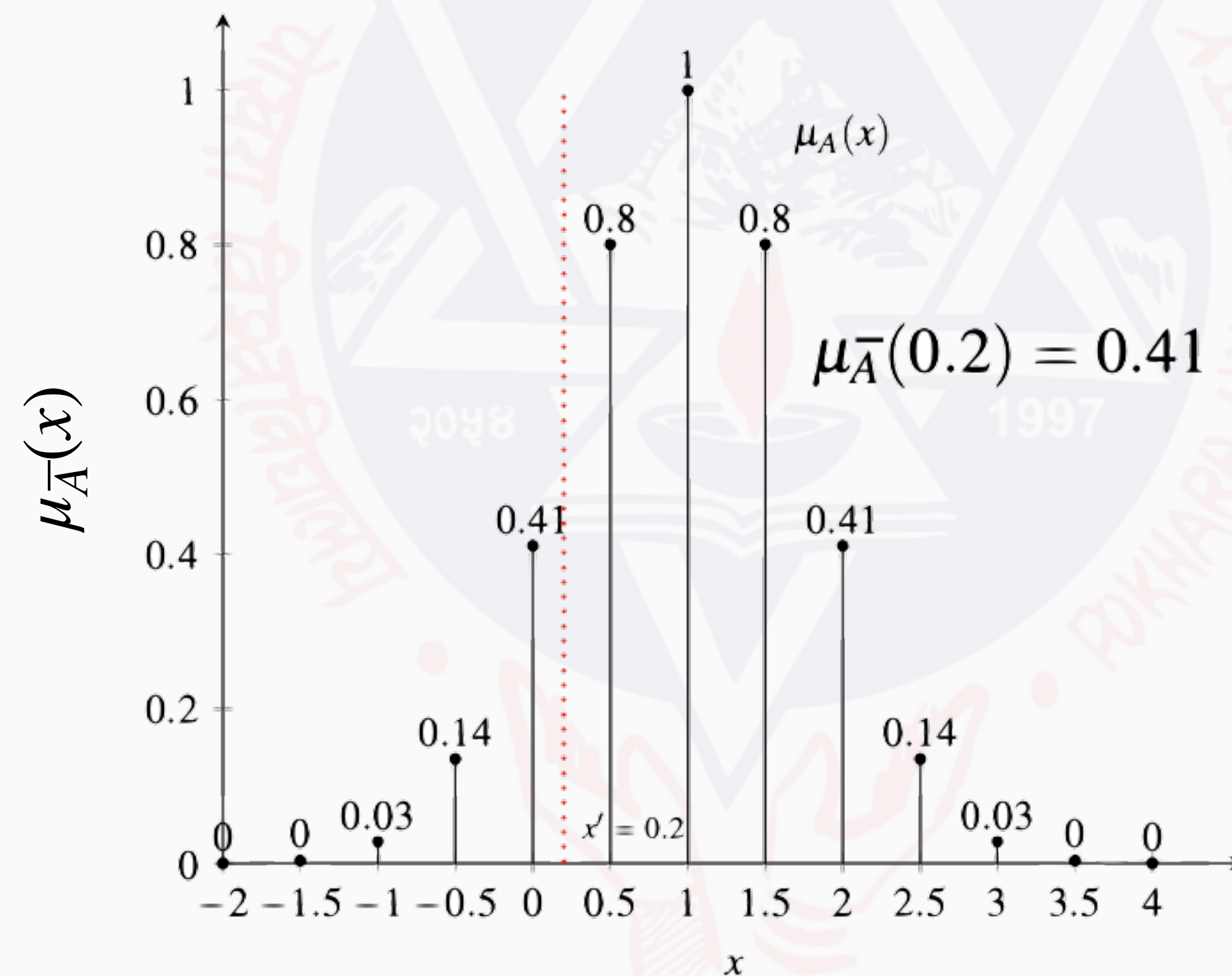
- Where, *note these points*
 - The *horizontal bar* is not a quotient but rather a delimiter.
 - The *summation symbol* is not for algebraic summation, but denotes the collection or aggregation of each element.
 - The “+” *signs* are not the algebraic “add” but are an aggregation or collection operator.



Fuzzy Sets- Membership Functions

$$\bar{A} = \left\{ \frac{0}{-2} + \frac{0}{-1.5} + \frac{0.03}{-1} + \frac{0.14}{-0.5} + \frac{0.41}{0} + \frac{0.8}{0.5} + \frac{1}{1} + \frac{0.8}{1.5} + \frac{0.41}{2} + \frac{0.14}{2.5} + \frac{0.03}{3} + \frac{0}{3.5} + \frac{0}{4} \right\}$$

Example



Fuzzy Sets- Membership Functions

- Similarly, a continuous fuzzy set $\bar{A}$ can be expressed as

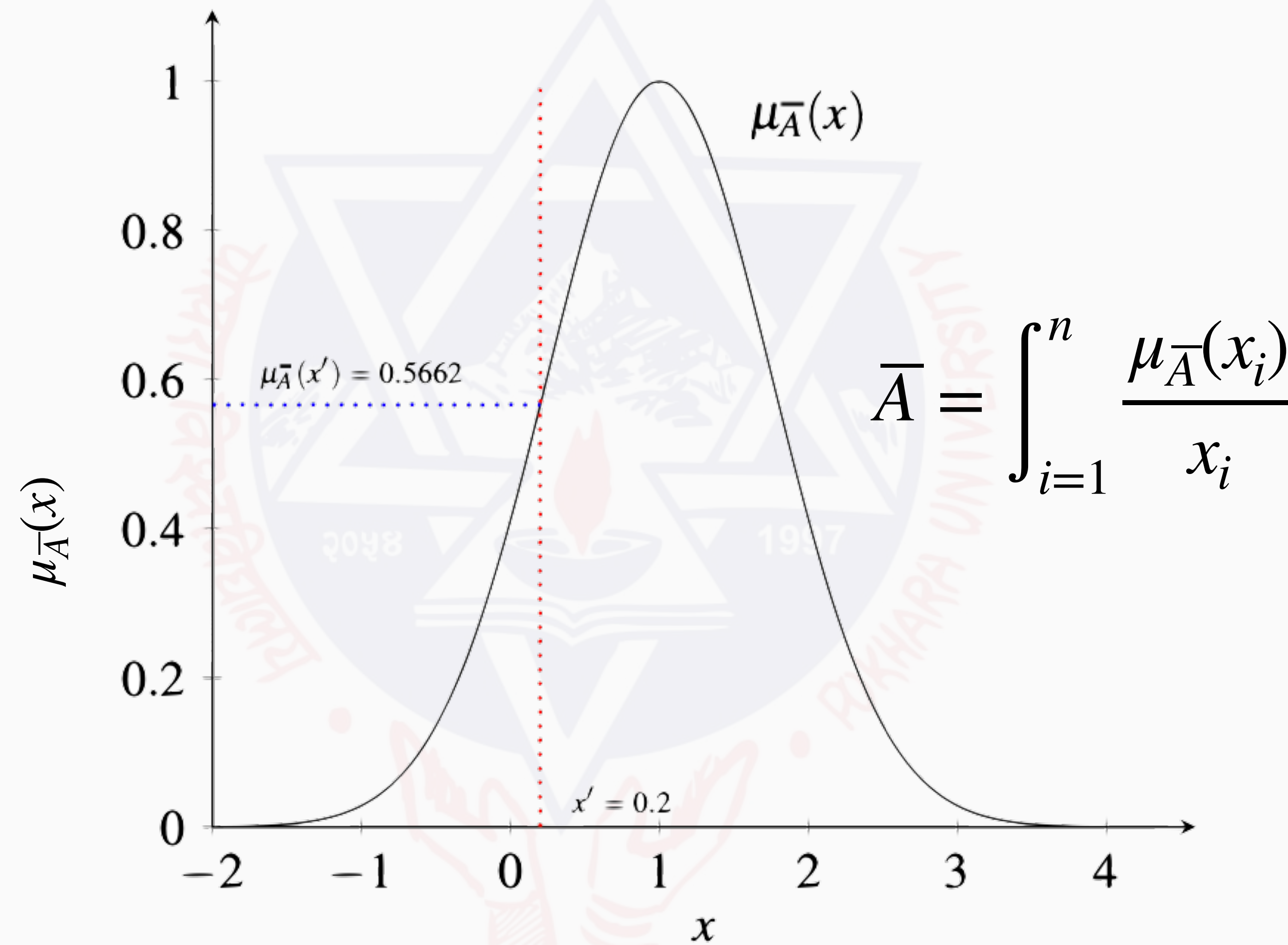
$$\bar{A} = \int_{i=1}^n \frac{\mu_{\bar{A}}(x_i)}{x_i}$$

- Where, *note these points*
 - The *horizontal bar* is not a quotient but rather a delimiter.
 - The *integral sign* is not for algebraic integral, but a continuous function-theoretic aggregation operator for continuous variables



Fuzzy Sets- Membership Functions

Example



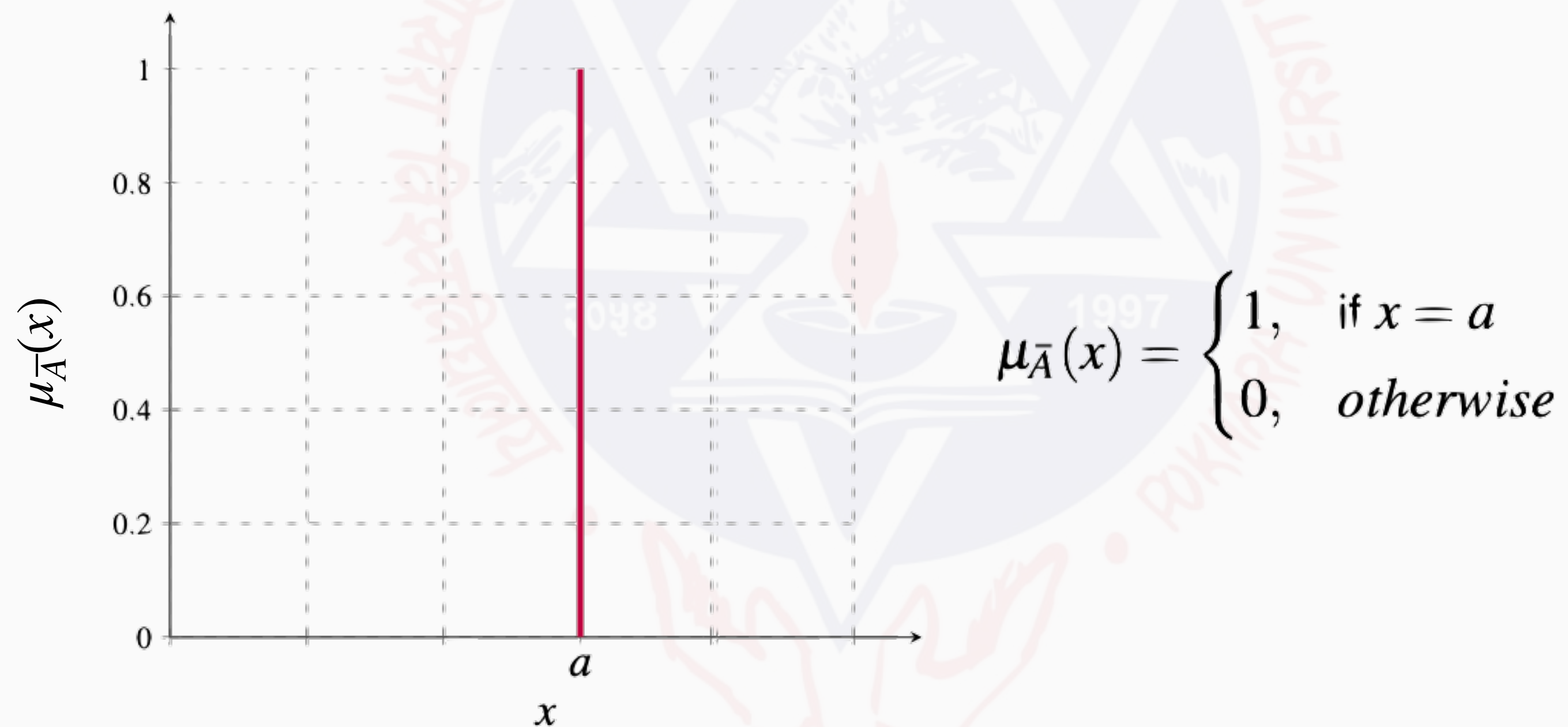
Fuzzy Sets- Membership Functions

- Membership functions
 - *transform crisp inputs into fuzzy values for processing.*
- Components of Membership Functions
 - Input domain:
 - It is *range of possible values* for a input variable
 - Membership Degree:
 - A value in the interval $[0,1]$ that indicates the degree of membership to the fuzzy logic.
 - Shape:
 - The graphical representation of the function, which determines how membership values are assigned.
- Types of Membership Functions:
 - Singleton, Triangular, Trapezoidal, Gaussian, Z-Shaped, S-Shaped Membership Function



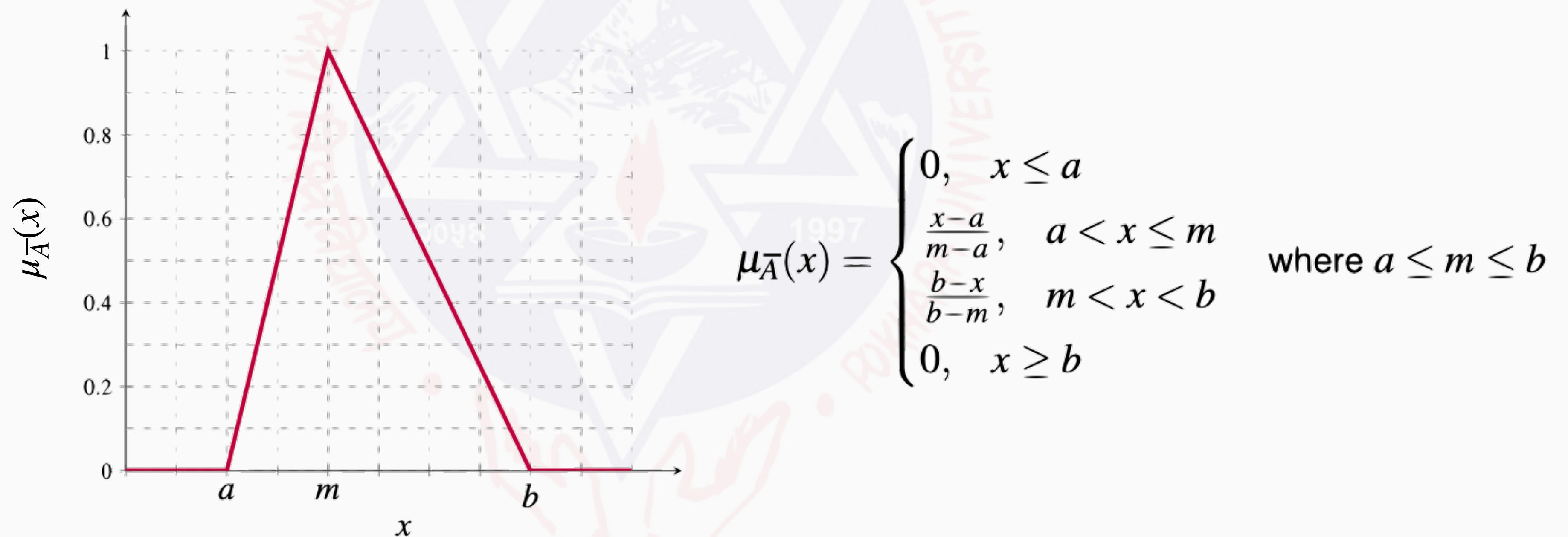
Fuzzy Sets- Membership Functions

- Singleton Membership Function



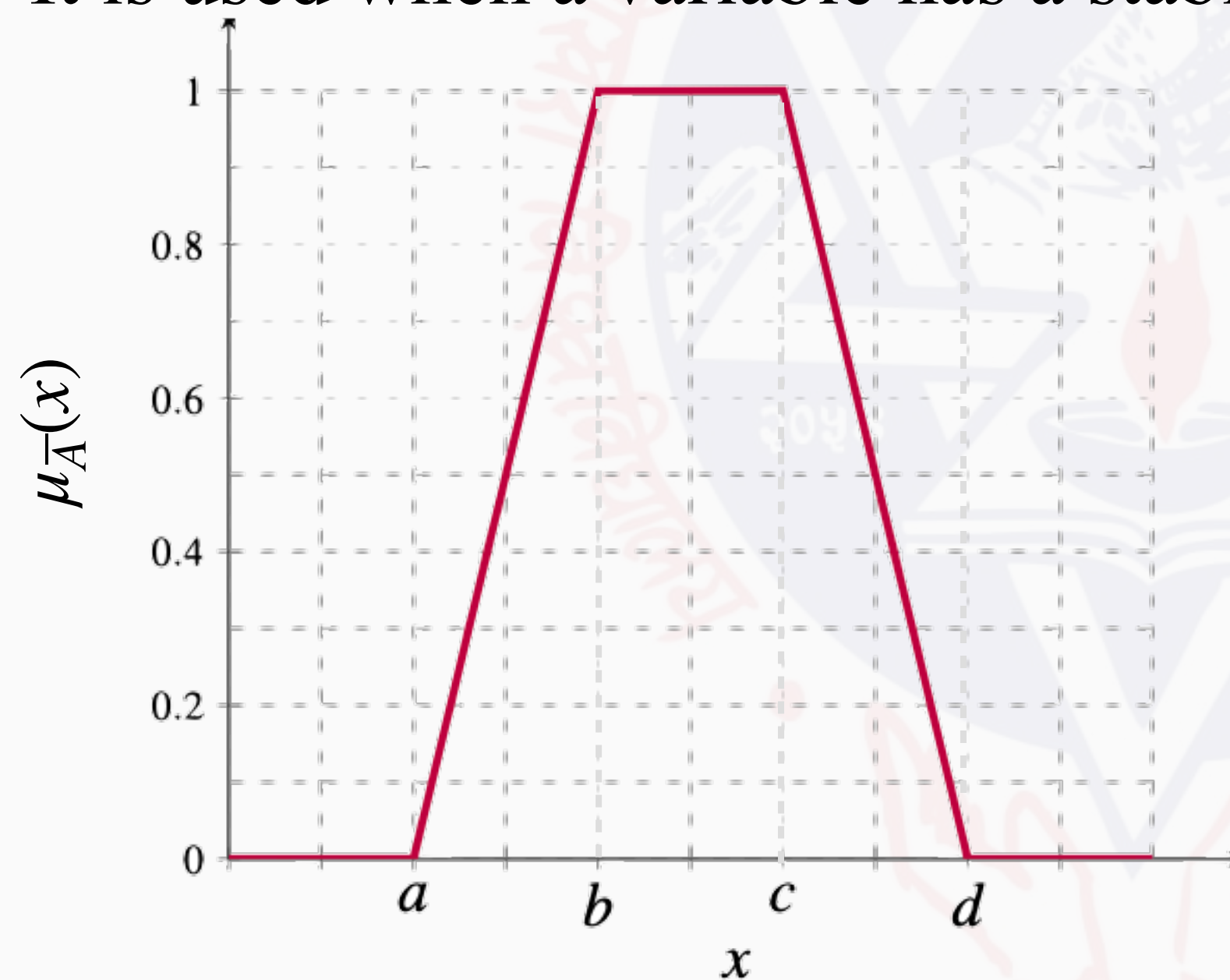
Fuzzy Sets- Membership Functions

- Triangular Membership Function
 - A triangle defined by three points: start (a), peak (m), and end (b).
 - It is used when data is approximately linear.



Fuzzy Sets- Membership Functions

- Trapezoidal Membership Function:
 - A trapezoid defined by four points: start (a), plateau start (b), plateau end (c), and end (d).
 - It is used when a variable has a stable range with full membership.



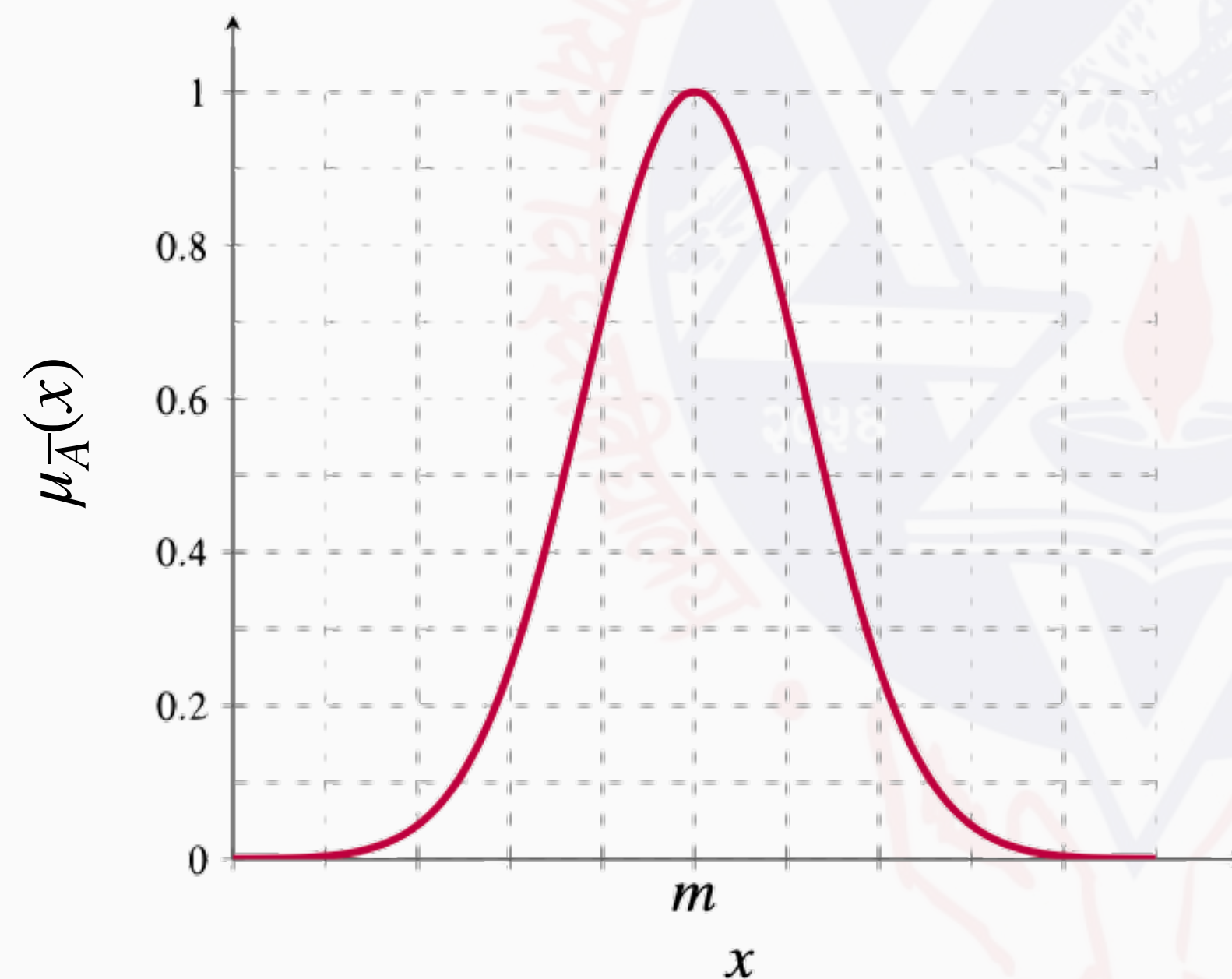
$$\mu_A(x) = \begin{cases} 0, & x < a \text{ or } x > d \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & b \leq x \leq c \\ \frac{d-x}{d-c}, & c \leq x \leq d \end{cases}$$

where $a \leq b \leq c \leq d$



Fuzzy Sets- Membership Functions

- Gaussian Membership Function:
 - A smooth, bell-shaped curve. It is used when smooth transitions are required, such as in control systems.



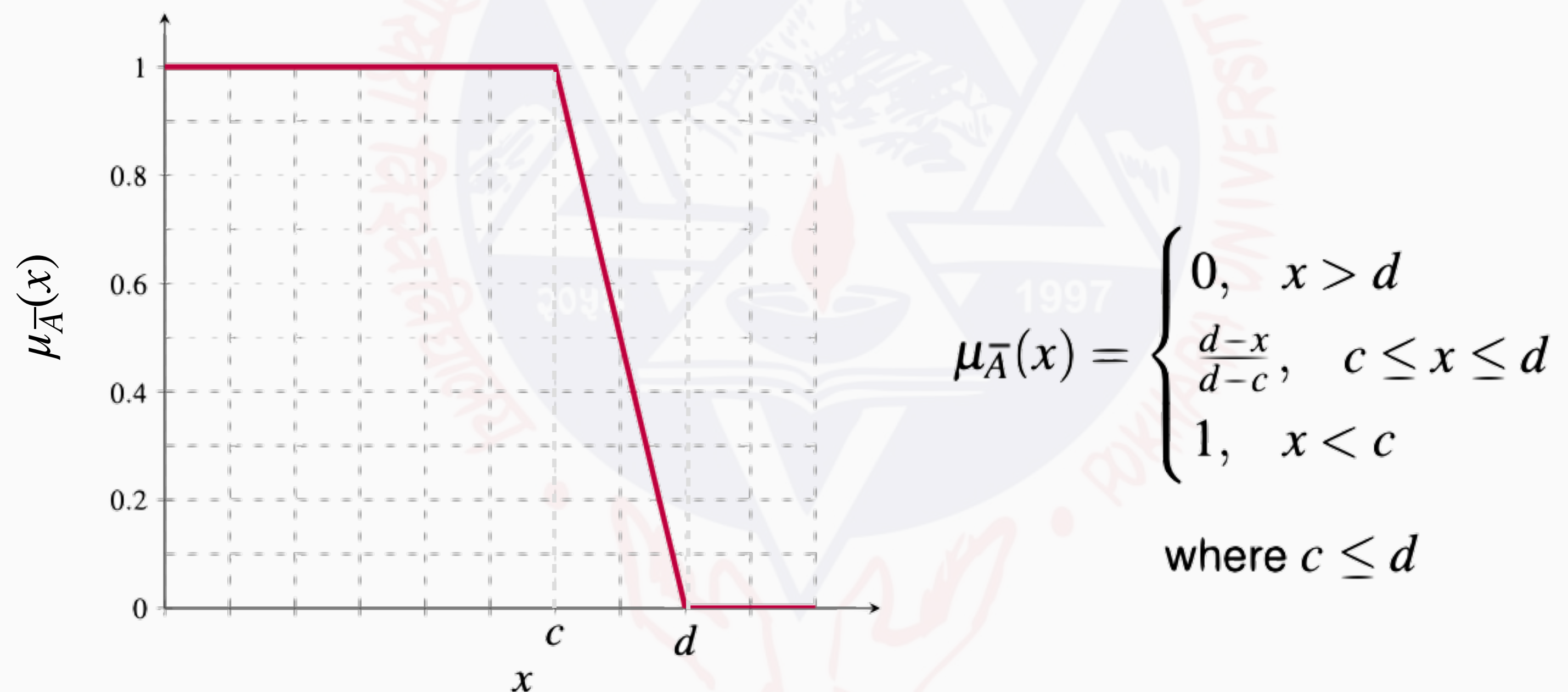
$$\mu_{\bar{A}}(x) = e^{-\frac{(x-m)^2}{2\sigma^2}}$$

Where m = center of gaussian curve
 σ = standard deviation, controlling the gaussian width.



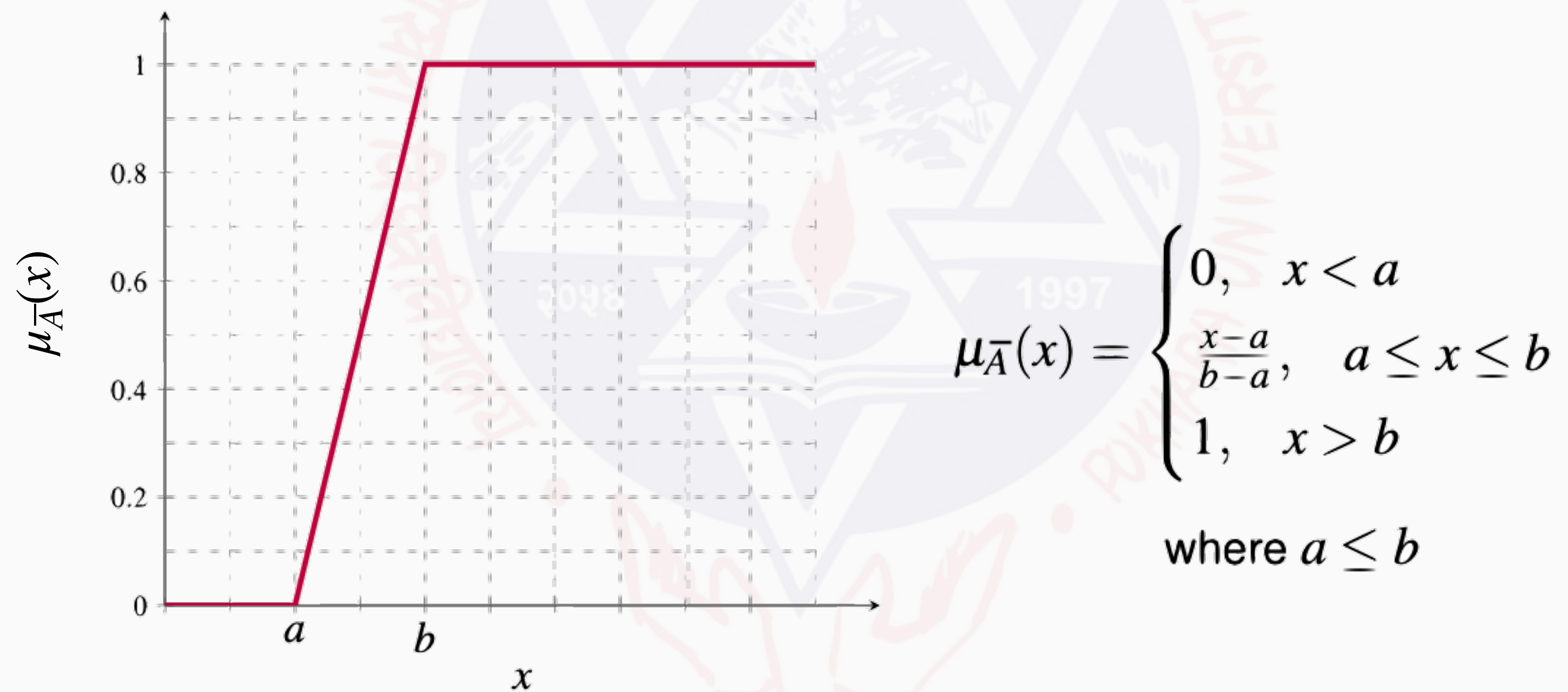
Fuzzy Sets- Membership Functions

- Z-Shaped Membership Function



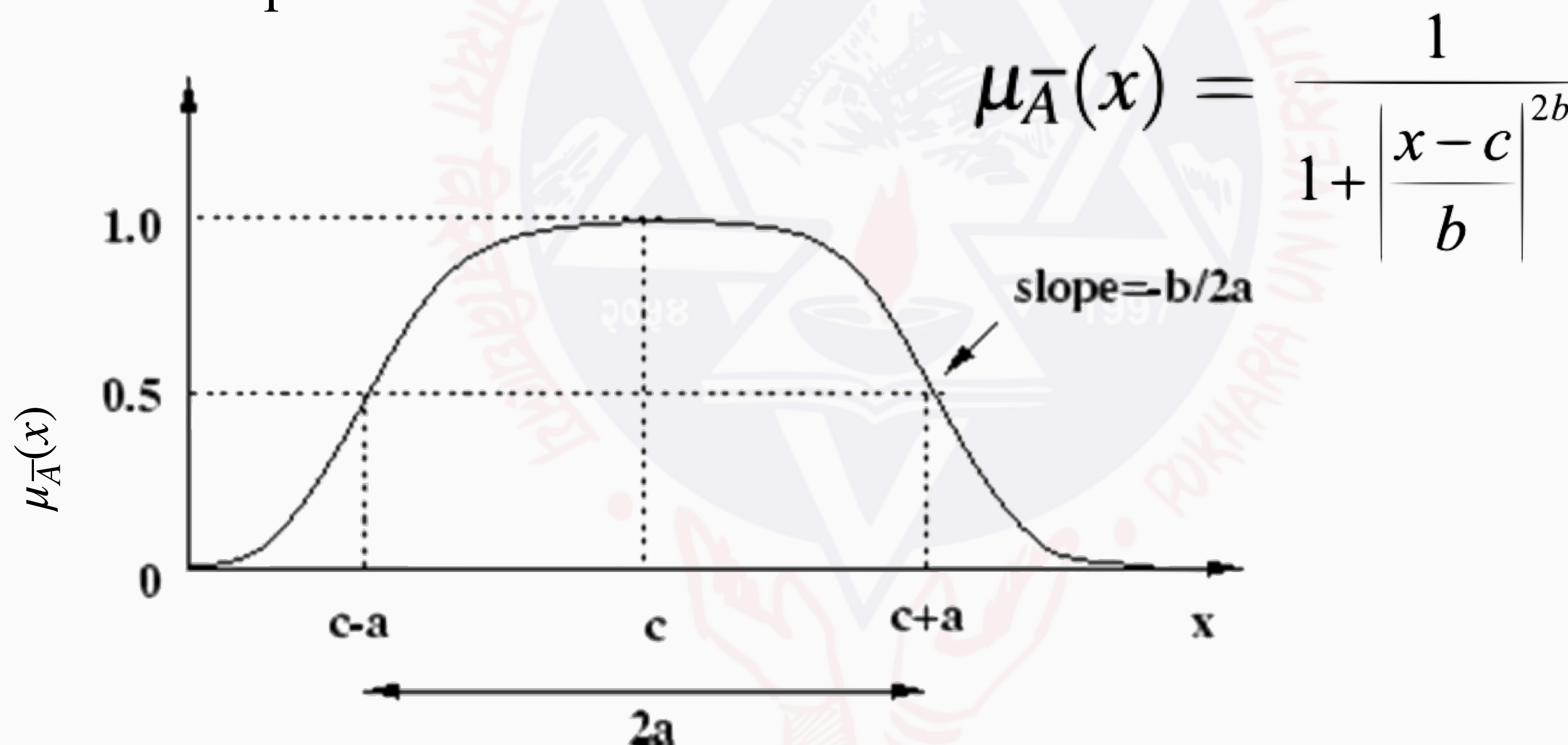
Fuzzy Sets- Membership Functions

- S-Shaped Membership Function



Fuzzy Sets- Membership Functions

- Generalized Bell Membership Function
 - A generalized bell membership function has three parameters: a – responsible for its width, c – responsible for its center and b – responsible for its slopes.



Fuzzy Sets- Membership Functions

- **Core**

- the core of a membership function $\mu_{\bar{A}}(x)$ of a fuzzy set $\bar{A}$ is the region of which $\mu_{\bar{A}}(x) = 1$ for all x in the universe U .

Mathematically, $Core(\bar{A}) = \{x \in U \mid \mu_{\bar{A}}(x) = 1\}$

- **Support**

- the support of the fuzzy set $\bar{A}$ is the set of all points $x \in U$ such that $\mu_{\bar{A}}(x) > 0$.

Mathematically, $Support(\bar{A}) = \{x \in U \mid \mu_{\bar{A}}(x) > 0\}$

- **Boundary**

- the boundary of the fuzzy set $\bar{A}$ is the set of all points $x \in U$ such that $0 < \mu_{\bar{A}}(x) < 1$.

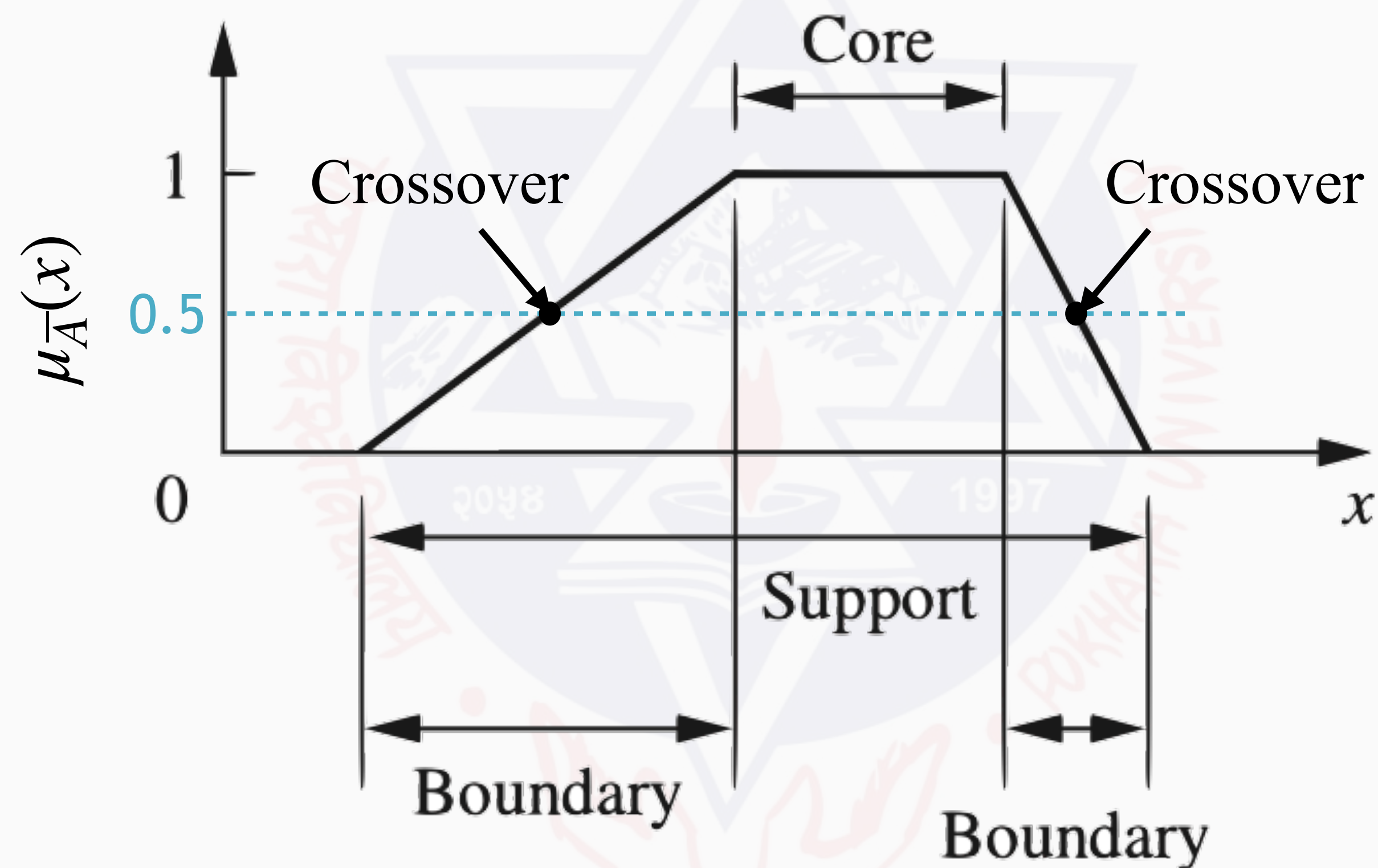
- **Crossover**

- A crossover point of a fuzzy set $\bar{A}$ is a point $x \in U$ at which $\mu_{\bar{A}}(x) = 0.5$.

Mathematically, $Crossover(\bar{A}) = \{x \in U \mid \mu_{\bar{A}}(x) = 0.5\}$



Fuzzy Sets- Membership Functions



Fuzzy Sets- Membership Functions

- We have multiple different types of membership functions.
 - the triangular waveform,
 - trapezoidal waveform,
 - Gaussian waveform,
 - bell-shaped waveform,
 - sigmoidal waveform and
 - S-curve waveform.
- Which one to choose?
 - It depends on the actual applications.
- If you need significant dynamic variation in a short period of time,
 - Use a triangular or trapezoidal waveform
- If you need very high control accuracy,
 - Use a Gaussian or S-curve waveform



Fuzzy Sets- Operations

- Consider, in the universe U , two fuzzy sets
- $\bar{A} = \{\frac{0.2}{a} + \frac{0.4}{b} + \frac{0.5}{c}\}$ and $\bar{B} = \{\frac{0.6}{a} + \frac{0.3}{b} + \frac{0.1}{c} + \frac{0.8}{d}\}$
- Let x is the element of U (i.e. $x \in U$). Let $\mu_{\bar{A}}(x)$ and $\mu_{\bar{B}}(x)$ are the membership values of x in fuzzy sets $\bar{A}$ and $\bar{B}$ respectively.
- Then we define the following fuzzy set operations in the next slides
 - Union
 - Intersection
 - Complement
 - Difference



Fuzzy Sets- Union Operation

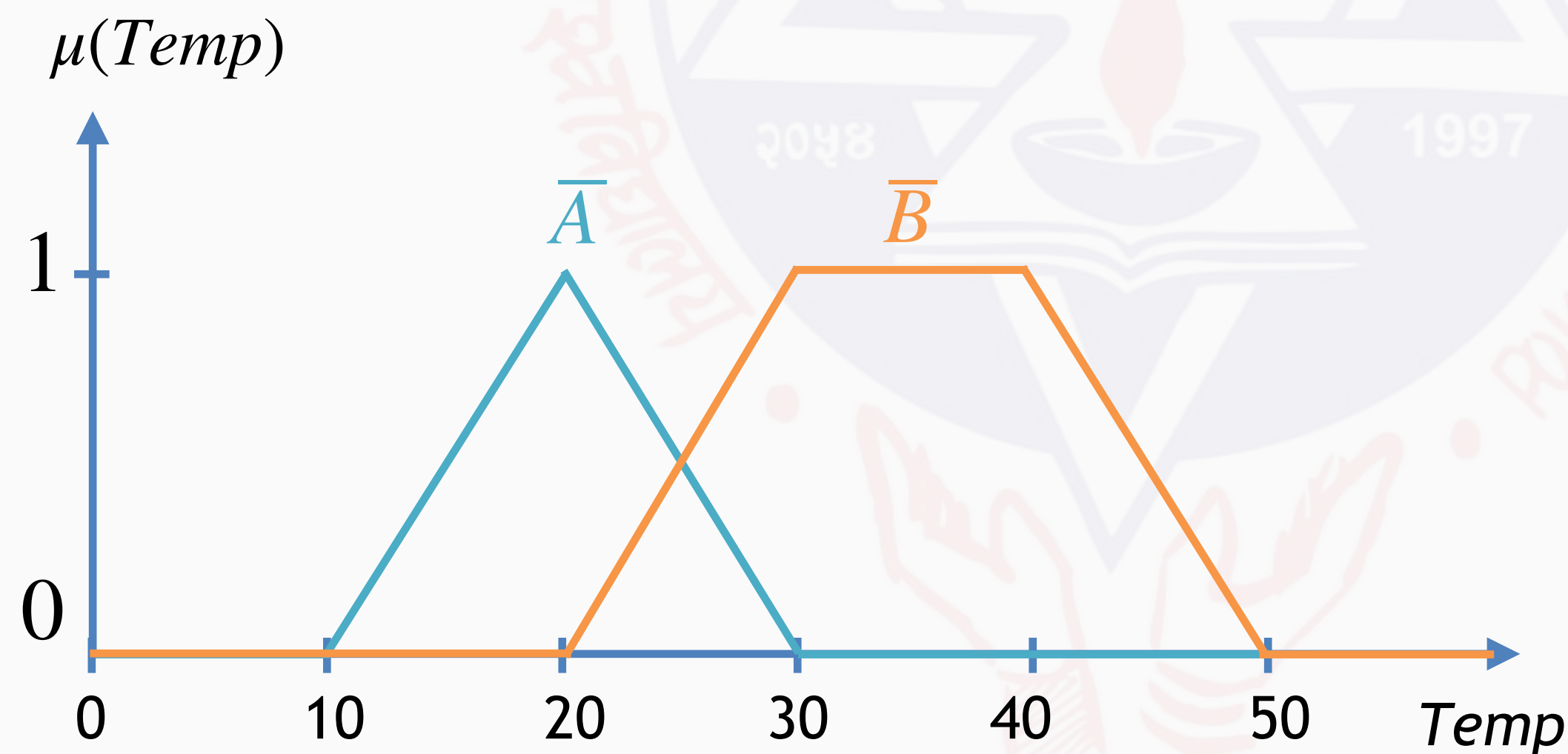
- Union of $\bar{A}$ and $\bar{B}$

$$\bar{A} \cup \bar{B} = \mu_{\bar{A}}(x) \cup \mu_{\bar{B}}(x)$$

$$= \max(\mu_{\bar{A}}(x), \mu_{\bar{B}}(x)) \quad \forall x \in U$$

$$\bar{A} = \left\{ \frac{0.2}{a} + \frac{0.4}{b} + \frac{0.5}{c} \right\}$$

$$\bar{B} = \left\{ \frac{0.6}{a} + \frac{0.3}{b} + \frac{0.1}{c} + \frac{0.8}{d} \right\}$$



Fuzzy Sets- Union Operation

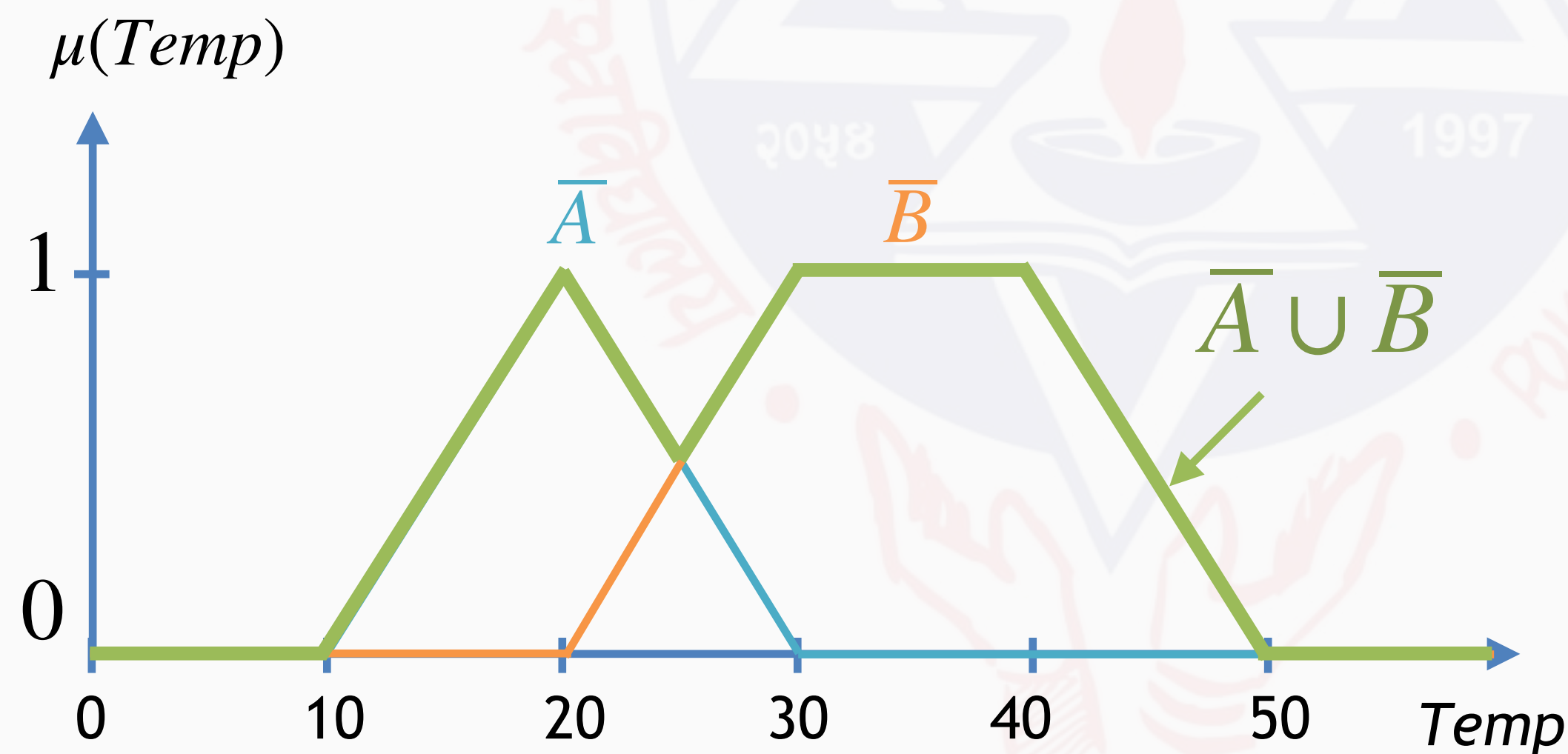
- Union of $\bar{A}$ and $\bar{B}$

$$\bar{A} \cup \bar{B} = \mu_{\bar{A}}(x) \cup \mu_{\bar{B}}(x)$$

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$$\bar{B} = \left\{ \frac{0.6}{a} + \frac{0.3}{b} + \frac{0.1}{c} + \frac{0.8}{d} \right\}$$



Fuzzy Sets- Union Operation

- Union of $\bar{A}$ and $\bar{B}$

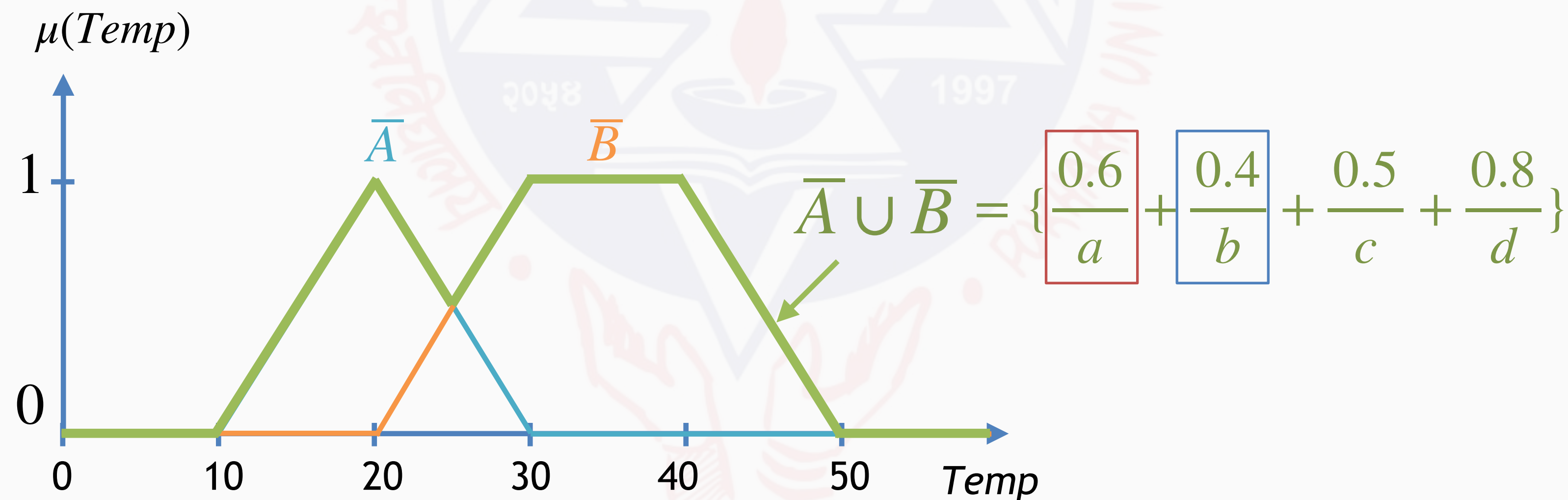
$$\begin{aligned}\bar{A} \cup \bar{B} &= \mu_{\bar{A}}(x) \cup \mu_{\bar{B}}(x) \\ &= \max(\mu_{\bar{A}}(x), \mu_{\bar{B}}(x)) \quad \forall x \in U\end{aligned}$$

$$\bar{A} = \left\{ \boxed{\frac{0.2}{a}} + \boxed{\frac{0.4}{b}} + \frac{0.5}{c} + \frac{0}{d} \right\}$$

max

$$\bar{B} = \left\{ \boxed{\frac{0.6}{a}} + \boxed{\frac{0.3}{b}} + \frac{0.1}{c} + \frac{0.8}{d} \right\}$$

max



Fuzzy Sets- Intersection Operation

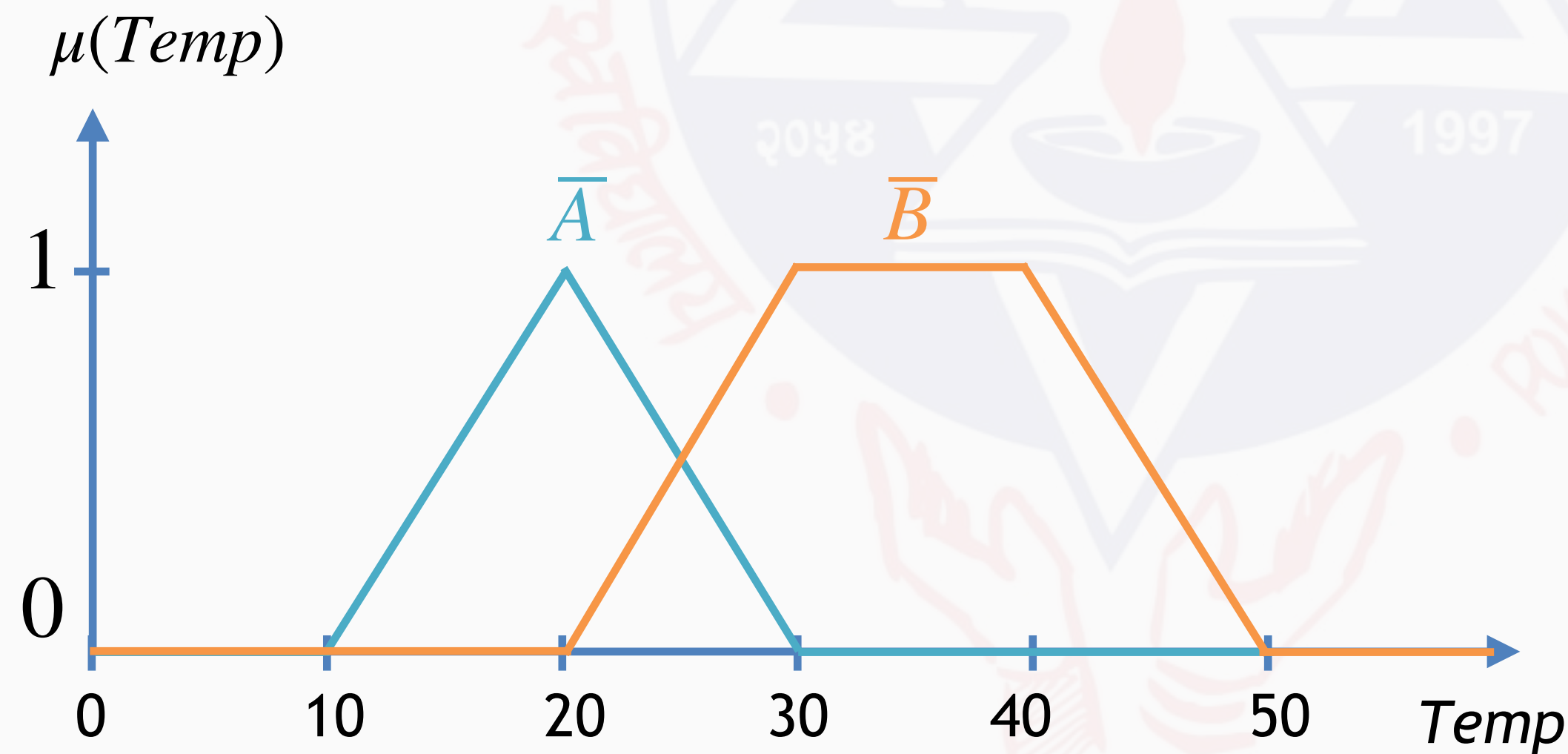
- Intersection of $\bar{A}$ and $\bar{B}$

$$\bar{A} \cap \bar{B} = \mu_{\bar{A}}(x) \cap \mu_{\bar{B}}(x)$$

$$= \min(\mu_{\bar{A}}(x), \mu_{\bar{B}}(x)) \quad \forall x \in U$$

$$\bar{A} = \left\{ \frac{0.2}{a} + \frac{0.4}{b} + \frac{0.5}{c} \right\}$$

$$\bar{B} = \left\{ \frac{0.6}{a} + \frac{0.3}{b} + \frac{0.1}{c} + \frac{0.8}{d} \right\}$$



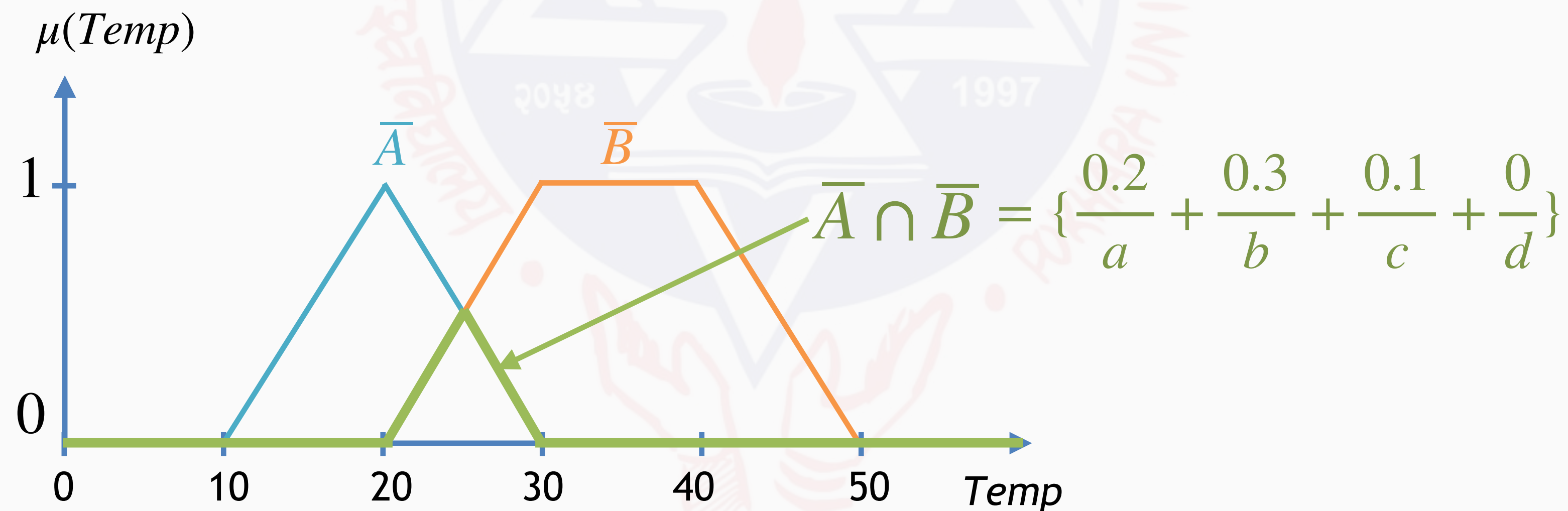
Fuzzy Sets- Intersection Operation

- Intersection of $\bar{A}$ and $\bar{B}$

$$\begin{aligned}\bar{A} \cap \bar{B} &= \mu_{\bar{A}}(x) \cap \mu_{\bar{B}}(x) \\ &= \min(\mu_{\bar{A}}(x), \mu_{\bar{B}}(x)) \quad \forall x \in U\end{aligned}$$

$$\bar{A} = \left\{ \frac{0.2}{a} + \frac{0.4}{b} + \frac{0.5}{c} + \frac{0}{d} \right\}$$

$$\bar{B} = \left\{ \frac{0.6}{a} + \frac{0.3}{b} + \frac{0.1}{c} + \frac{0.8}{d} \right\}$$



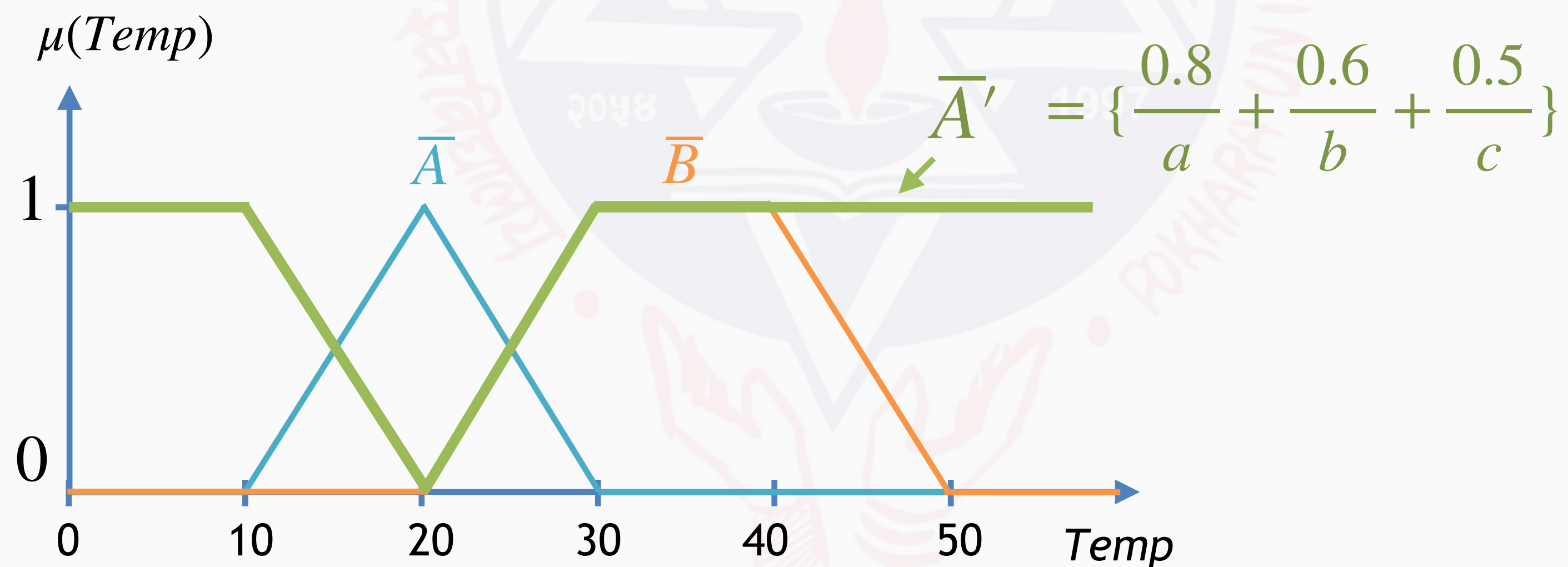
Fuzzy Sets- Complement Operation

- Complement of $\bar{A}$

$$\bar{A} = \left\{ \frac{0.2}{a} + \frac{0.4}{b} + \frac{0.5}{c} \right\}$$

$$\bar{A}' = \mu_{\bar{A}'}(x)$$

$$= 1 - \mu_{\bar{A}}(x) \quad \forall x \in U$$

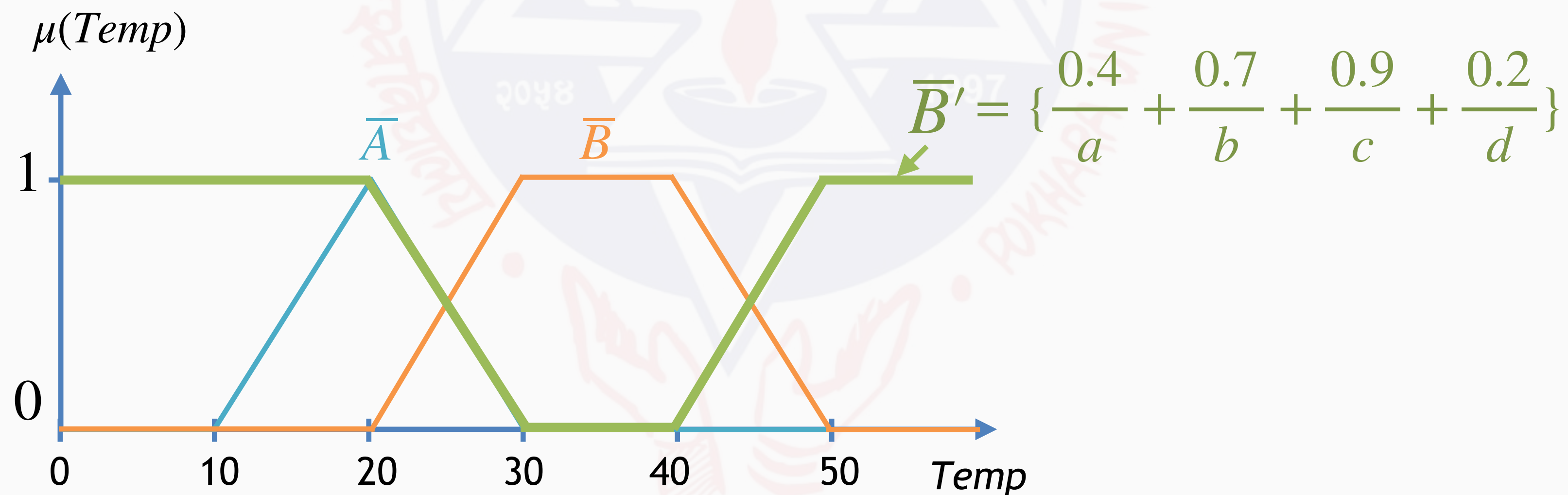


Fuzzy Sets- Complement Operation

- Complement of $\bar{B}$

$$\begin{aligned}\bar{B}' &= \mu_{\bar{B}'}(x) \\ &= 1 - \mu_{\bar{B}}(x) \quad \forall x \in U\end{aligned}$$

$$\bar{B} = \left\{ \frac{0.6}{a} + \frac{0.3}{b} + \frac{0.1}{c} + \frac{0.8}{d} \right\}$$



Fuzzy Sets- Difference Operation

- Difference of $\bar{A}$ and $\bar{B}$

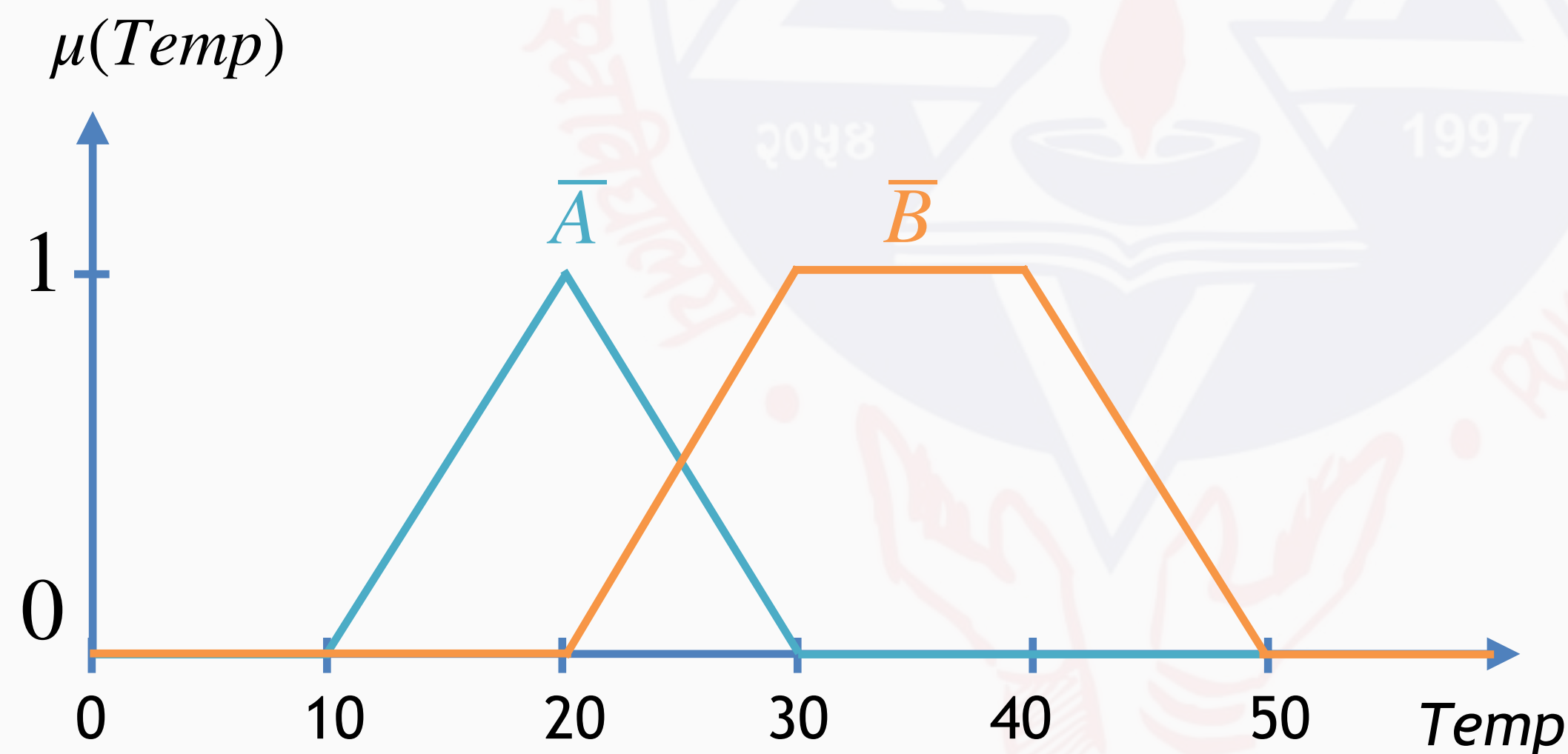
$$\bar{A} - \bar{B} = \mu_{\bar{A}}(x) - \mu_{\bar{B}}(x)$$

$$= \min(\mu_{\bar{A}}(x), 1 - \mu_{\bar{B}}(x)) \quad \forall x \in U$$

$$= \min(\mu_{\bar{A}}(x), 1 - \mu_{\bar{B}}(x))$$

$$\bar{A} = \left\{ \frac{0.2}{a} + \frac{0.4}{b} + \frac{0.5}{c} \right\}$$

$$\bar{B} = \left\{ \frac{0.6}{a} + \frac{0.3}{b} + \frac{0.1}{c} + \frac{0.8}{d} \right\}$$



Fuzzy Sets- Difference Operation

- Difference of $\bar{A}$ and $\bar{B}$

$$\bar{A} - \bar{B} = \mu_{\bar{A}}(x) - \mu_{\bar{B}}(x)$$

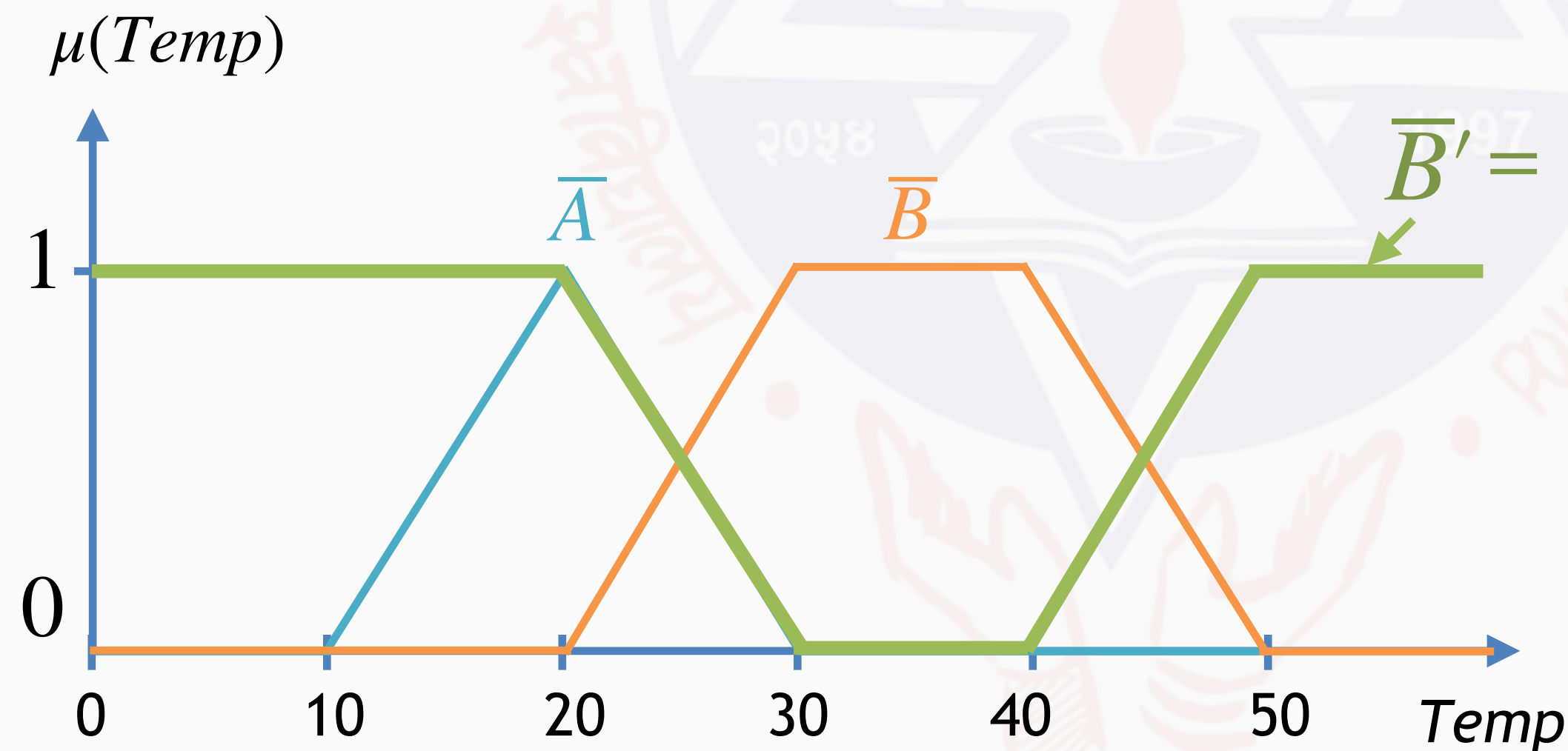
$$= \min(\mu_{\bar{A}}(x), 1 - \mu_{\bar{B}}(x)) \quad \forall x \in U$$

$$= \min(\mu_{\bar{A}}(x), \mu_{\bar{B}'}(x))$$

$$\bar{A} = \left\{ \frac{0.2}{a} + \frac{0.4}{b} + \frac{0.5}{c} \right\}$$

$$\bar{B} = \left\{ \frac{0.6}{a} + \frac{0.3}{b} + \frac{0.1}{c} + \frac{0.8}{d} \right\}$$

$$\bar{B}' = \left\{ \frac{0.4}{a} + \frac{0.7}{b} + \frac{0.9}{c} + \frac{0.2}{d} \right\}$$



Fuzzy Sets- Difference Operation

- Difference of $\bar{A}$ and $\bar{B}$

$$\bar{A} - \bar{B} = \mu_{\bar{A}}(x) - \mu_{\bar{B}}(x)$$

$$= \min(\mu_{\bar{A}}(x), 1 - \mu_{\bar{B}}(x)) \quad \forall x \in U$$

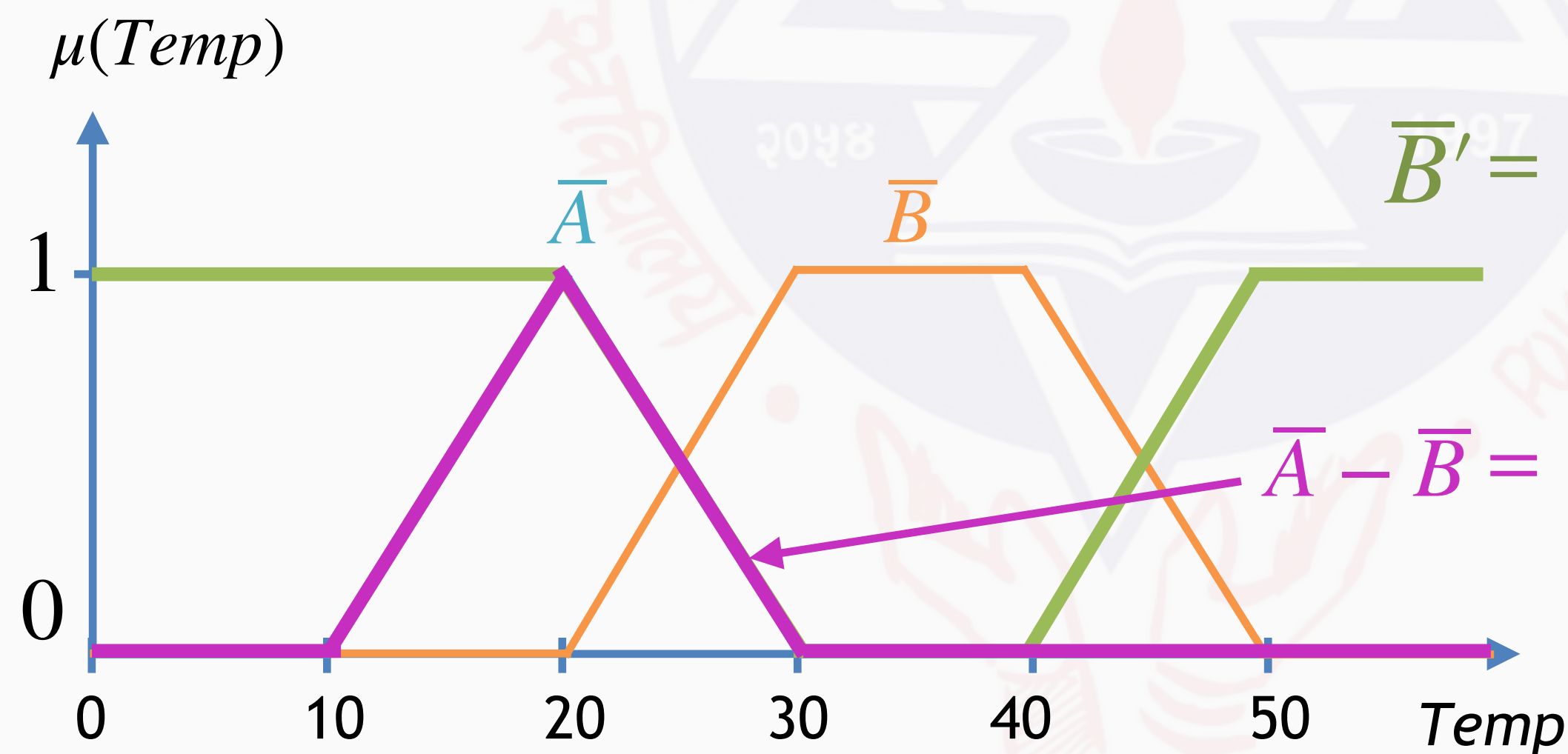
$$= \min(\mu_{\bar{A}}(x), \mu_{\bar{B}'}(x))$$

$$\bar{A} = \left\{ \frac{0.2}{a} + \frac{0.4}{b} + \frac{0.5}{c} \right\}$$

$$\bar{B} = \left\{ \frac{0.6}{a} + \frac{0.3}{b} + \frac{0.1}{c} + \frac{0.8}{d} \right\}$$

$$\bar{B}' = \left\{ \frac{0.4}{a} + \frac{0.7}{b} + \frac{0.9}{c} + \frac{0.2}{d} \right\}$$

$$\bar{A} - \bar{B} = \left\{ \frac{0.2}{a} + \frac{0.4}{b} + \frac{0.5}{c} + \frac{0}{d} \right\}$$



Fuzzy Set Operations

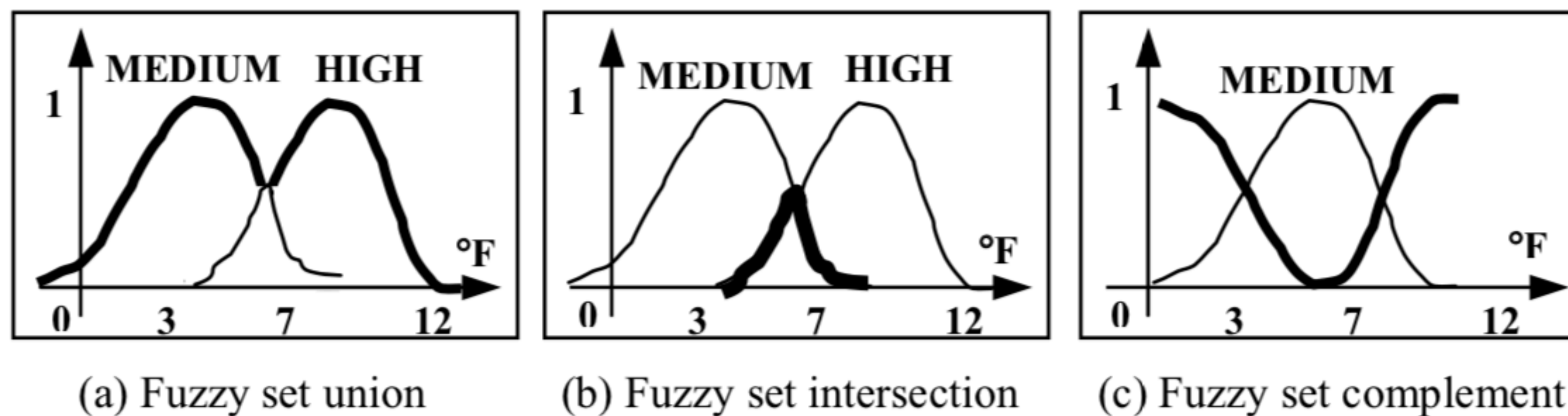


Figure 2.4. Fuzzy set operations



Fuzzy Control Rules

- A *Fuzzy rule*
 - describes how to make decisions based on fuzzy inputs.
 - is represented by a sequence of *IF- THEN* form.
- A fuzzy *IF-THEN* rule in format- **IF conditions THEN actions**
 - associates a *condition* (described using linguistic variables and fuzzy sets) to an *output* (or a conclusion).
 - The IF part is used to capture knowledge and
 - the THEN part provides the conclusion in linguistic variable.
- Example:
 - IF the temperature is **LOW**, THEN the heater motor should be **FAST**.
 - IF the temperature is **LOW**, and the change rate of the temperature is **LOW**, THEN the heater motor speed should be **FAST**
 - "If the room is *slightly warm* and the humidity is *high*, then set the fan to *medium*."



Fuzzy Control Rules

- General Rule Format:

Rule i : *IF* x_1 is $\bar{A}_{i1}$ *AND/OR* x_2 is $\bar{A}_{i2}$ *AND/OR...* *THEN* y is $\bar{B}_i$ $i = 1, 2, 3 \dots r$

$\underbrace{\hspace{15em}}_{\text{Antecedent}} \qquad \underbrace{\hspace{5em}}_{\text{Consequent}}$

- $x_1, x_2 \dots$ are the fuzzy/linguistic variables
- y is the output of the fuzzy inference system
- *and* and *or* are the fuzzy operators
- $\bar{A}_{i1}, \bar{A}_{i2} \dots$ are the fuzzy sets (associated with linguistic variables representing i^{th} antecedents pair
- $\bar{B}_i$ is the fuzzy set (associated with linguistic variables) representing the i^{th} consequent.



Fuzzy Control Rules

- A Fuzzy Operators in Fuzzy Rule

- **AND (intersection)**

- The degree of truth is calculated as the **minimum** of the input membership values.

- $\mu_{AND}(x) = \mu_{A \cap B} = \mu_A(x) \wedge \mu_B(x) = \min(\mu_A(x), \mu_B(x))$

- **OR (Union)**

- The degree of truth is calculated as the **maximum** of the input membership values.

- $\mu_{OR}(x) = \mu_{A \cup B} = \mu_A(x) \vee \mu_B(x) = \max(\mu_A(x), \mu_B(x))$

- **NOT (Complement)**

- The degree of truth is calculated as the complement of the input membership value.

- $\mu_{NOT}(x) = \mu_{A'}(x) = 1 - \mu_A(x)$



Fuzzy Control Rules

- **Air Conditioning System**

- *Input*: Temperature

- *Output*: Cooling Power

- *Rules*:

1. *IF* temperature is cold **THEN** cooling power is low.

2. *IF* temperature is warm **THEN** cooling power is medium.

3. *IF* temperature is hot **THEN** cooling power is high.



Fuzzy Control Rules

- **Washing Machine**

- *Input:* Dirt Level and Load Weight

- *Output:* Washing Time

- *Rules:*

1. IF dirt level is low AND load weight is light THEN washing time is short.
2. IF dirt level is medium AND load weight is heavy THEN washing time is long.
3. IF dirt level is high OR load weight is very heavy THEN washing time is extra long.



Fuzzy Control Rules

- **Traffic Control System**
 - *Input:* Vehicle Density
 - *Output:* Traffic Light Duration
 - *Rules:*
 - IF vehicle density is low THEN green light duration is long.
 - IF vehicle density is medium THEN green light duration is medium.
 - IF vehicle density is high THEN green light duration is short.



Fuzzy Control Rules

- *Lookup Table*

- For a system with two inputs and one output, a rule based table, also called *look up table*, can be constructed.
- The rows and columns represent two inputs, (For example, the temperature and the change rate of the temperature). These inputs are related to IF parts in IF- THEN rules.
- The conclusion or control output can be expressed as a third dimensional variable that is located at the cross point of each row and each column
- The conclusion is associated with the THEN part in IF-THEN rules.



Fuzzy Control Rules

- *Example 1:*

- Here a system has two inputs- *Temperature (T)* and *Change Rate of Temperature T'* and one output- *heater motor speed*).

Table 2.1. An example of fuzzy rules

$T' \backslash T$	LOW	MEDIUM	HIGH
LOW	FAST	MEDIUM	MEDIUM
MEDIUM	FAST	SLOW	SLOW
HIGH	MEDIUM	SLOW	SLOW

IF the temperature is LOW, **AND** the change rate of the temperature is LOW,
THEN the output (heater motor speed) should be FAST



Fuzzy Control Rules

- *Example 2:*

- Here a system with two inputs- *Temperature* and *Humidity* and one output- *Cooling Power*), a rule based table can be constructed as

Humidity → Temperature ↓	LOW	MEDIUM	HIGH
COLD	Low Cooling	Low Cooling	Medium Cooling
WARM	Medium Cooling	Medium Cooling	High Cooling
HOT	Medium Cooling	High Cooling	High Cooling

IF the temperature is HOT, **AND** the humidity is MEDIUM,
THEN the output (Cooling Power) should be HIGH COOLING.



Fuzzy Inference System

- A ***Fuzzy Inference System*** (FIS) is also known as ***fuzzy-rule-based system, fuzzy expert system, fuzzy logic system, fuzzy model*** etc.
- It mimics the human being's feeling and inference process.
- It is a computing framework based on fuzzy set theory, fuzzy (If-Then) rules and fuzzy reasoning
- To implement it to a real application requires the following steps:
 - ***Fuzzification***
 - ***Fuzzy Inference Process***
 - ***Defuzzification***
- A fuzzy process is a process of crisp-fuzzy-crisp for a real system.



Fuzzy Inference System

- **Fuzzification**

- is the process of converting crisp (precise) inputs into fuzzy values by mapping them to ***fuzzy sets*** using ***membership functions***.
- This step allows the system to handle uncertain or imprecise data.

- **Fuzzy Inference Process**

- apply the fuzzy logic rules to the fuzzified inputs to produce fuzzy outputs.
- The fuzzy logic rules are typically based on if-then statements.

- **Defuzzification**

- is the process of converting the fuzzy output back into a crisp, actionable value.



Fuzzy Inference System

- A **Fuzzy Inference System** (FIS) consists of 4 components:

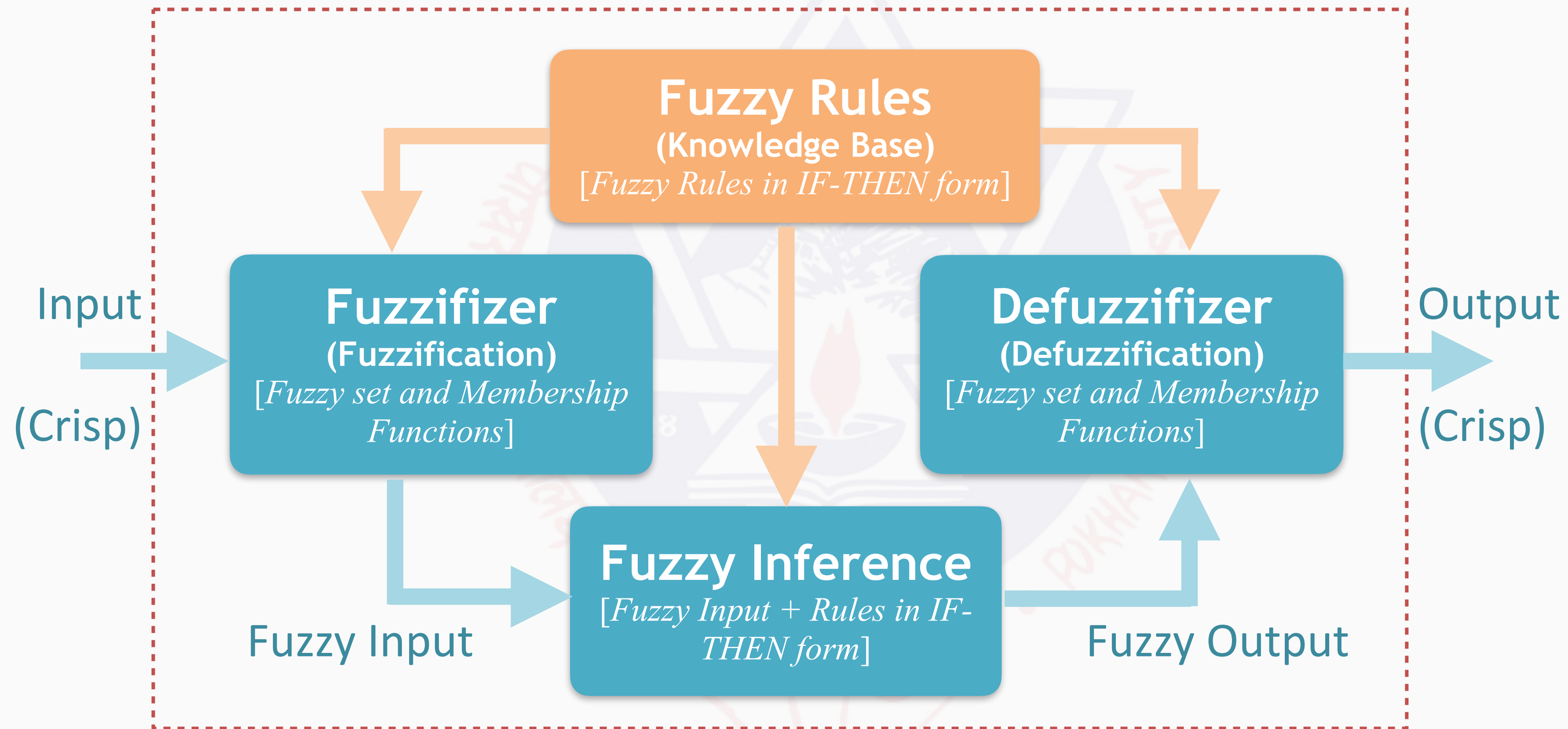


Figure: Fuzzy Inference System



Fuzzy Inference System

- **Types of Fuzzy Inference System**

- Mamdani FIS
 - Min-Max Mamdani
 - Max-Product Mamdani
- Sugeno FIS
- Tsukamoto FIS



Defuzzification

- The conclusion or control output derived from fuzzy inference engine is still a vague or fuzzy element. The fuzzy conclusion or output is still a linguistic variable.
- This linguistic variable (or fuzzy output) needs to be converted to the crisp output via a defuzzification process.
- The ***defuzzification process*** converts the fuzzy output back to the crisp or classical output.
- Commonly used defuzzification methods
 - Mean of Maximum method (MOM)
 - Center of Gravity method (COG)
 - Height method

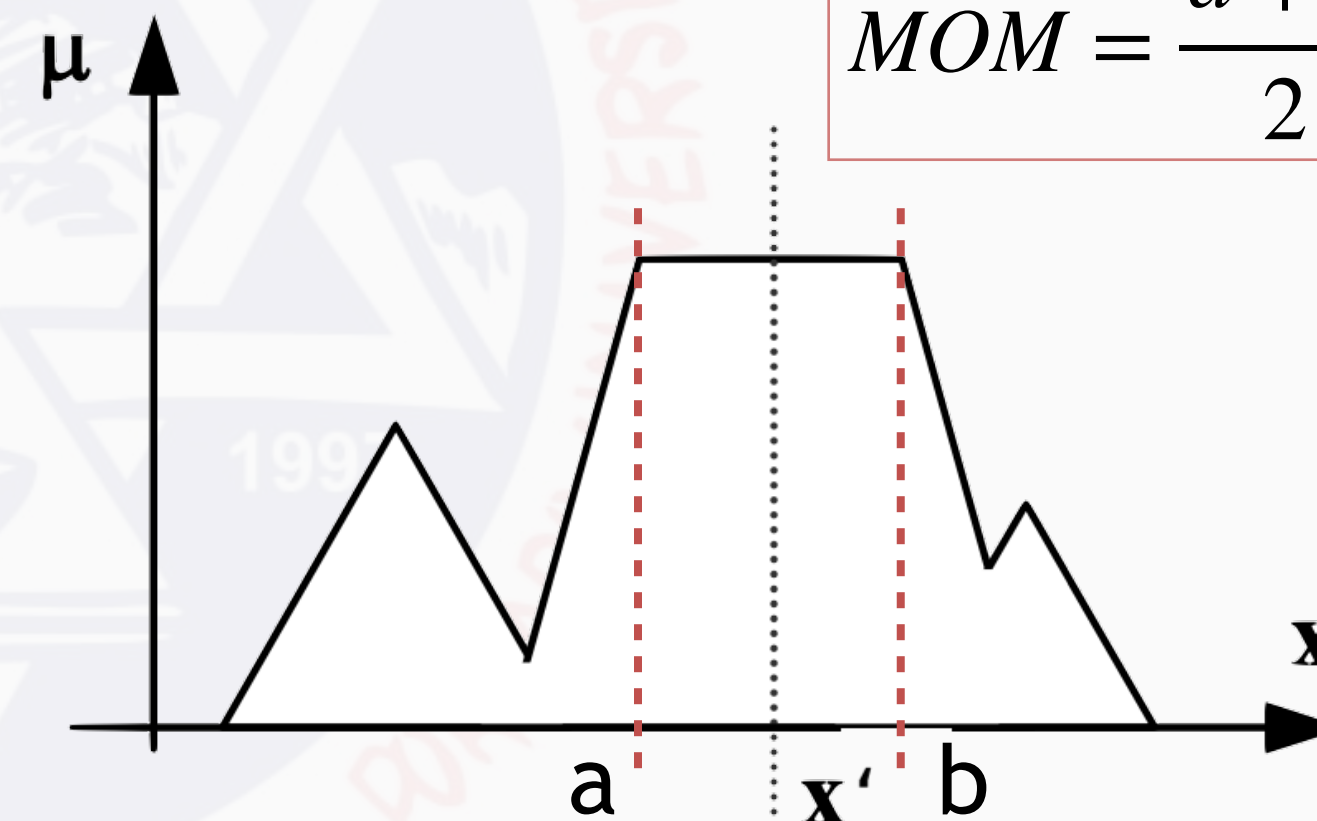


Defuzzification Methods

- The Mean of Maximum (MOM) defuzzification method
 - computes the average of those fuzzy conclusions or outputs that have the highest degrees.
 - For example, the fuzzy conclusion is: *the heater motor x is rotated FAST*. By using the MOM method, this defuzzification can be expressed as

$$MOM(FAST) = \frac{\sum_{x' \in T} x'}{T}$$

$$T = \{x' \mid \mu_{FAST}(x') = \text{Support}_{\mu_{FAST}}(x)\}$$



Or Simply,

$$MOM = \frac{a + b}{2}$$

- Here, the horizontal line is algebraic division.
- T is the set of output x that has the highest degrees in the set FAST.
- **Drawback:** it does not consider the entire shape of the output membership function. It only takes care of the points that have the highest degrees.



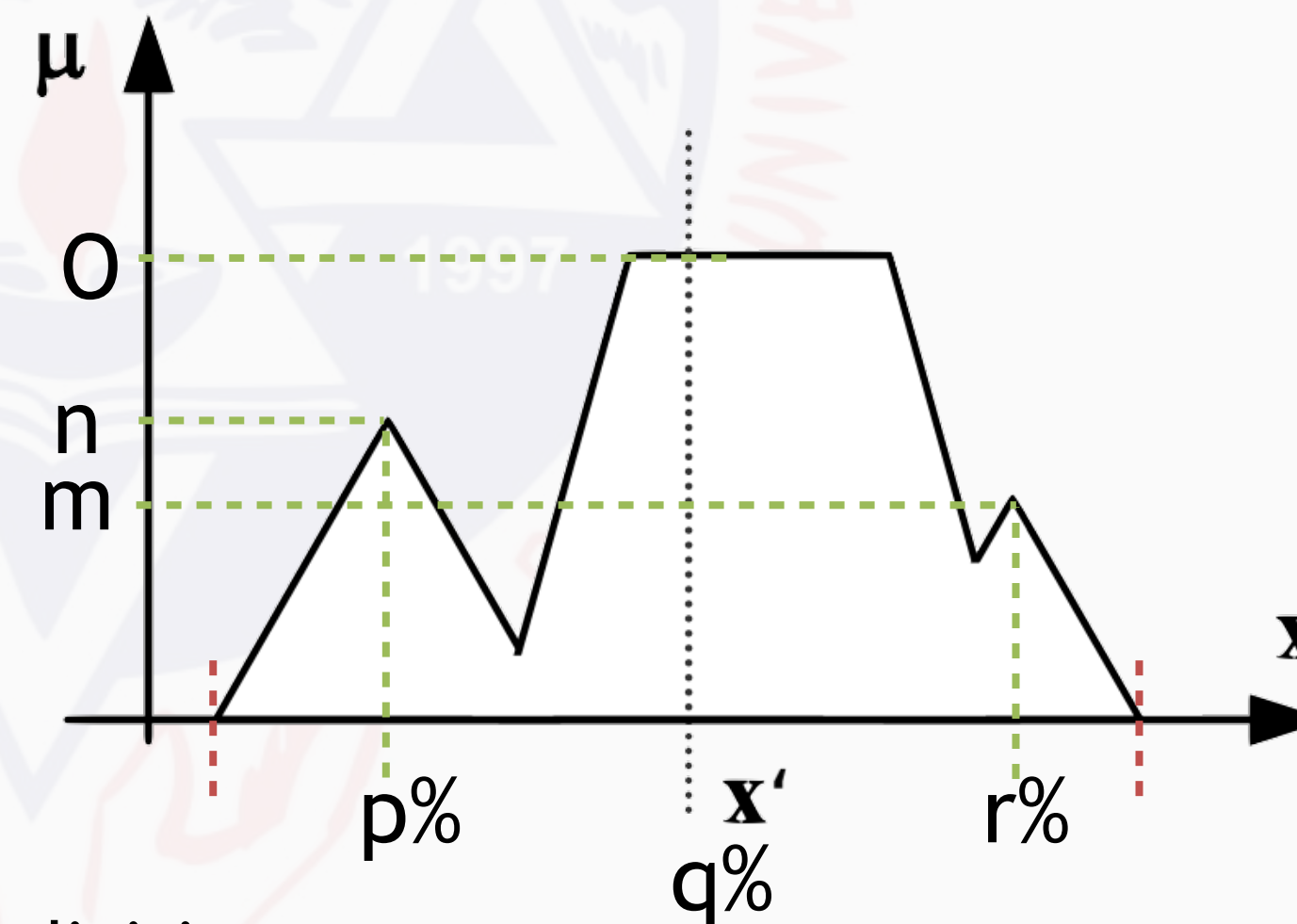
Defuzzification Methods

- **The Center of Gravity (COG) Method**

- is the most popular defuzzification method
- This method is similar to the formula for calculating the center of gravity in physics.
- Computes the weighted average of the membership function of the area bounded by the membership function curve.
- The COG output can be expressed as

$$COG = \frac{\sum_i \mu(x_i)x_i}{\sum_i \mu(x_i)}$$

$$\therefore COG = \frac{n * p + o * q + m * r}{n + o + m}$$

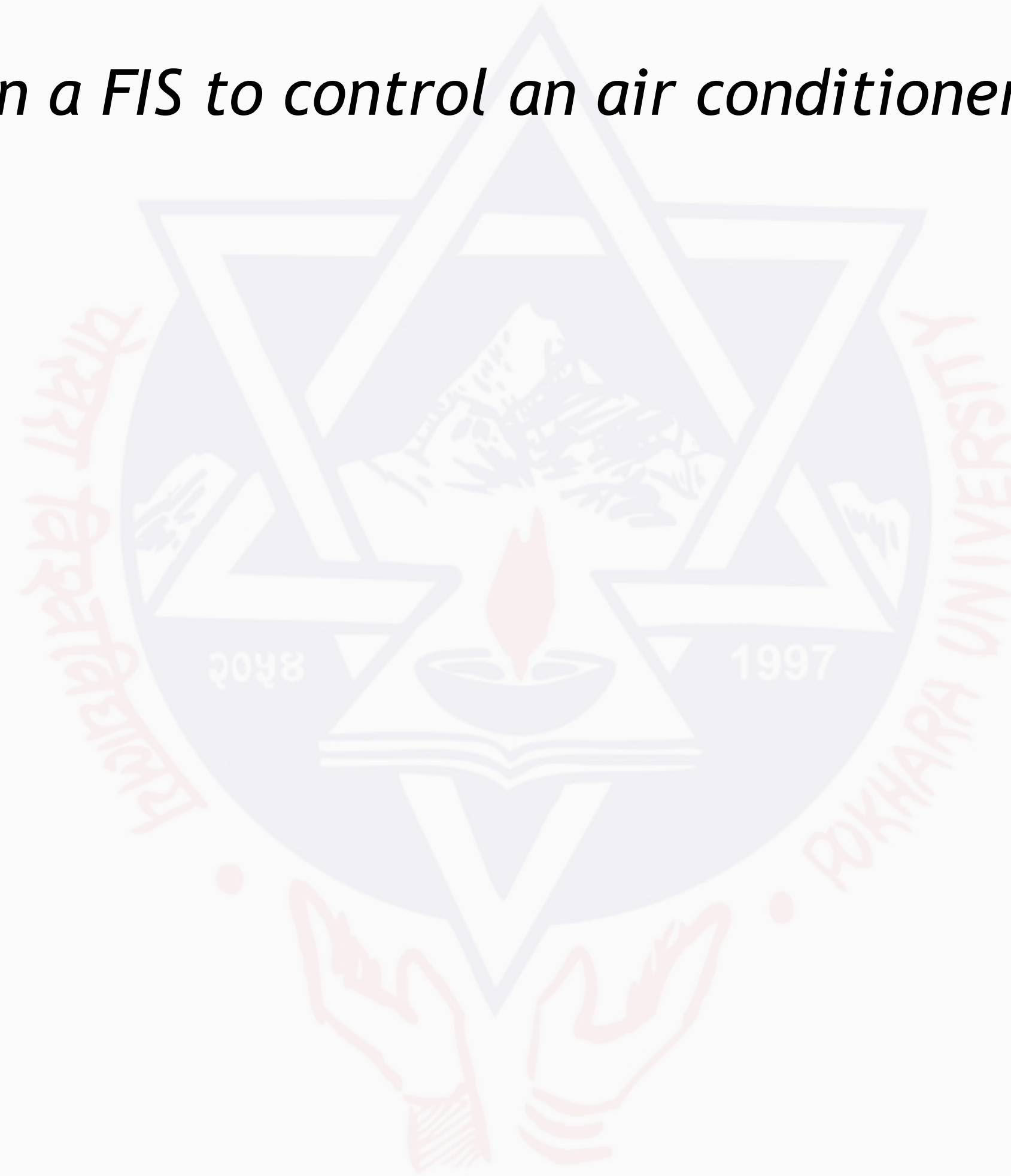


- Here, the horizontal line is algebraic division.



Air Conditioner Control System

- **Problem:** *Design a FIS to control an air conditioner based on room temperature.*



Air Conditioner Control System

- 1. Choose *membership* function and define *Linguistic* variables and *input/output ranges*

- Membership function: *Triangular*

- Input: Temperature in °C

- COLD (0-15-25)

- WARM (15-25-35)

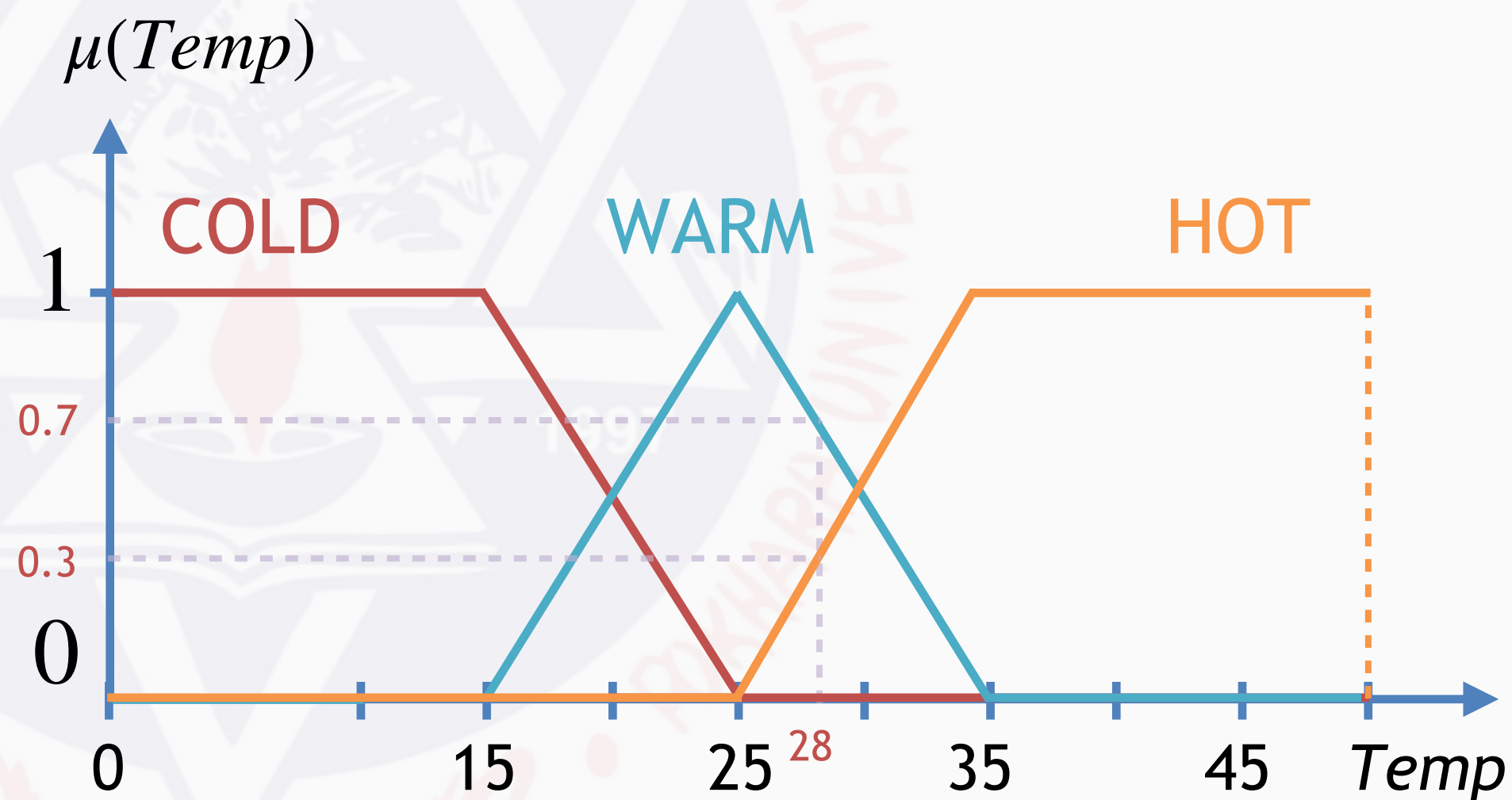
- HOT (25-35-50)

- Output: Cooling Power in %

- LOW (0-25-50)

- MEDIUM (25-50-75)

- HIGH (50-75-100)



Air Conditioner Control System

- 1. Define Membership Functions

- COLD:**

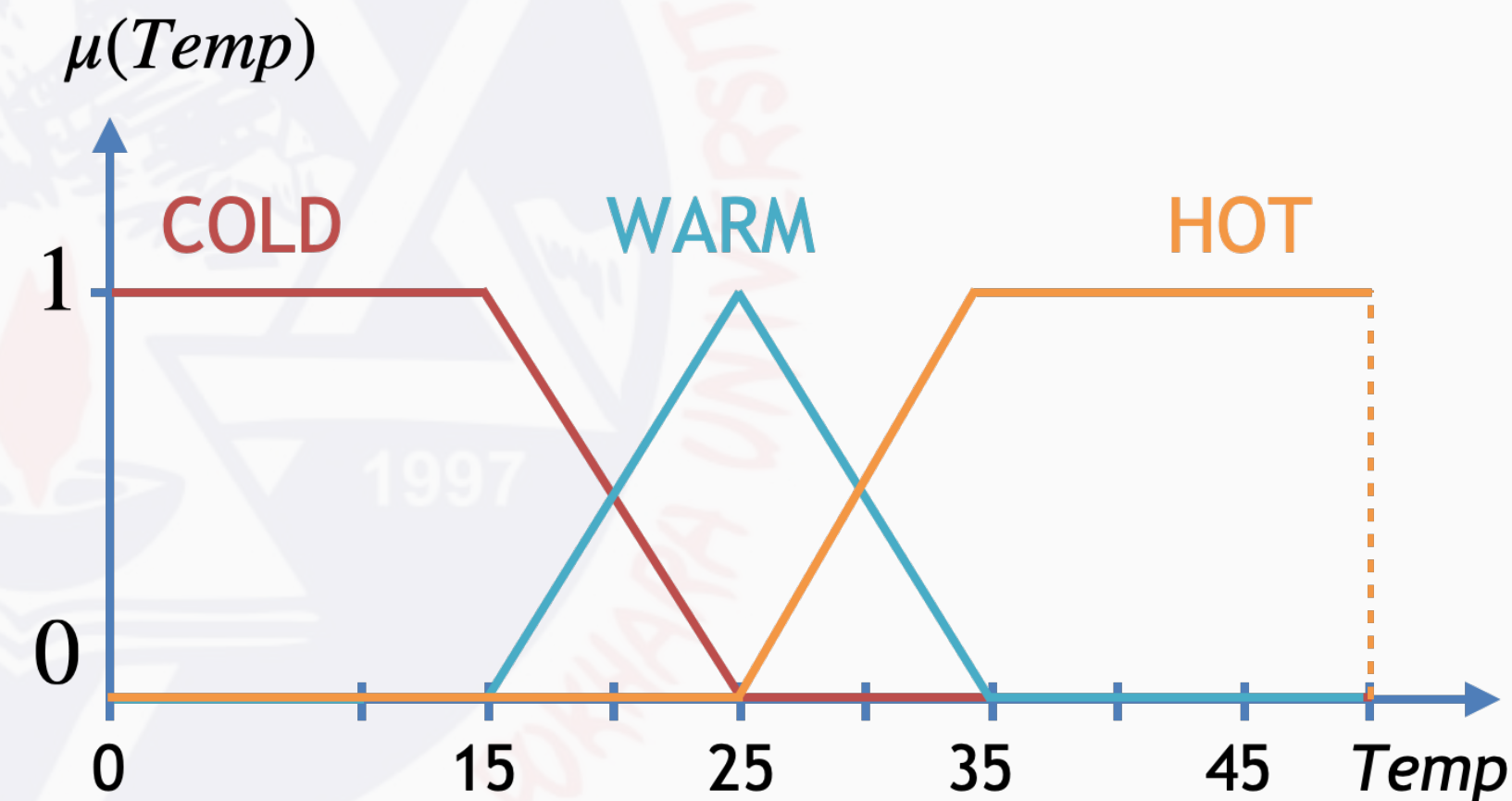
$$\mu_{COLD}(x) = \begin{cases} 1 & \text{if } x \leq 15 \\ \frac{25-x}{10} & \text{if } 15 < x \leq 25 \\ 0 & \text{if } x > 25 \end{cases}$$

- WARM:**

$$\mu_{WARM}(x) = \begin{cases} \frac{x-15}{10} & \text{if } 15 < x \leq 25 \\ \frac{35-x}{10} & \text{if } 25 < x \leq 35 \\ 0 & \text{if } x > 35 \end{cases}$$

- HOT:**

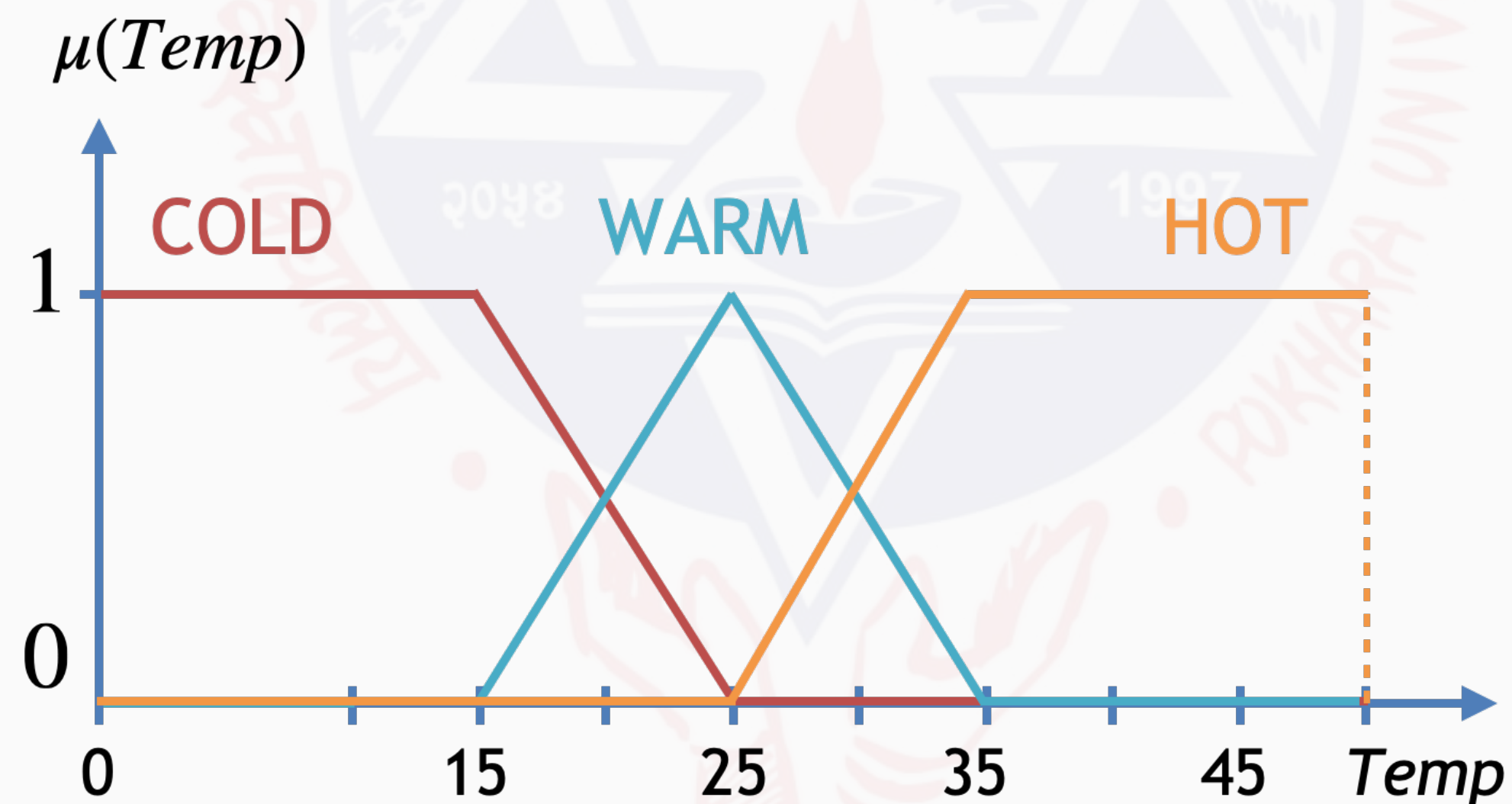
$$\mu_{HOT}(x) = \begin{cases} 0 & \text{if } x \leq 25 \\ \frac{x-25}{10} & \text{if } 25 < x \leq 35 \\ 1 & \text{if } x > 35 \end{cases}$$



Air Conditioner Control System

- 2. Define Fuzzy Rules

- IF Temperature is **COLD**, THEN Cooling Power is **LOW**.
- IF Temperature is **WARM**, THEN Cooling Power is **MEDIUM**.
- IF Temperature is **HOT**, THEN Cooling Power is **HIGH**.



Air Conditioner Control System

- 3.Fuzzification- converting crisp data into fuzzy data

- Crisp data = 28° C

$$\mu_{COLD}(x) = \begin{cases} 1 & \text{if } x \leq 15 \\ \frac{25-x}{10} & \text{if } 15 < x \leq 25 \\ 0 & \text{if } x > 25 \end{cases} \quad \mu_{WARM}(x) = \begin{cases} \frac{x-15}{10} & \text{if } 15 < x \leq 25 \\ \frac{35-x}{10} & \text{if } 25 < x \leq 35 \\ 0 & \text{if } x > 35 \end{cases} \quad \mu_{HOT}(x) = \begin{cases} 0 & \text{if } x \leq 25 \\ \frac{x-25}{10} & \text{if } 25 < x \leq 35 \\ 1 & \text{if } x > 35 \end{cases}$$

- Convert the crisp data 28° C into fuzzy membership value for COLD, WARM and HOT

- COLD membership, $\mu_{COLD}(28) = 0$

- WARM membership, $\mu_{WARM}(28) = \frac{35-28}{10} = 0.7$

- HOT membership, $\mu_{HOT}(28) = \frac{28-25}{10} = 0.3$

- *We can convert every crisp data into membership value for COLD, WARM and HOT*



Air Conditioner Control System

- 4. Apply fuzzy rules

1. *IF* Temperature is **COLD**, *THEN* Cooling Power is **LOW**.
2. *IF* Temperature is **WARM**, *THEN* Cooling Power is **MEDIUM**.
3. *IF* Temperature is **HOT**, *THEN* Cooling Power is **HIGH**.

- Since the rules in our example contains single antecedent, these fuzzy rules produce the Cooling Power as follows:

- Rule 1: $\mu_{COLD}(28) = 0 \implies CoolingPower(Low) = 0$
- Rule 2: $\mu_{WARM}(28) = 0.7 \implies CoolingPower(MEDIUM) = 0.7$
- Rule 3: $\mu_{HOT}(28) = 0.3 \implies CoolingPower(HIGH) = 0.3$



Air Conditioner Control System

- **5. Aggregation**

- Combines the outputs of all rules into a single fuzzy set using max operation.
 - LOW: $\max(0) = 0$
 - MEDIUM: $\max(0.7) = 0.7$
 - HIGH: $\max(0.3) = 0.3$



Air Conditioner Control System

- **5. Defuzzification**

- Using Center of Gravity, computes the weighted average of the membership function of the area bounded by the membership function curve.

- Outputs of:

- LOW: $\max(0) = 0$
- MEDIUM: $\max(0.7) = 0.7$
- HIGH: $\max(0.3) = 0.3$

- Center Values of:

- LOW: 25 %
- MEDIUM: 50 %
- HIGH: 75 %

- Cooling Power = $\frac{0.7 * 50 + 0.3 * 75}{0.7 + 0.3} = 57.5 \%$

- Therefore, at 28° C, the air conditioner will set cooling power to **57.5%**.



Air Conditioner Control System

- Implementation using C++ (Mamdani FIS)

```
float triangularMembership(float x, float a, float b, float c) {  
    if (x <= a || x >= c) return 0.0;  
    else if (x == b) return 1.0;  
    else if (x < b) return (x - a) / (b - a);  
    else return (c - x) / (c - b);  
}
```

```
float defuzzify(float low_mem, float medium_mem, float high_mem) {  
    float numerator = (low_mem * 25) + (medium_mem * 50) + (high_mem * 75);  
    float denominator = low_mem + medium_mem + high_mem;  
    if (denominator == 0) return 0; // Avoid division by zero  
    return numerator / denominator;  
}
```



Air Conditioner Control System

```
int main() {
    // Input: Temperature
    float temperature;
    cout << "Enter temperature (in Celsius): ";
    cin >> temperature;

    // Fuzzification: Calculate membership values for "Cold", "Warm", "Hot"
    float coldMembership = triangularMembership(temperature, 0, 15, 25);
    float warmMembership = triangularMembership(temperature, 15, 25, 35);
    float hotMembership = triangularMembership(temperature, 25, 35, 50);

    // Output: Rule evaluation
    // Rule 1: IF Cold THEN Low Cooling Power
    float lowMembership = coldMembership;
    // Rule 2: IF Warm THEN Medium Cooling Power
    float mediumMembership = warmMembership;
    // Rule 3: IF Hot THEN High Cooling Power
    float highMembership = hotMembership;

    // Defuzzification
    float coolingPower = defuzzify(lowMembership, mediumMembership, highMembership);

    // Output the result
    cout << "Cooling Power: " << coolingPower << "%" << endl;

    return 0;
}
```



Fuzzy Inference System- Mamdani

- The Fuzzy Inference process you used in the example of the Air Conditioner Control system is the ***Mamdani Fuzzy Inference System***.
- The Mamdani is a fuzzy inference system.
- It was proposed by Ebhasim Mamdani in 1975.
- It was designed to control a steam engine and boiler combination by a set of linguistic control rules obtained from the experienced human operators.
- In a Mamdani system, the output of each rule is a fuzzy set.
- They are well-suited to expert system applications where the rules are created from human expert knowledge, such as ***medical diagnostics***.



Mamdani FLS in Basic Tipping Problem

- **Illustration:** *How fuzzy logic is used to determining the correct amount to tip a waitperson at a restaurant (using Mamdani).*
- **Basic Tipping Problem:—**
 - Given a number from 0 through 10 that represents the quality of service at a restaurant, where 10 is excellent and another number from 0 through 10 that represents the quality of the food, where 10 is delicious, **what should the tip be?**
 - Tipping Behavior in US: An average tip for a meal is 15%. A generous tip could be as high as 25% and a cheap tip could be 5%.
 - The **actual amount of the tip** can vary depending on the **quality of the service and food**.

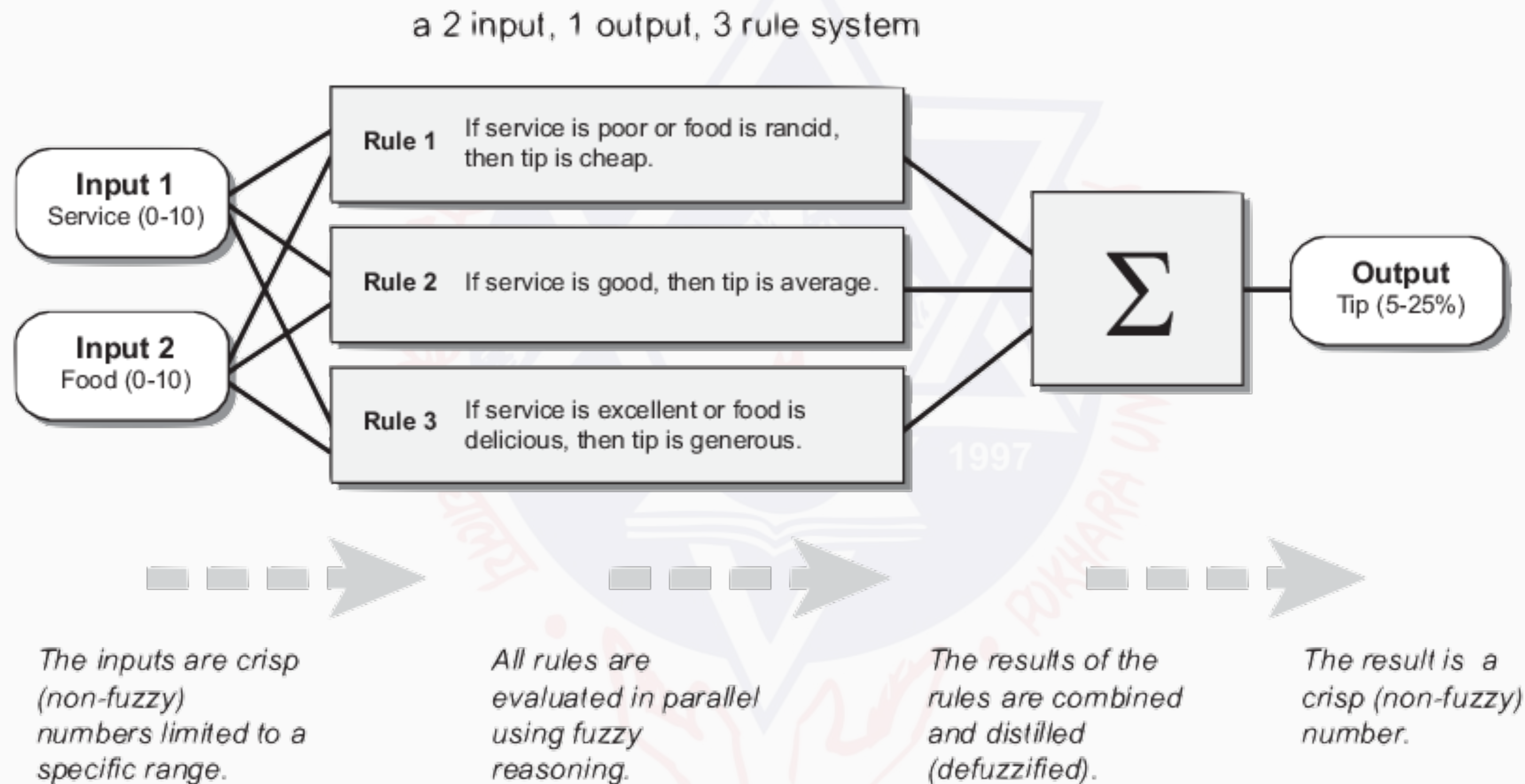


Mamdani FIS in Basic Tipping Problem

- We describe the fuzzy inference process for the two-input, one-output, three-rule tipping problem:
- **Find the Input and Linguistic Variables:**
 1. Service Quality (POOR, GOOD, EXCELLENT)
 2. Food Quality (RANCID, DELICIOUS)
- **Find the Output and Linguistic Variables:**
 1. Tip Percent (CHEAP, AVERAGE, GENEROUS)
- **Define Fuzzy Rules:**
 1. If the service is POOR or the food is RANCID, then tip is CHEAP.
 2. If the service is GOOD, then tip is AVERAGE.
 3. If the service is EXCELLENT or the food is DELICIOUS, then tip is GENEROUS.



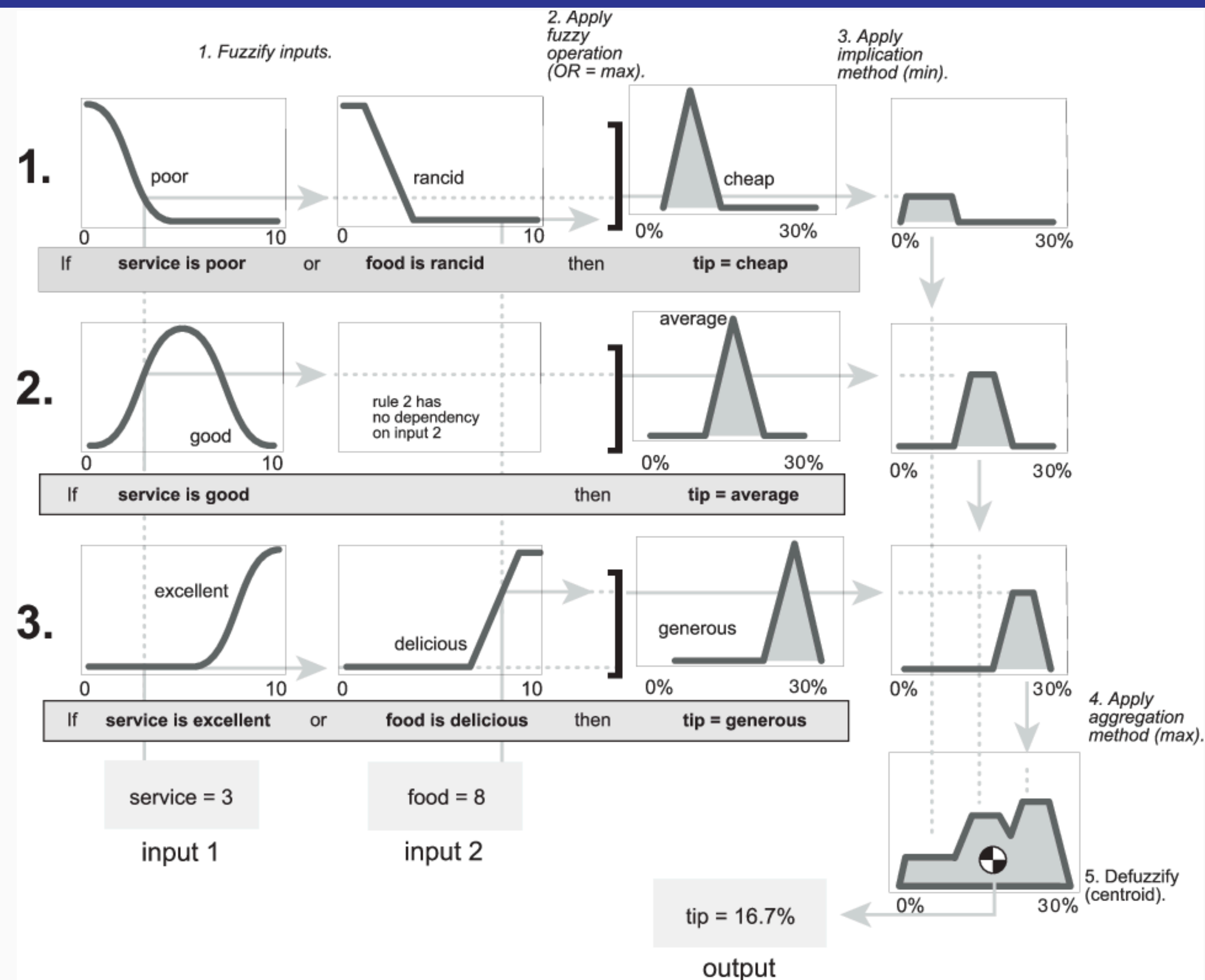
Mamdani FIS in Basic Tipping Problem



Mamdani FIS in Basic Tipping Problem

For detail explanation, click link

<https://www.mathworks.com/help/fuzzy/inference-process.html>



Implementation of mamdani for controlling a fan based on temperature

// This code implements a simple Mamdani fuzzy inference system for controlling a fan based on temperature input.

```
1  #include <iostream>
2  #include <algorithm>
3  using namespace std;
4
5  // Membership functions
6  float tempLow(float temp) {
7      if (temp <= 20) return 1;
8      else if (temp >= 30) return 0;
9      else return (30 - temp) / 10.0;
10 }
11
12 float tempHigh(float temp) {
13     if (temp <= 20) return 0;
14     else if (temp >= 30) return 1;
15     else return (temp - 20) / 10.0;
16 }
17
18 // Defuzzify using centroid method (simple 2-point average)
19 float defuzzify(float degreeFast, float degreeSlow) {
20     float fast = 90, slow = 30;
21     return (degreeFast * fast + degreeSlow * slow) / (degreeFast + degreeSlow);
22 }
23
24 int main() {
25     float temp;
26     cout << "Enter temperature (°C): ";
27     cin >> temp;
28
29     float low = tempLow(temp);
30     float high = tempHigh(temp);
31
32     float slow = low; // Rule: IF temp is Low → Fan is Slow
33     float fast = high; // Rule: IF temp is High → Fan is Fast
34
35     float output = defuzzify(fast, slow);
36
37     cout << "Fan speed: " << output << " %" << endl;
38     return 0;
39 }
```

Output

```
Enter temperature (°C): 28
Fan speed: 78 %
```



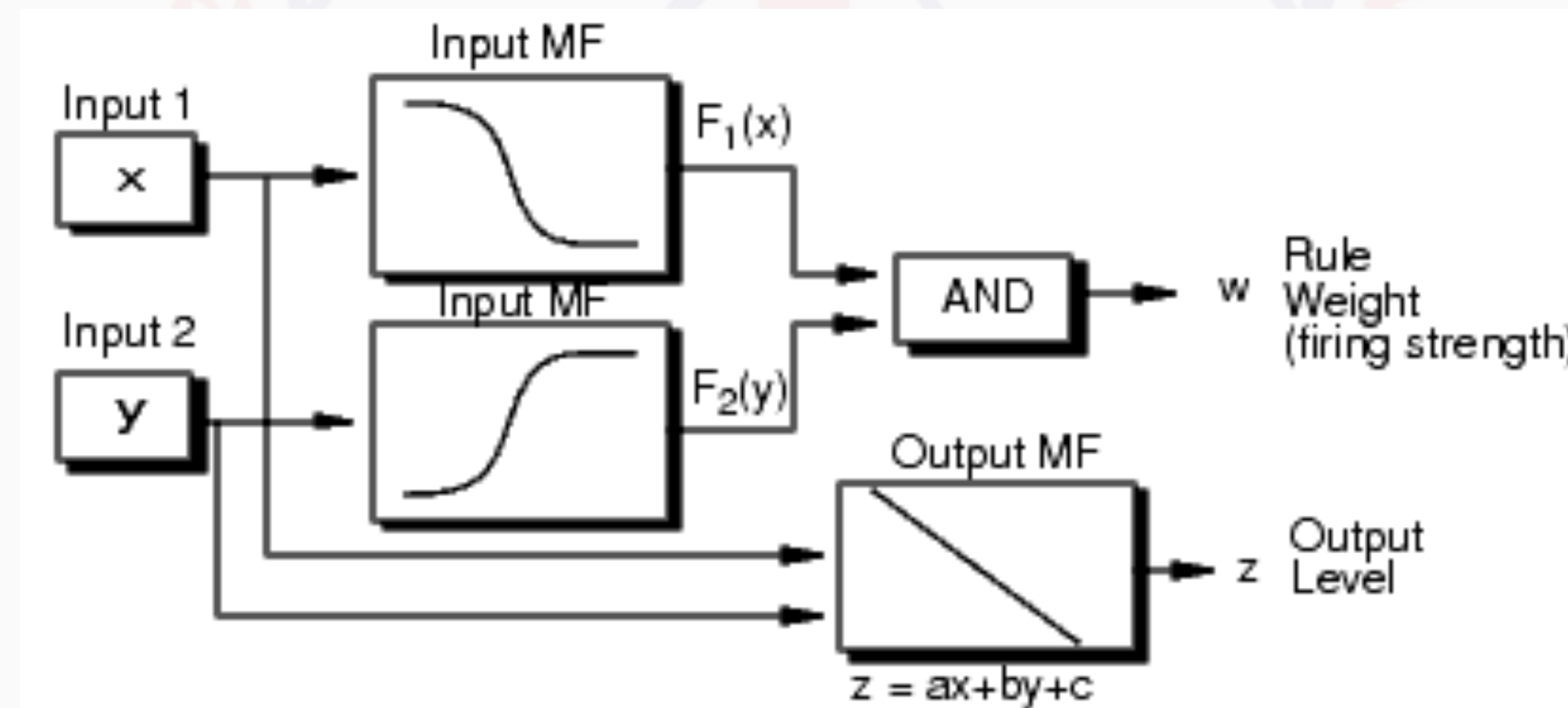
Sugeno FIS

- Sugeno fuzzy inference is also referred to as
 - Takagi-Sugeno-Kang fuzzy inference
- It uses ***singleton*** output membership functions
 - that are either ***constant*** or a ***linear function of the input values***.
- The defuzzification process
 - is more ***computationally efficient*** compared to that of a Mamdani system
 - Because → it uses a weighted average or weighted sum of a few data points rather than compute a centroid of a two-dimensional area.



Sugeno FIS

- Each rule generates: z_i and w_i (**Inputs**: x and y)
 - z_i = Rule output level (which is either a constant value or a linear function of the input values):
 - $z_i = a_i x + b_i y + c_i$
 - w_i = Rule firing strength derived from the rule antecedent:
 - $w_i = \text{AndMethod}(F_1(x), F_2(y))$
 - Here, $F_1(x)$ and $F_2(y)$ are the membership functions for inputs 1 and 2, respectively.



Sugeno FIS

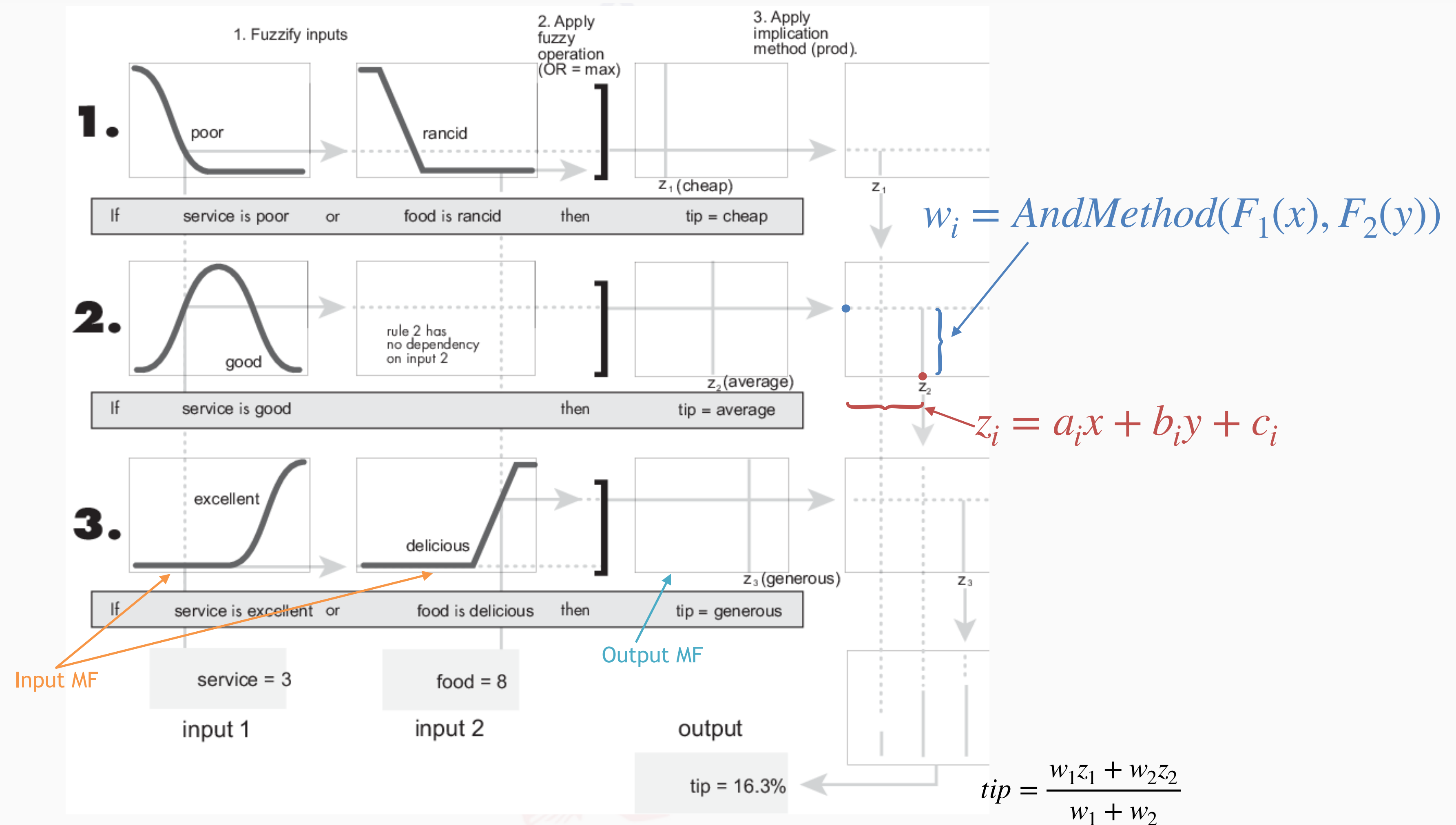
- Visualization of **first-order Sugeno systems** (a and b are nonzero):
 - the singleton output spikes can move around in a linear fashion within the output space, depending on the input values.
 - The rule firing strength then defines the size of the singleton spike.
- The final output of the system is the weighted average over all rule outputs:

$$\text{Final Output} = \frac{\sum_{i=1}^N w_i z_i}{\sum_{i=1}^N w_i}$$

- Where N is number of rules.



Sugeno FIS



Sugeno FIS- SUMMARY

- The **Sugeno Fuzzy Inference System** widely used fuzzy logic model, especially in control systems and data-driven modeling due to its efficiency and simplicity in generating crisp outputs.
- Rule Format:
 - Zero order Sugeno
 - IF x is A AND y is B THEN $z = c$
 - First Order Sugeno
 - IF x is A AND y is B THEN $z = a*x + b*y + c$

Where a , b , and c are real coefficients.



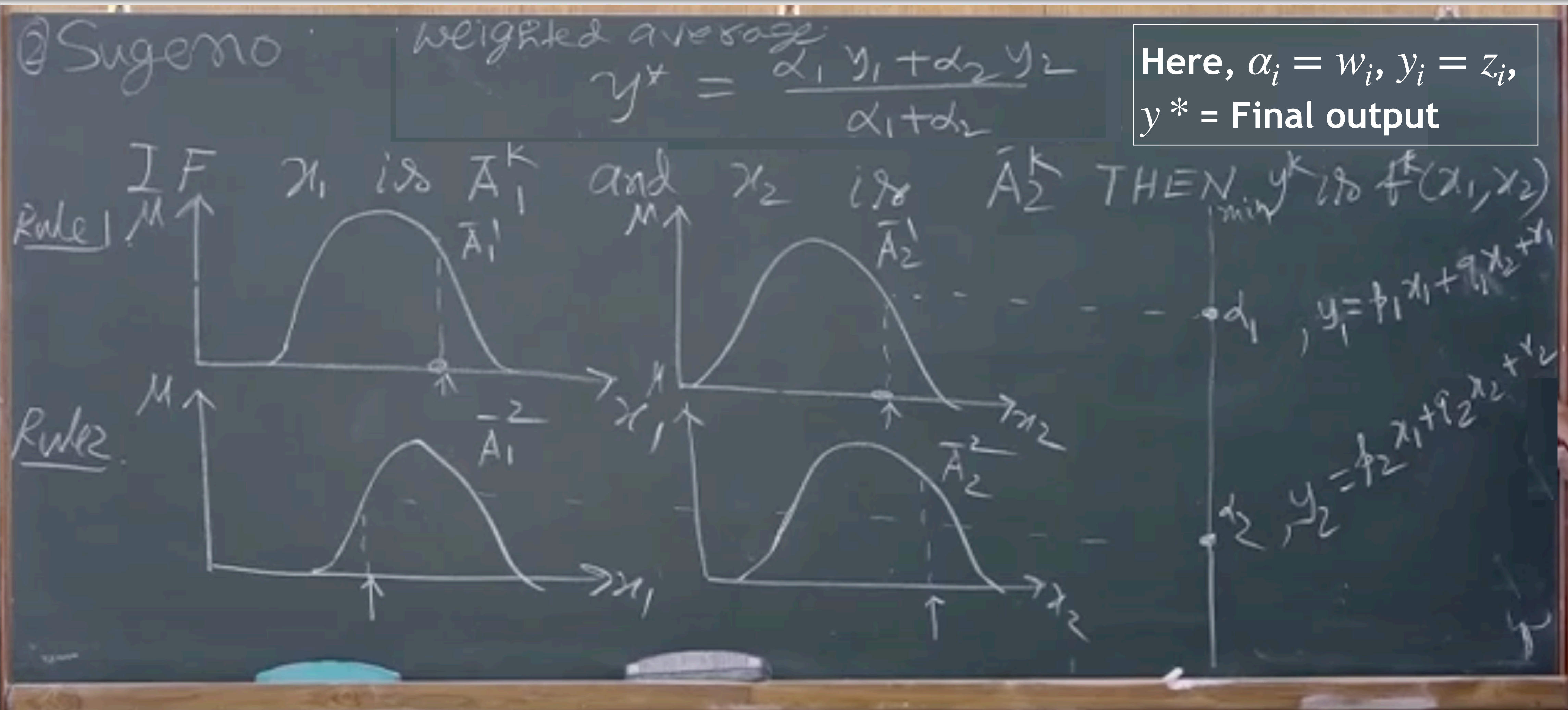
Sugeno FIS- Inference (Summary)

- Fuzzification
 - Compute membership degrees $\mu_A(x)$, $\mu_B(y)$
- Rule Evaluation
 - Compute rule firing strength, $w_i = \mu_{A_i}(x) \cdot \mu_{B_i}(y)$
- Rule Output
 - Compute $z_i = f_i(x, y)$ for each rule
- Aggregation
 - Compute the final output using **weighted average**:

- $$\text{Final Output } z^* = \frac{\sum_{i=1}^N w_i \cdot z_i}{\sum_{i=1}^N w_i}$$



Sugeno FIS- Inference (Summary)



Implementation of Sugeno FIS for controlling fan speed based on temperature

// This code implements a simple Sugeno fuzzy inference system for controlling a fan based on temperature input.

```
1  #include <iostream>
2  using namespace std;
3
4  // Membership functions
5  float tempLow(float temp) {
6      if (temp <= 20) return 1;
7      else if (temp >= 30) return 0;
8      else return (30 - temp) / 10.0;
9  }
10
11 float tempHigh(float temp) {
12     if (temp <= 20) return 0;
13     else if (temp >= 30) return 1;
14     else return (temp - 20) / 10.0;
15 }
16
17 int main() {
18     float temp;
19     cout << "Enter temperature (°C): ";
20     cin >> temp;
21
22     float w1 = tempLow(temp);
23     float w2 = tempHigh(temp);
24
25     float z1 = 20;           // Constant output
26     float z2 = 0.5 * temp + 10; // Linear output
27
28     float output = (w1 * z1 + w2 * z2) / (w1 + w2);
29
30     cout << "Fan speed: " << output << " %" << endl;
31     return 0;
32 }
```

Output

```
Enter temperature (°C): 28
Fan speed: 23.2 %
```



Tsukamoto FIS

- The Tsukamoto Fuzzy Inference System (FIS) is proposed by Y. Tsukamoto in 1979.
- It combines elements of Mamdani-style rule structure with a distinct method of generating crisp outputs.
- Rule format: IF x is A AND y is B THEN z is C
 - A and B are fuzzy sets for inputs.
 - C is a monotonically increasing or decreasing fuzzy set (output).
 - The rule computes a **single crisp value** z by inverting C using the rule's firing strength.



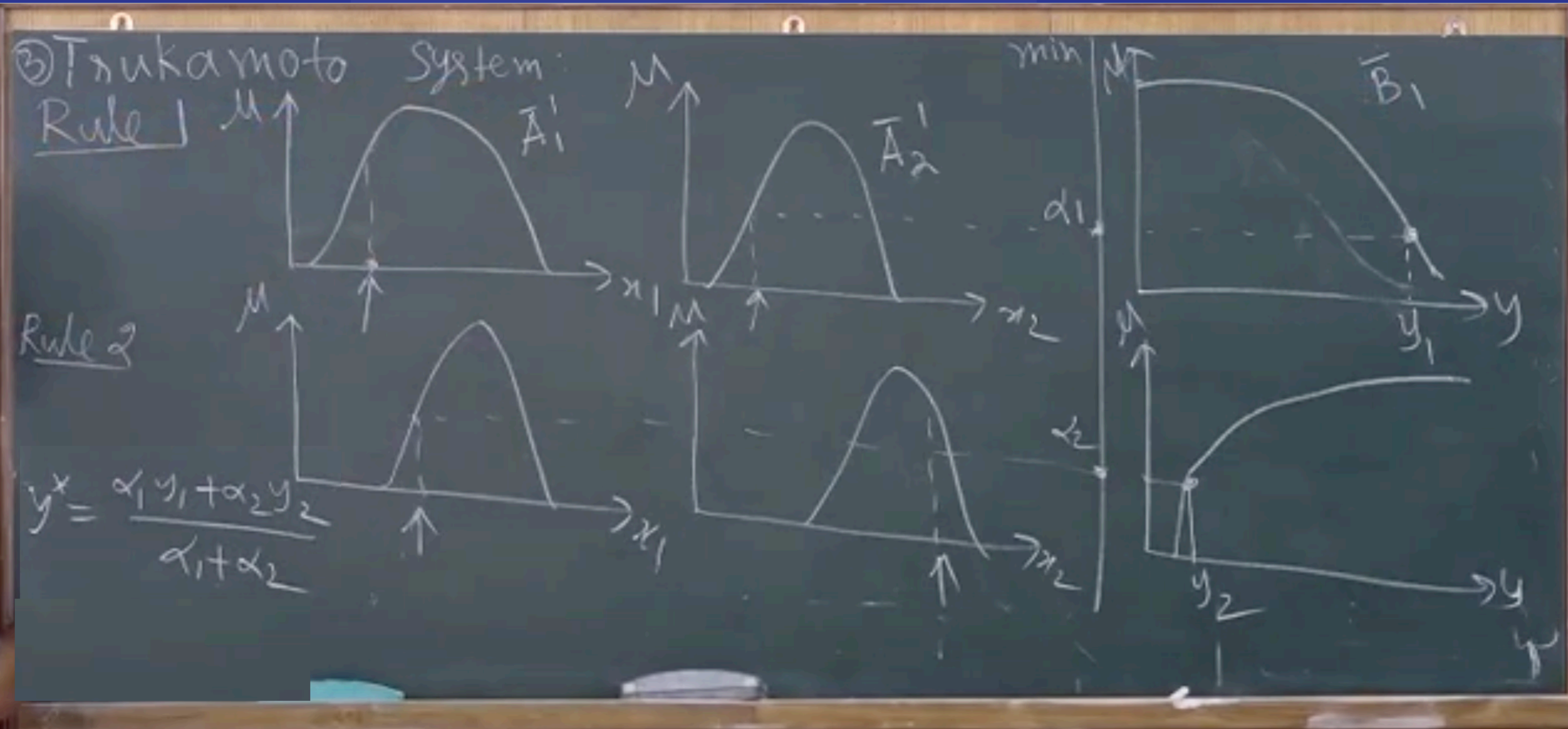
Tsukamoto FIS- Inference Steps

- Fuzzification
 - Determine the membership value of each input variable using its respective membership function
- Rule Evaluation (Firing Strength)
 - Combine antecedents using a t-norm (e.g., min or product) to get the rule firing strength α .
- Output Calculation (y)
 - Invert the monotonic output membership function $\mu_C(y) = \alpha$ to get the crisp y value.
- Aggregation of Output
 - Combine all rule outputs using weighted average:

- $$Final\ Output = \frac{\sum_{i=1}^N \alpha_i \cdot y_i}{\sum_{i=1}^N \alpha_i}$$



Tsukamoto FIS



Implementation of Tsukamoto FIS for controlling fan speed based on temperature

```
// This code implements a simple Tsukamoto fuzzy inference system for
controlling a fan based on temperature input.
1  #include <iostream>
2  #include <cmath>
3  using namespace std;
4
5  // Input Membership Functions
6  float tempLow(float t) {
7      if (t <= 20) return 1;
8      else if (t >= 30) return 0;
9      else return (30 - t) / 10.0;
10 }
11
12 float tempHigh(float t) {
13     if (t <= 20) return 0;
14     else if (t >= 30) return 1;
15     else return (t - 20) / 10.0;
16 }
17
18 // Inverse output membership functions
19 float inverseSlow(float mu) { // Fan speed: 0 (slow) to 50
20     return 50 - mu * 50;
21 }
22
23 float inverseFast(float mu) { // Fan speed: 50 to 100
24     return mu * 50 + 50;
25 }
26
27 int main() {
28     float temp;
29     cout << "Enter temperature (°C): ";
30     cin >> temp;
31
32     float w1 = tempLow(temp);
33     float w2 = tempHigh(temp);
34
35     float z1 = inverseSlow(w1);
36     float z2 = inverseFast(w2);
37
38     float output = (w1 * z1 + w2 * z2) / (w1 + w2);
39
40     cout << "Fan speed: " << output << " %" << endl;
41     return 0;
42 }
```

Output

```
Enter temperature (°C): 28
Fan speed: 80 %
```



Case Study

- Consider a simple tipping problem. Design and Implement the mamdani, sugeno and Tsukamoto inference systems for this problem (Use c++ or python)



Fuzzy Logic

The idea of fuzzy logic was invented by Professor L. A. Zadeh of the University of California at Berkeley in 1965.

This invention was not well recognized until Dr. E. H. Mamdani, who is a professor at London University, applied the fuzzy logic in a practical application to control an automatic steam engine in 1974, which is almost ten years after the fuzzy theory was invented.

Then, in 1976, Blue Circle Cement and SIRA in Denmark developed an industrial application to control cement kilns. That system began to operation in 1982.

More and more fuzzy implementations have been reported since the 1980s, including those applications in industrial manufacturing, automatic control, automobile production, banks, hospitals, libraries and academic education.

Fuzzy logic techniques have been widely applied in all aspects in today's society.

References (Fuzzy Logic)

1. Bai, Ying & Wang, Dali. (2007). Fundamentals of Fuzzy Logic Control – Fuzzy Sets, Fuzzy Rules and Defuzzifications. 10.1007/978-1-84628-469-4_2.
2. <https://www.cs.princeton.edu/courses/archive/fall07/cos436/HIDDEN/Knapp/fuzzy004.htm>
3. [https://en.wikipedia.org/wiki/Membership_function_\(mathematics\)](https://en.wikipedia.org/wiki/Membership_function_(mathematics))
4. <https://cse.iitkgp.ac.in/~dsamanta/courses/archive/sca/Archives/Chapter%203%20Fuzzy%20Membership%20Functions.pdf>
5. Fuzzy rule based system and FISs- mamdani, surgeon and tsukamoto <https://www.youtube.com/watch?v=Qf8xVriv4ik>
6. FIS- Mamdani <https://www.youtube.com/watch?v=fqZvzQayx7A>
7. [https://nms.kcl.ac.uk/hk.lam/HKLam/images/HKLam/Introduction to Fuzzy Inference Systems.pdf](https://nms.kcl.ac.uk/hk.lam/HKLam/images/HKLam/Introduction%20to%20Fuzzy%20Inference%20Systems.pdf)
8. <https://www.mathworks.com/help/fuzzy/fuzzy-inference-process.html>
9. <https://www.mathworks.com/help/fuzzy/types-of-fuzzy-inference-systems.html#>





THANK YOU